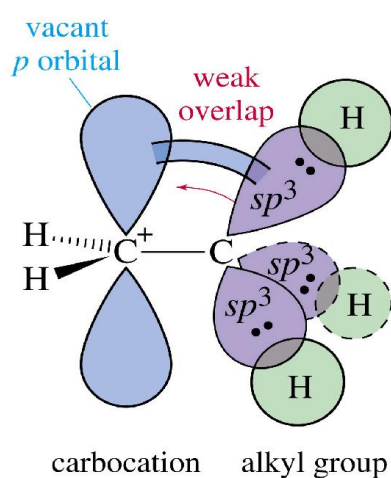


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1. GENERAL ORGANIC CHEMISTRY

1.1 Introduction

In 1807, Berzelius proposed the term 'Organic Chemistry' for the study of compounds derived from natural sources. This was based on the theory of **vitalism** which said that all living systems possessed a 'vital force' which was absent in non-living systems. Compounds derived from living natural sources (organic) were thought to be fundamentally different from inorganic compounds.

Vital
Force

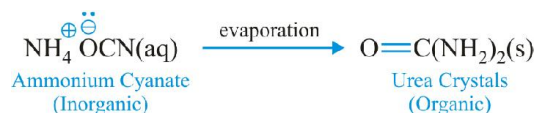
Living Systems

No Vital
Force

Non-living Systems

The vital force could be philosophically thought as the mysterious force God instilled in the living systems.

In 1823, Friedrich Wohler joined Berzelius as his student. In 1828, Wohler made a discovery which changed the definition of organic chemistry. Wohler conducted the following experiment.



Wohler successfully synthesized an organic compound starting from an inorganic compound. Following this, many others synthesized organic compounds starting from inorganic compounds. Thus, the theory of vitalism and the definition of organic chemistry lost its meaning.

But what was common in all the above compounds synthesized was the presence of carbon. Carbon shows a special property **catenation**. Carbon can connect with other carbon atoms to form long chains and rings (self-catenation) and can connect with atoms of many other elements in the periodic table (cross-catenation). Because of this reason, carbon can form a wide variety of compounds. Therefore, **the modern definition of organic chemistry is the study of carbon compounds.**

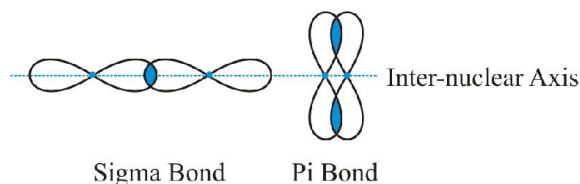
Probably, the vital force can be explained by the fact that most of the life-giving and life-sustaining functions are performed by carbon compounds, for example, the human tissues and skin are formed by proteins, respiration is possible due to haemoglobin, the information in our genes is carried out in the form of DNA/RNA etc.

General Organic Chemistry is the detailed study of the basic concepts and factors that govern the progress and outcome of reactions.



The making and breaking of bonds usually occurs in several discrete steps before transforming into products. The detailed sequential description of all the steps is called the **mechanism** of the reaction.

2.1 Sigma and Pi Bonds - Comparison



Property	Sigma (σ) Bond	Pi (π) Bond
Overlap	Axial/Head-on	Parallel/Lateral/Side-ways
Electron Cloud	Along the inter-nuclear axis	Perpendicular to the inter-nuclear axis
Bond Strength	Stronger	Weaker

2.2 Structural Formulas

Several kinds of formulas are used by organic chemists to represent organic compounds.

2.2.1 Complete Formulas

Complete formulas are lewis structures which shows all bond pair of electrons as a dash (–). Lone pair of electrons are shown as a pair of dots.

2.2.2 Condensed Formulas

Condensed formulas are written without showing all the individual bonds. Each central atom is shown together with the atoms that are bonded to it.

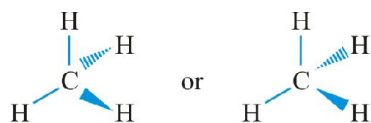
2.2.3 Line-Angle Formulas

These are also called **skeletal structures** or a **stick figure**. Line-angle formulas are often used for cyclic compounds and occasionally for non-cyclic ones. Bonds are represented by lines, and carbon atoms are assumed to be present where two lines meet or a line begins or ends. Hydrogens are generally implicit in these drawings.

2.2.4 Tetrahedral Representation

This is generally the three-dimensional (3-D) representation of molecules. Dashed Wedge (▤) or solid wedge (◼) are used to indicate bonds projecting behind the plane

(away from the observer) and out of the plane (towards the observer) respectively. Bonds lying in the plane of paper are depicted by using a normal line (—).

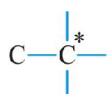


2.3 Degrees of Carbon

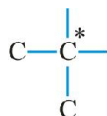
It is defined as the number of carbons attached to carbon under observation.



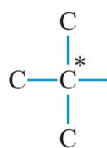
Methyl



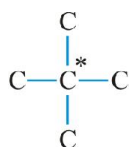
1° (Primary)



2° (Secondary)



3° (Tertiary)



4° (Quaternary)

2.4 Hybridization

Hybridisation is a process in which two or more atomic orbitals of comparable energy of the valence-shell of an atom (**central atom** of the molecule or ion) either in its ground state or in its excited state mix together and give rise to the formation of new degenerate orbitals which are called **hybrid orbitals**.

2.5 Applications of Hybridization

Hybridization	% s-character
sp^3	25.0
sp^2	33.3
sp	50.0

2.5.1 Size of Hybrid Orbitals

As % s-character increases, size of hybrid orbital decreases. Therefore

$$\text{Size of Hybrid Orbital : } sp^3 > sp^2 > sp$$

2.5.2 Electronegativity of Hybrid Orbitals

As % s-character increases, electronegativity of hybrid orbital increases. Therefore

$$\text{EN of Hybrid Orbital : } sp > sp^2 > sp^3$$

2.6 Dienes

Dienes are organic compounds containing two double bonds. There are three types of dienes :

(a) Isolated (b) Conjugated (c) Cumulated

2.6.1 Isolated Diene

In this case, double bonds are separated by atleast one sp^3 carbon.

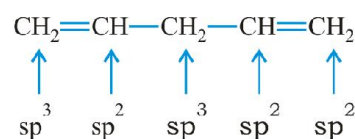
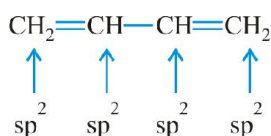


Table : Hybridization of Common Molecules.

Molecule	σ	π	LP	Hybridization	Geometry	Shape
$*CH_4$	4	0	0	sp^3	Tetrahedral	Tetrahedral
$*NH_3$	3	0	1	sp^3	Tetrahedral	Trigonal Pyramidal
$H_2\ddot{O}^*$	2	0	2	sp^3	Tetrahedral	V-shape or Bent
$*CH_2=CH_2$	3	1	0	sp^2	Trigonal Planar	Trigonal Planar
$*CH\equiv CH$	2	2	0	sp	Linear	Linear

2.6.2 Conjugated Diene

Double bonds are separated by only one single bond (or 4 sp^2 carbons in a row).



2.6.3 Cumulated Diene

Both sets of double bonds are at the same carbon.



A substituted **allene**

An allene is $\text{CH}_2 = \text{C} = \text{CH}_2$

2.6.4 Stability of Dienes

The relative stabilities of dienes follows the order

Conjugated > Isolated > Cumulated

Important: Stability $\propto \frac{1}{\text{Energy Content of the molecule}}$

2.7 Commonly Occurring Forms of Carbon

The commonly occurring forms of carbon are

- (a) Diamond (b) Graphite (c) Carbides
(d) Fullerenes (e) Charcoal



Diamond - Each C is sp^3 . Tetrahedral solid.
Graphite - Each C is sp^2 . Layered solid with weak van der Waal's forces between layers.
Calcium Carbide - Each C is sp .
Fullerene - Each C is sp^2 .

3. BREAKING OF BONDS

In organic chemistry, the bond that is important for the study of reactions is covalent bond. We, therefore, study ways in which a covalent bond can be broken.

- (a) Homolytic Fission (b) Heterolytic Fission

3.1 Homolytic Fission or Homolytic Cleavage

In this kind of bond breaking, each atom separates with one electron, leading to the formation of highly reactive species known as radicals (or free radicals).



The bond breaking is shown by two **half-headed or fish-hook** arrows. A half-headed arrow shows the movement of one electron.

Radicals are **neutral** and are **odd electron** species.

3.2 Heterolytic Fission or Heterolytic Cleavage

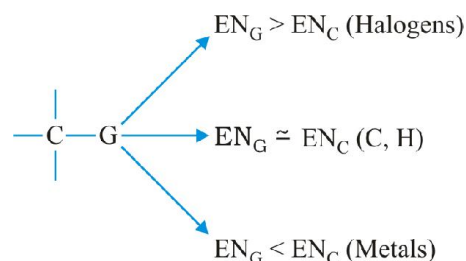
In this type of covalent bond breaking, the shared pair of electrons are transferred to the more electronegative part. Therefore, this fission leads to the formation of **a cation and an anion (ion-pair)**.



The bond breaking is shown by a **full-headed** arrow. A full headed arrow shows the movement of a pair of electrons. In organic chemistry, the movement of electrons is always shown by curved arrows - half-headed or full-headed arrows.

4. INDUCTIVE EFFECT

When two unlike atoms form covalent bond, the electron-pair forming the sigma bond is never shared equally between the two atoms but is shifted slightly towards the more electronegative species.



There are broadly three types of groups/atoms that may be attached to carbon as illustrated. Although C is more electronegative than H, the electronegativity difference is small and the bond is generally considered non-polar.

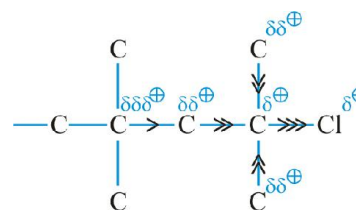
4.1 Nature of Inductive Effect

Inductive effect is a **permanent effect** and can be directly correlated to its dipole moment.

It is a weak effect as the shifting of electrons takes place only through sigma bonds.

4.2 Effect of branched carbon chain

An illustration has been marked for operation of inductive effect which is self-explanatory.



More the number of δ , lesser the effect

4.3 Electron Donating and Electron withdrawing Groups

Inductive effect may be due to single atom or a group of atoms. Relative inductive effects are measured with reference to hydrogen. Those which donate electrons to carbon chain are called **electron-donating groups (EDG)** or **electron-releasing groups (ERG)** and are said to exert **+I effect**. Those which withdraw electrons from carbon chain are called **electron-withdrawing groups (EWG)** and are said to exert **-I effect**.

Important :

1. I.E. of alkyl groups: $3^\circ > 2^\circ > 1^\circ > \text{CH}_3-$
2. In general, greater is the number of carbons in an alkyl group, greater is its +I effect.
3. For problem-solving, we take electronegativity of sp^3 hybridized carbon to be more than sp^3 hybridized nitrogen.

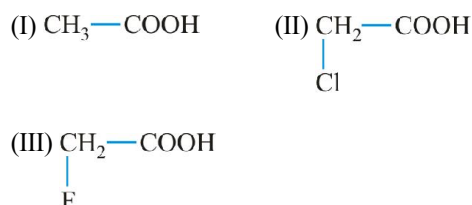
4.4 Applications of Inductive Effect

4.4.1 Effect on Acidic/Basic Strength

EWG increases acidic strength and decreases basic strength. ERG decreases acidic strength and increases basic strength.

Example - 1

Compare the acidic strength :

**Solution :**

An alkyl group is donating only if no other EWG is present on it. Therefore, groups like $-\text{CH}_2\text{Cl}$ and $-\text{CH}_2\text{F}$ become electron withdrawing groups.

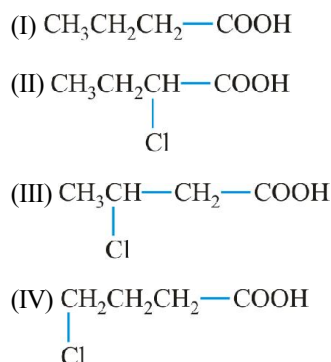
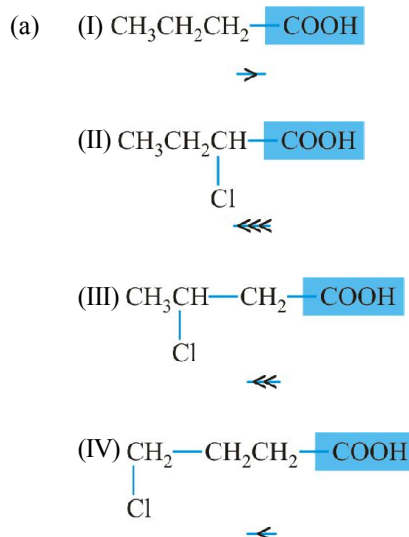
Order of Acidic Strength : $\text{III} > \text{II} > \text{I}$

4.4.2 Effect of Distance

If the ERG/EWG moves away, the inductive effect diminishes.

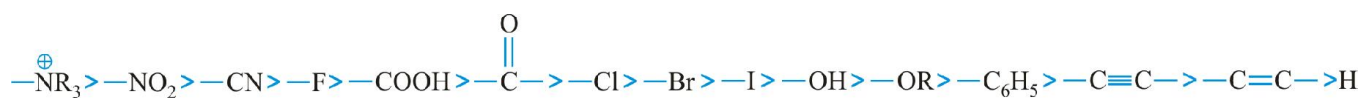
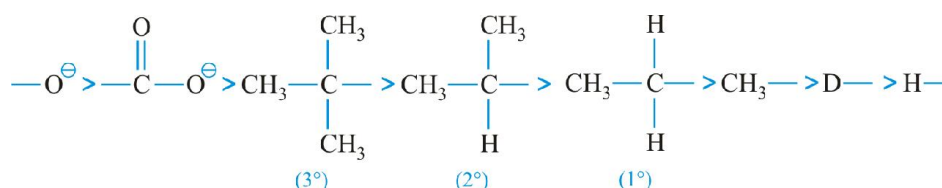
Example - 2

- (a) Compare the acidic strength of :

**Solution :**

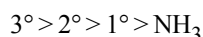
Order of acidic strength : $\text{II} > \text{III} > \text{IV} > \text{I}$

Series of +I and -I groups in order of their strength

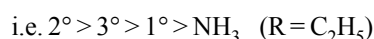
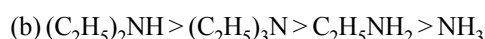
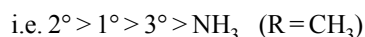
-I Series (EWG)**+I Series (ERG)**

4.4.3 Basicity of Amines

The order in **gaseous or liquid phase** is :



To determine the basic strength of amines in **aqueous phase**. We have to consider **inductive effect, solvation effect and steric hinderance**. The order of basic strength is therefore experimental in aqueous state as we can't give priority to stability provided by any one factor. Two results are important for aqueous phase :



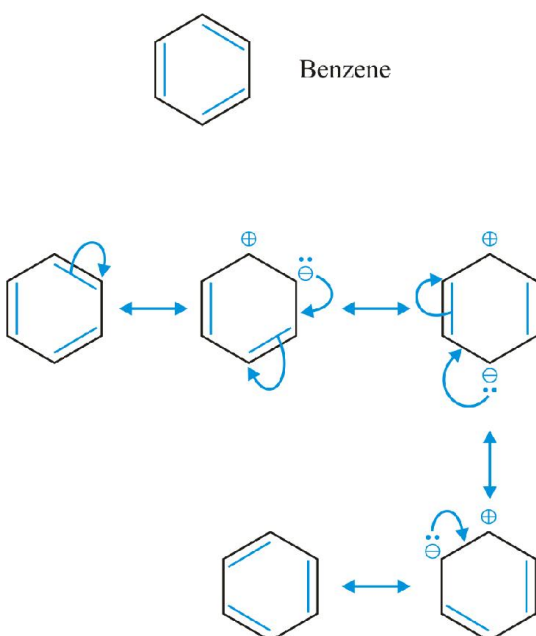
5. RESONANCE

Molecules are generally represented by simple lewis structures but some molecules can not be represented by just one Lewis structure. This led to the discovery of resonance. Resonance refers to the delocalization of electrons (generally π -electrons).

5.1 Conjugated Systems

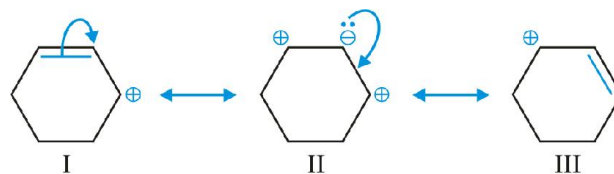
5.1.1 Pi alternate Pi

Example - 3



5.1.2 Pi alternate Positive Charge

Example - 4



5.1.3 Pi alternate Negative Charge

Example - 5



5.1.4 Pi alternate Odd Electron

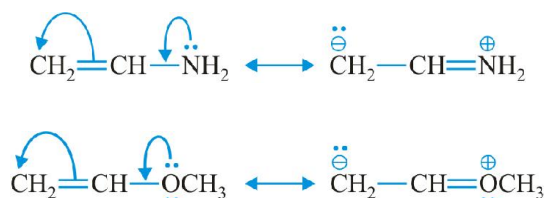
Example - 6



5.1.5 Pi alternate Lone Pair

This case is similar to 'pi alternate negative charge' as lone pair and negative charge are treated similarly.

Example - 7



5.1.6 Lone Pair and Positive Charge on Adjacent Atoms

Example - 8



5.2 Rules for Validity of Lewis Structures

Rule-1 :

All the lewis structures must conform to lewis octet rule.

Rule-2 :

Position of atoms in all resonating structures must be the same. Only the electrons move.

Rule-3 :

All the resonating structures must have the same number of paired and unpaired electrons, i.e. sum of bond pairs and lone pairs must be constant.

Rule-4 :

All the atoms participating in resonance in a molecule must be coplanar. This is required for the effective overlap of p orbitals and the delocalization of electrons, for eg, buta-1,3-diene.

5.3 Criteria for Major/Minor Contributors

Resonance forms can be compared using the following criteria in the following order :

- As many octets as possible (a neutral molecule is always more stable in which its octet is complete).
- As many π bonds as possible.
- Negative charge on more electronegative atom is stable.
- Charge separation.
 - Similar charges - Keep them as FAR as possible to minimize repulsion and instability.
 - Opposite charges - Keep them as NEAR as possible to maximize attraction and stability.

Example - 9

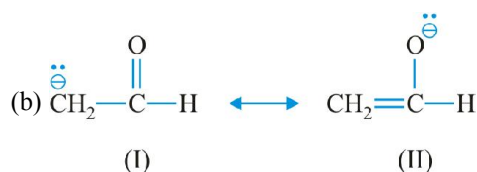
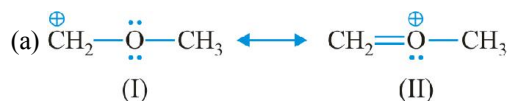
Which of the following structures is more stable ?

**Solution :**

II is more stable as all the octets are complete.

Example - 10

Which of the following is more stable in the following pairs ?

**Solution :**

- (a) In II, all octets are complete. Therefore, II is more stable.

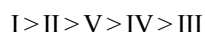
- (b) I and II are tied on octets and number of π bonds but negative charge is more stable on more electronegative atom. Hence, II is more stable.

Example - 11

Give the order of stability of following resonating structures

**Solution :**

In (I), there are maximum number of pi bonds. Therefore, it is most stable. In (II) and (V), the number of pi bonds is equal but charge separation is greater in (V). Therefore, (II) is more stable than (V). In (III) and (IV), there is maximum charge separation but (III) is highly unstable due to electrostatic repulsion. Hence, the order of stability is :

**6. MESOMERIC EFFECT**

The permanent polarization, due to a group conjugated with a π bond or a set of alternate π bonds, is transmitted through the π electrons of the system-resulting in a different distribution of electrons in the unsaturated chain.

This kind of electron redistribution in unsaturated compounds conjugated with electron-releasing or electron-withdrawing groups (or atoms) is called **Mesomeric Effect** or **Resonance Effect**.

This effect is **permanent** and is indicated by the dipole moment.

6.1 Electron-Releasing and Electron-Withdrawing Groups

Groups which release or withdraw electrons by resonance are said to exert M or R effect.

6.1.1 Electron-Releasing Groups (+R or +M effect)

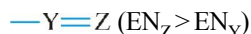
The common thing about all the groups listed is that the atom connected with the conjugated system has a lone pair to donate. Therefore, a generic representation can be



6.1.2 Electron-Withdrawing Groups (–R or –M effect)



The common thing about all the groups listed is that the atom connected with the conjugated system has a π bond with another more electronegative atom which withdraws the electrons or directly has a positive charge on them. Therefore, a generic representation can be



6.1.3 Dual Behaviour

Groups such as $-\text{N}=\text{O}$ are both electron-releasing and electron-withdrawing as illustrated.

Example - 12

As electron releasing group



As electron withdrawing group



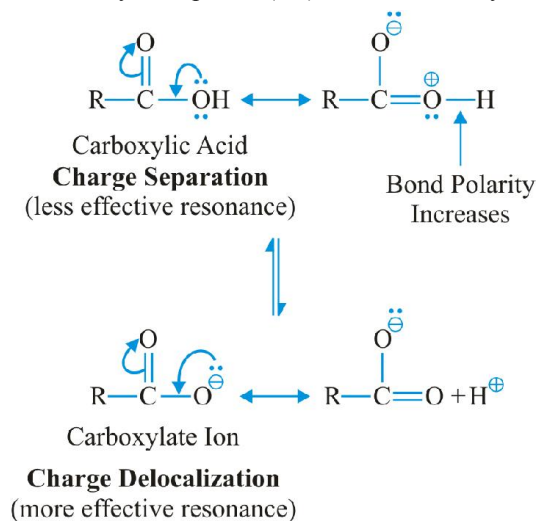
Which behaviour dominates and which is used in a particular context will be discussed later in **Electrophilic Aromatic Substitution** later.

Resonance Effect does NOT depend upon distance unlike inductive effect.

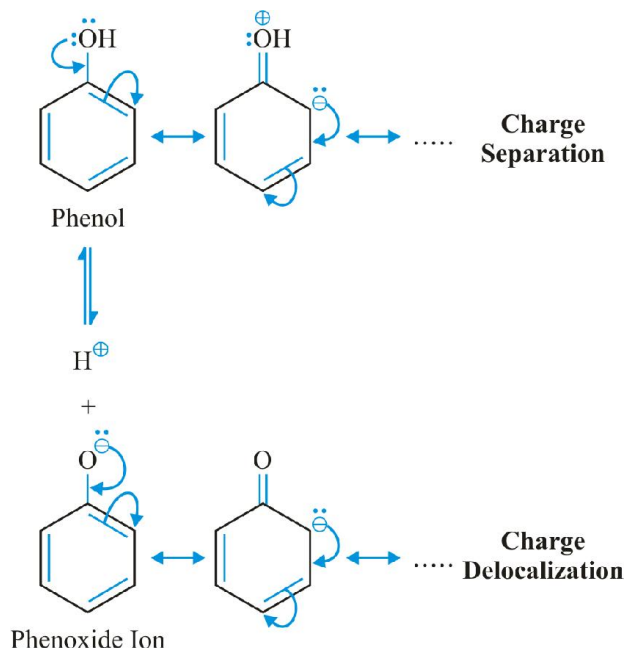
6.2 Applications of Mesomeric Effect

6.2.1 Effect on Acidic Strength of Carboxylic Acids and Phenols

The resonating structure of carboxylic acid leads to charge-separated structure which is less stable than the carboxylate ion in which charge is delocalized. Therefore, carboxylic acid readily loses proton (H^+) to form a carboxylate ion.

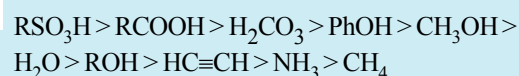


Similarly, in phenol, resonance leads to charge separation which increases the rate of ionization and forms phenoxide ion which is stabilized by charge delocalization.



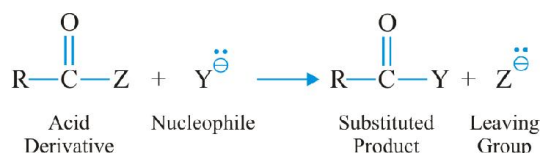
Note...

Order of acidic strength



6.2.2 Effect on Reactivity of Carboxylic Acid Derivatives

A typical nucleophilic reaction is represented as :



The stronger is the bond between C and Z, the difficult it is for a nucleophile to break a bond and therefore, lower reactivity.



Reactivity order of carboxylic acid derivatives towards nucleophilic acyl substitution is :

Acyl Chloride > Acid Anhydride > Ester > Amide

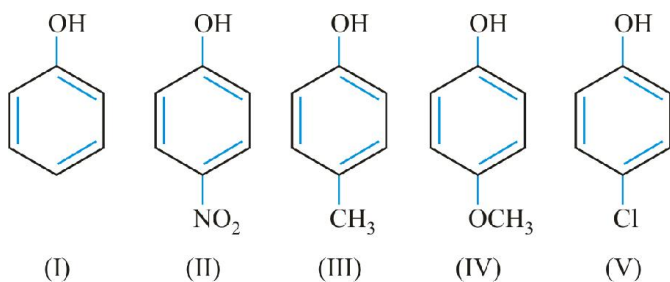
6.2.3 Effect of ERG/EWG on Acidic/Basic Strength

EWG increases the acidic strength and decreases the basic strength.

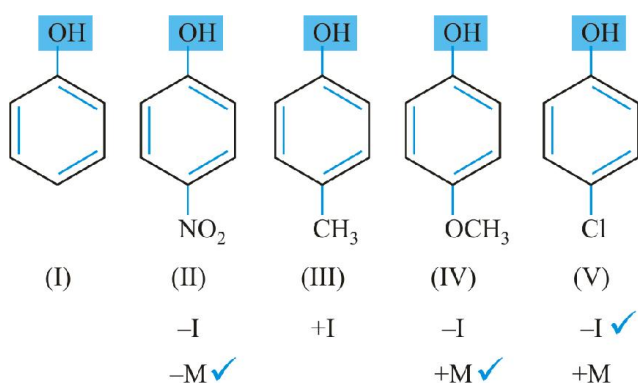
ERG decreases the acidic strength and increases the basic strength.

Example - 13

Arrange the following in the order of decreasing acidic strength :

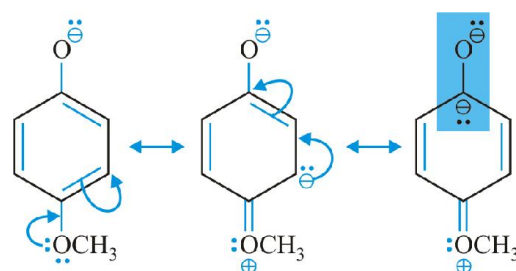
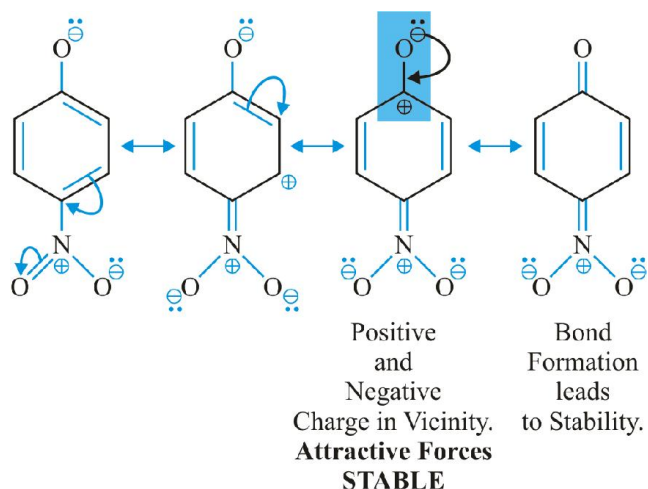


Solution :



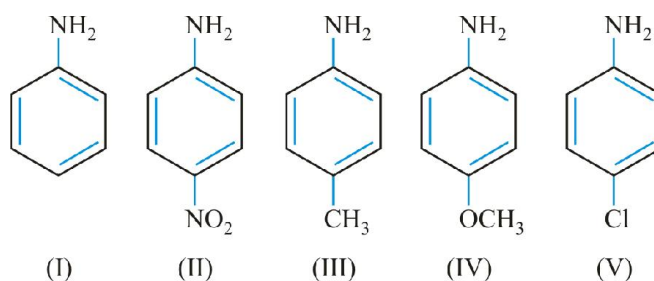
The order of acidic strength is : II > V > I > III > IV

In the previous example, let's also discuss the stability of phenoxide ions corresponding to (II) and (IV).

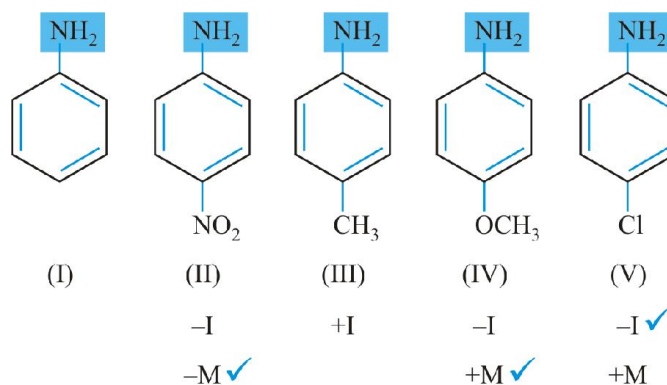


Example - 14

Arrange the following in decreasing order of basic strength

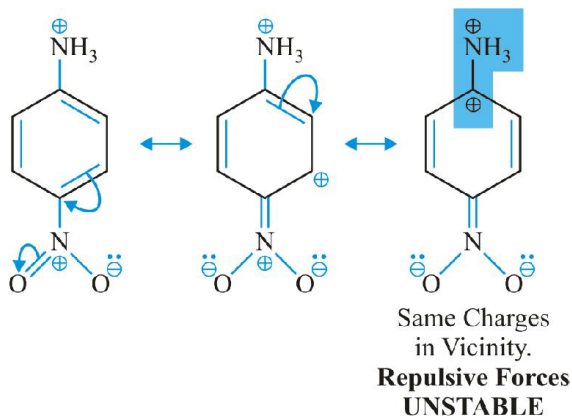


Solution :

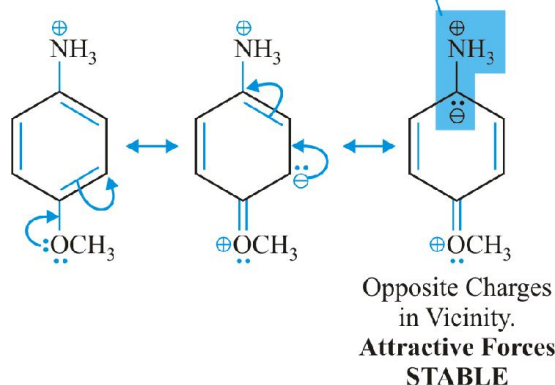


Therefore, order of basic strength is : IV > III > I > V > II

Let's also discuss the stability of anilinium ions corresponding to (II) and (IV).



Do NOT form a bond between N & C which renders nitrogen pentavalent.

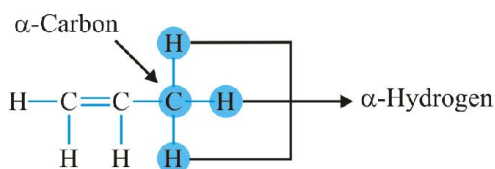


7. HYPERCONJUGATION

Hyperconjugation is the ability of the σ bond electrons of an α C—H bond to undergo conjugation with the adjacent π electrons. It is also known as **Baker-Nathan Effect**, **No-Bond Resonance** and **σ - π Effect**.

7.1 α -Carbon and α -Hydrogen

We have already discussed the α , β , γ nomenclature. Let's take an example :



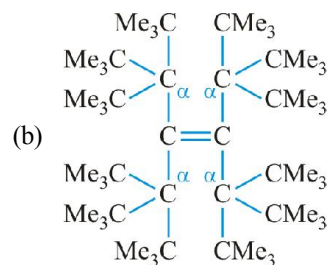
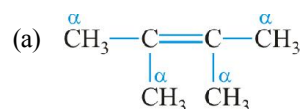
α -Carbon is the carbon attached to a functional group such as $C=C$. The hydrogen attached to α -carbon is called α -hydrogen. **For an α C—H bond to be eligible for hyperconjugation, α C must be sp^3 hybridized.**

Example - 15

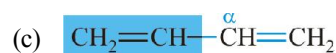
Mark the number of α -C and α -H in the given compounds

- $(CH_3)_2C=C(CH_3)_2$
- $[((CH_3)_3C)_3C]_2C=C[C(C(CH_3)_3)_3]_2$
- $CH_2=CH-CH=CH_2$
- $CH_2=CH-CH_2-\underset{\text{CH}_3}{\text{CH}}-CH=CH_2$

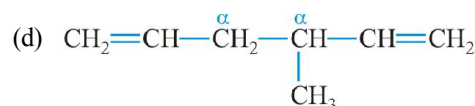
Solution :



α C = 4, α H = 0

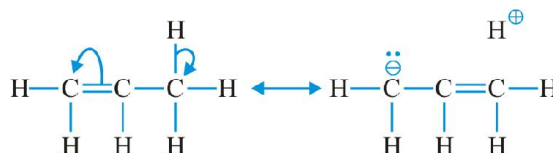


α C = 1, α H = 1 but since α C is sp^2 hybridized, therefore, it won't participate in hyperconjugation. Therefore, α H = 0 that will participate in hyperconjugation.

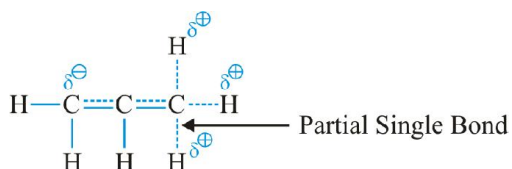


α C = 2, α H = 3

7.2 Mechanism of Electron Donation in Hyperconjugation

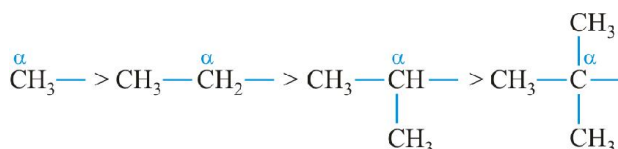


The hybrid formed by these resonating structures better known as hyperconjugating structures is :



Now, greater the number of α -H, greater the number of hyperconjugating structures and more is the electron donation of alkyl group to π bond.

The order of electron-donation of alkyl groups based on hyperconjugation is :



Methyl > 1° > 2° > 3°

Note... More is the number of α -H, more is the π bond delocalized. This implies that more will be the stability of alkene and less will be the heat of hydrogenation and more is the **no-bond resonance energy**.

7.3 Applications of Hyperconjugation

7.3.1 Stability of Alkenes

More is the number of α -hydrogen, more is the number of hyperconjugating structure and therefore more stability and greater no bond resonance.

Example - 16

Which alkene is more stable ?

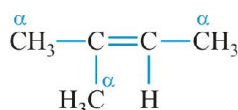


(I)



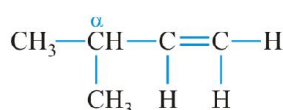
(II)

Solution :



(I)

$\alpha\text{C} = 3, \alpha\text{H} = 9$



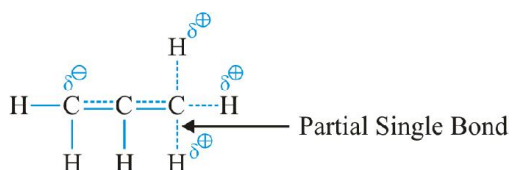
(II)

$\alpha\text{C} = 1, \alpha\text{H} = 1$

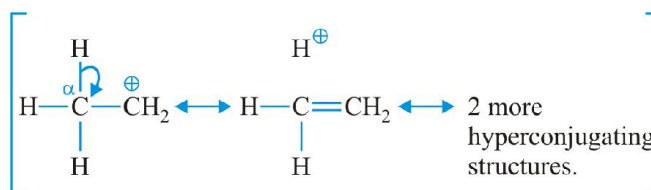
I is more stable than II.

7.3.2 Acidic Character of Alkenes

Hyperconjugation weakens the $\alpha\text{C}-\text{H}$ bond in hyperconjugation hybrid (partial single bond) and therefore αH can be lost easily.

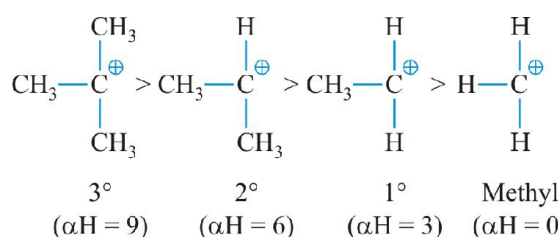


7.3.3 Stability of Carbocations



The positive charge on C is delocalized over αH to give stability to the carbocation. More is the number of αH , more is the stability of carbocations.

Stability of Carbocations



8. ELECTROMERIC EFFECT

Electromeric effect is observed only in the presence of a reagent and is therefore, a temporary effect. When a reagent approaches a molecule, the multiple bond such as $\text{C}=\text{C}$ or $\text{C}=\text{O}$ is polarized by the complete transfer of π electrons.



When the multiple bond is between two unlike atoms, the shift of electrons takes place towards more electronegative atom.



9. COMPARISON OF INDUCTIVE, HYPERCONJUGATION AND RESONANCE EFFECTS

Inductive Effect is a $\sigma-\sigma$ interaction and acts through **strong sigma bonds**.

Resonance/Mesomeric Effect is a $\pi-\pi$ interaction and acts through **weak pi bonds**.

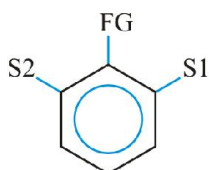
Hyperconjugation is a $\sigma-\pi$ interaction and acts through a **strong sigma and a weak pi bond**.

Therefore, the order of importance is :

Resonance > Hyperconjugation > Inductive

10. STERIC INHIBITION OF RESONANCE (SIR)

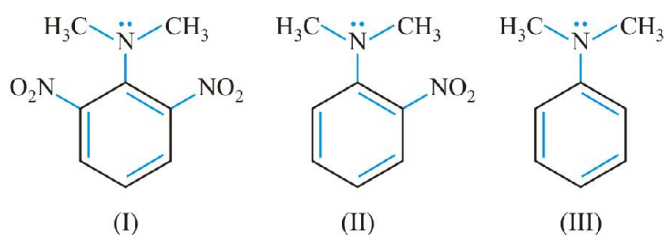
When both the **ortho** positions of a **bulky functional group** are occupied by **bulky substituents**, all the three groups are out of plane of the benzene ring.



FG, S1, S2 - Bulky Groups

Example - 17

Mark the order of basic strength :



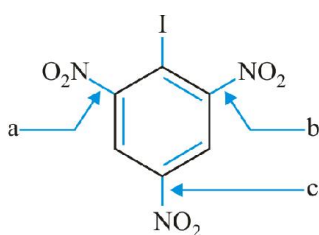
Solution :

In (II) and (III), the lone pair of N is in conjugation with the benzene ring and is not available for donation. (II) is less basic than (III) due to $-I$ and $-M$ of $-\text{NO}_2$ group. It may seem that (I) is least basic due to presence of 2 $-\text{NO}_2$ groups but $-\text{NO}_2$ and $-\text{N}(\text{CH}_3)_2$ are all bulky groups. This is a case of steric inhibition of resonance due to which the lone pair of N is not in conjugation and is readily available for electron donation. Hence, the order of basic strength is :

(I) > (III) > (II)

Example - 18

Mark the order of bond lengths in the given molecule.



Solution :

$-\text{I}$, $-\text{NO}_2$ are bulky groups and is case of steric inhibition of resonance. Therefore, the $-\text{NO}_2$ groups ortho to $-\text{I}$ are out of conjugation while the $-\text{NO}_2$ group para to $-\text{I}$ will be in conjugation with the benzene ring. Therefore, bonds 'a' and 'b' will always have single bond character while 'c' has double bond character. Therefore :

$c < a = b$

11. CARBOCATION

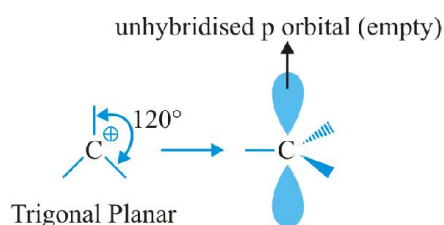
11.1 Definition

Carbocation is the intermediate of carbon containing positive charge. It has six electrons in the valence shell.

11.2 Geometry and Hybridization

Hybridization of $\text{C}^{\oplus} = \text{sp}^2$

Geometry of $\text{C}^{\oplus} = \text{Trigonal Planar}$



11.3 Classification of Carbocations

This classification will also be used for carbanions and carbon free radical and will be studied only in this section.

- $\oplus\text{CH}_3$ Methyl Carbocation
- $\text{R}-\oplus\text{CH}_2$ 1° Primary Carbocation
- $\text{R}_2-\oplus\text{CH}$ 2° Secondary Carbocation
- $\text{R}_3\text{C}^{\oplus}$ 3° Tertiary Carbocation

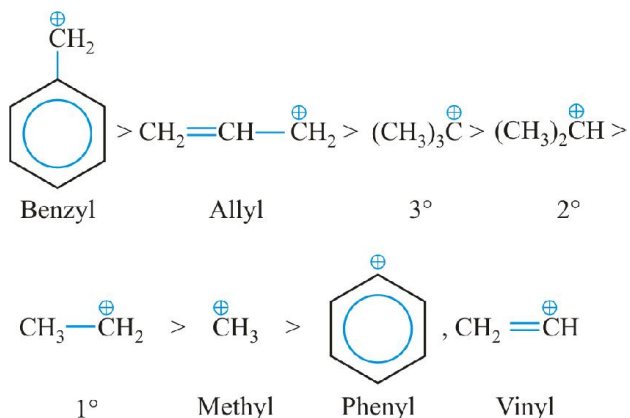
Type	Generic	Example
Allylic Carbocation		$\text{CH}_2=\text{CH}-\oplus\text{CH}_2$ Allyl Carbocation
Benzylic Carbocation		
Vinylic Carbocation		$\text{CH}_2=\oplus\text{CH}$ Vinyl Carbocation
Aryl Carbocation		

11.4 Stability of Carbocations

There are three factors contributing to the stability of carbocations :

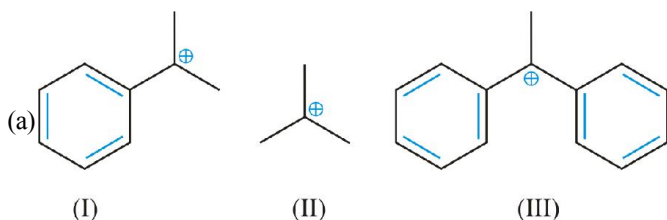
- Inductive Effect
- Hyperconjugation
- Resonance

Order of stability :

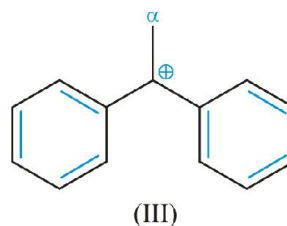
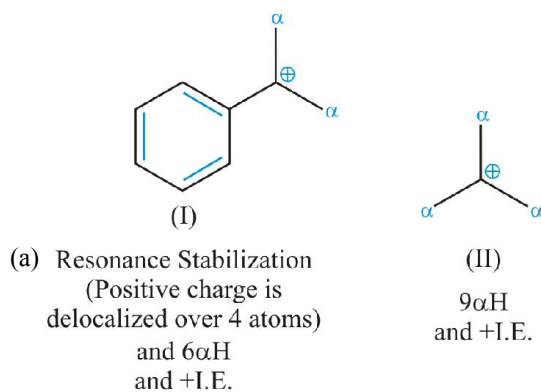


Example - 19

Rank the stability of carbocations in each case :



Solution :



Resonance Stabilization
(Positive charge is
delocalized over 7 atoms
and 3 αH
and +I.E.)

Order of stability : III > I > II

11.5 Formation of Carbocations

11.5.1 Ionization of Carbon-Leaving Group Bond

In this method :

- Bond between carbon and leaving group ionizes.
- Leaving group accepts the pair of electrons that were shared in the covalent bond.

Rate of formation of carbocation depends on :

- The stability of carbocation formed.
- The nature of the leaving group. **Weaker the base better the leaving group.** This is because weaker leaving group implies a stable compound and its formation will therefore be favoured.

11.5.2 Addition of Proton to a π bond

Rate of carbocation formation depends on :

- Stability of carbocation formed.
- Strength of the electrophile.

11.6 Reactions of Carbocations

There are three important reactions of carbocations :

- Capture a Nucleophile
- Lose a proton to form a π bond.
- Rearrangement

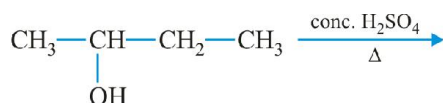
11.6.1 Capture a Nucleophile



11.6.2 Loose a proton to form a π bond

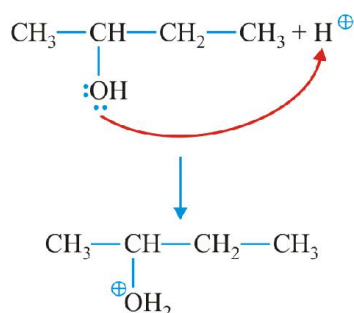
Example - 20

Form the products from the following reaction :

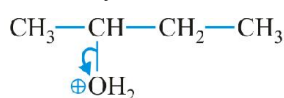


Solution :

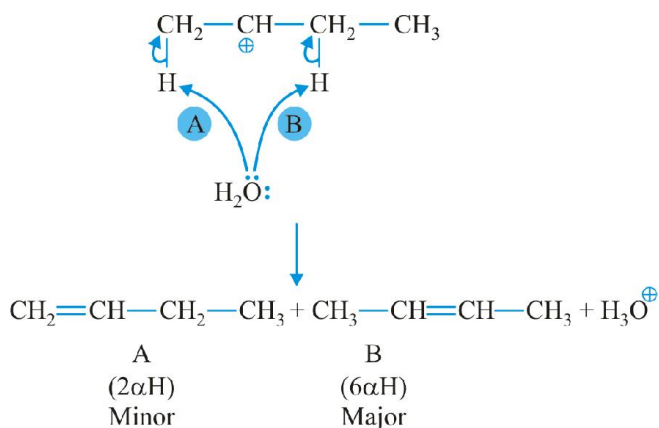
Step-1 : Protonation



Step-2 : Formation of Carbocation



Step-3 : Deprotonation



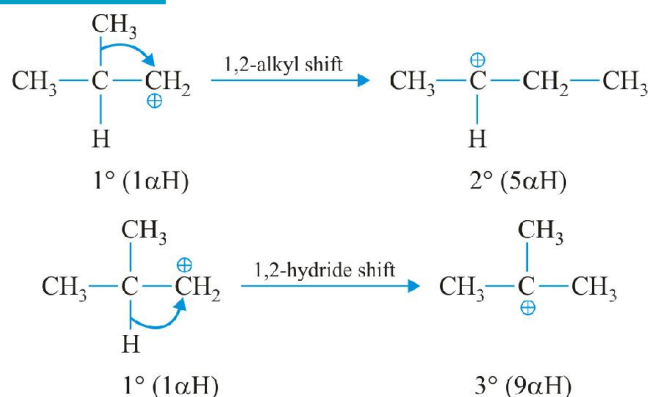
When carbocation deprotonation can lead to more than one product, all products are formed and the most stable product is the major product.

11.6.3 Carbocation Rearrangement

A carbocation can become more stable by rearrangement. Bonding electrons of carbocation may shift between adjacent atoms to form more stable carbocation. There are two kinds of shifts that take place in order to gain stability.

- (a) Hydride Shift (b) Alkyl Shift

Example - 21



In the above example, both hydride and methyl shifts are possible leading to more stable carbocation but **only that shift is preferred which leads to more stable carbocation**. In this example, hydride shift will take place.

Important : The shift takes place in the form of H^- or CH_3^- .

12. CARBANIONS

12.1 Definition

Carbanion is the intermediate of carbon containing negative charge. It has eight electrons in the valence shell.

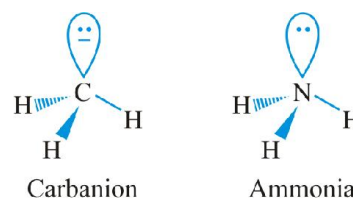


12.2 Geometry and Hybridization

Hybridization of C^- : sp^3

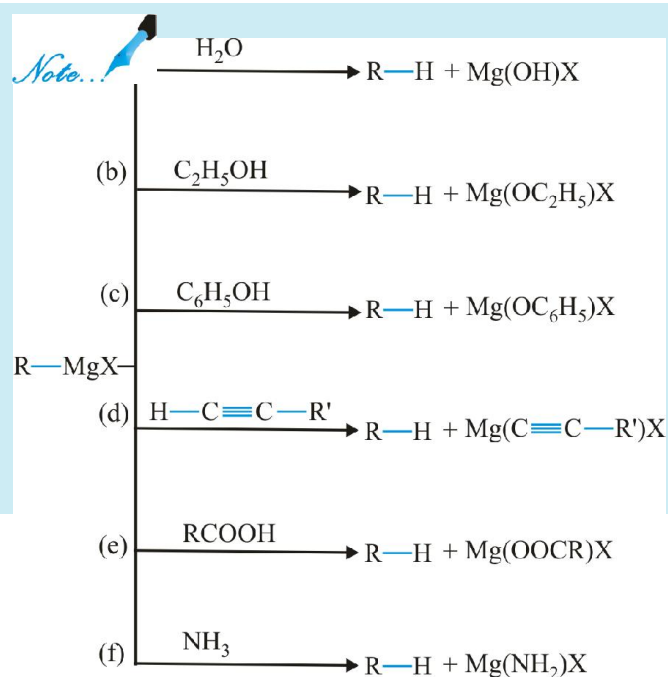
Geometry : Trigonal Pyramidal

Carbanion and ammonia are isoelectronic species having same structure



12.3 Stability

ERG will increase the electron density at carbon and will make it unstable. EWG will decrease the electron density at carbon and will make it stable.

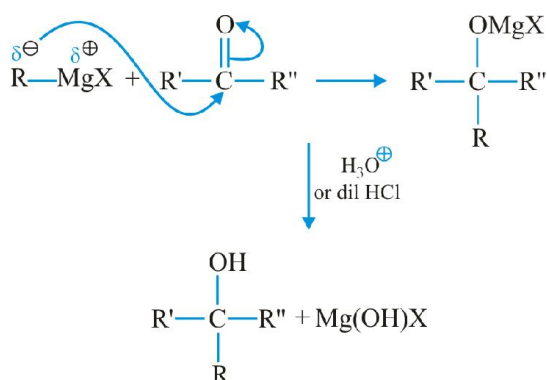


In (d), the reaction with terminal alkyne also takes place as sp -hybridized carbon is highly electronegative and therefore H attached to it is fairly acidic.

12.5.2 Grignard Reagent as a Nucleophile

Grignard Reagent reacts with carbonyl compounds to form alcohols. This is a very important method for the synthesis of alcohols.

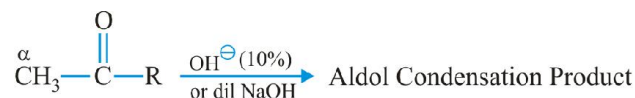
Mechanism :



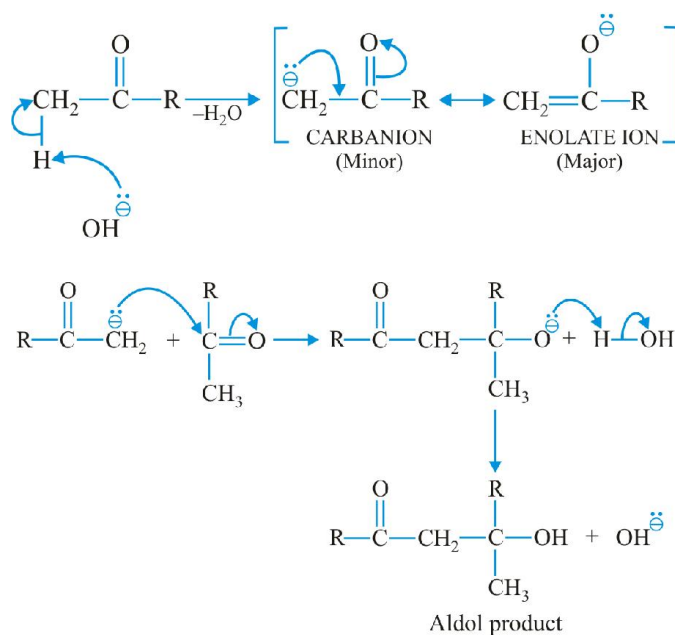
H_3O^+ is never added along with the first step as Grignard Reagent will react with H^+ to give $\text{R}-\text{H}$ as in previous reaction.

12.5.3 Aldol Condensation

This reaction is shown by carbonyl compounds containing at least one αH in presence of dilute base such as dilute NaOH.



Mechanism :



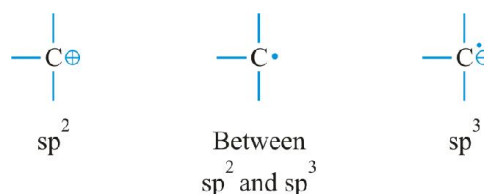
13. CARBON FREE RADICALS

13.1 Definition

Carbon Free Radical is the intermediate of carbon having an odd electron. It is neutral and has seven electrons in the valence shell. It is highly reactive as it requires only one electron to complete its octet and therefore, is short-lived.



13.2 Geometry and Hybridization



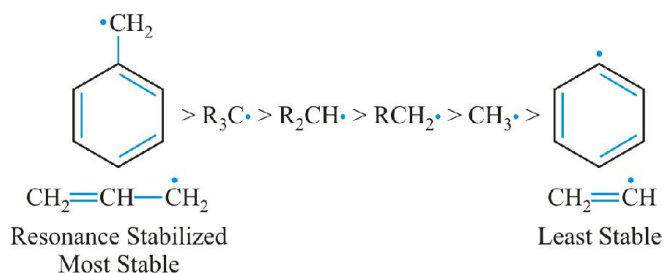
The hybridization of carbon free radical was proposed after experimental verification of structure of different radicals.

It was proposed that when ERG are placed on $C^\bullet$, it has sp^2 hybridization and when EWG are placed on $C^\bullet$, it has sp^3 hybridization.

Molecule	Hybridization	Location of odd electron
$\bullet CH_3$	sp^2	p
$\bullet CF_3$	sp^3	sp^3

13.3 Stability

ERG increase stability while EWG decrease stability.



13.4 Formation of Carbon Free Radical

Carbon Free Radicals are formed by homolytic cleavage of bonds. They are formed :

- at high temperature in the gas phase
- in non-polar solvents
- by ultraviolet light
- by addition of other radicals

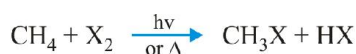
13.5 Reactions of Carbon Free Radical

The most common reactions in which free radical is involved are :

- Halogenation of alkanes.
- Addition of HBr in the presence of peroxides to alkenes. (Anti-Markonikov Rule)
- α -Halogenation of alkenes.
- Wurtz Reaction
- Decarboxylation reaction

13.5.1 Free Radical Halogenation of Alkanes

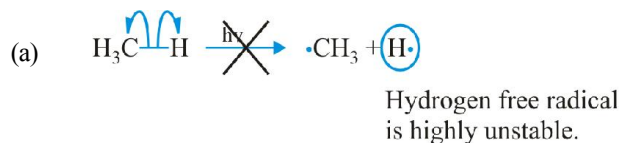
The typical reactions of free radical are chain reaction mechanisms. There are three steps in a chain reaction mechanism : initiation, propagation and termination.



Mechanism

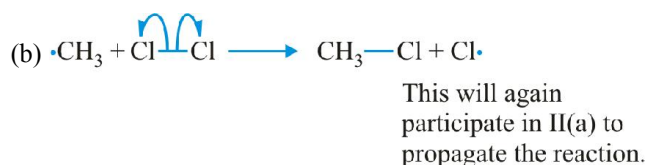
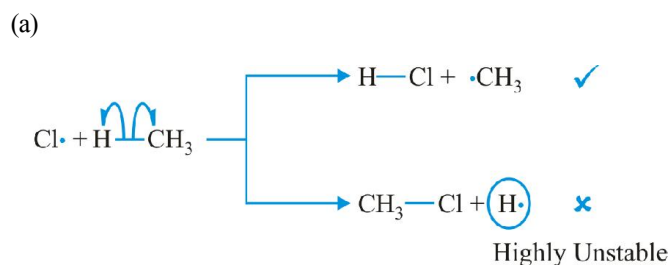
(I) Chain Initiation

There are two choices :



(I) (b) will take place as it is energetically feasible.

(II) Chain Propagation



(III) Chain Termination

The various free radicals in the progress of reaction are : $Cl\bullet$, $CH_3\bullet$. There are ways in which free radicals from different chains may combine to terminate the reaction.



There is a side-product CH_3-CH_3 that is formed in this reaction. Besides these, a number of other side-products such as CH_2Cl_2 , $CHCl_3$ and CCl_4 have also been observed. These get formed since carbon free radical is highly reactive. It randomly reacts with other species that make it stabler.

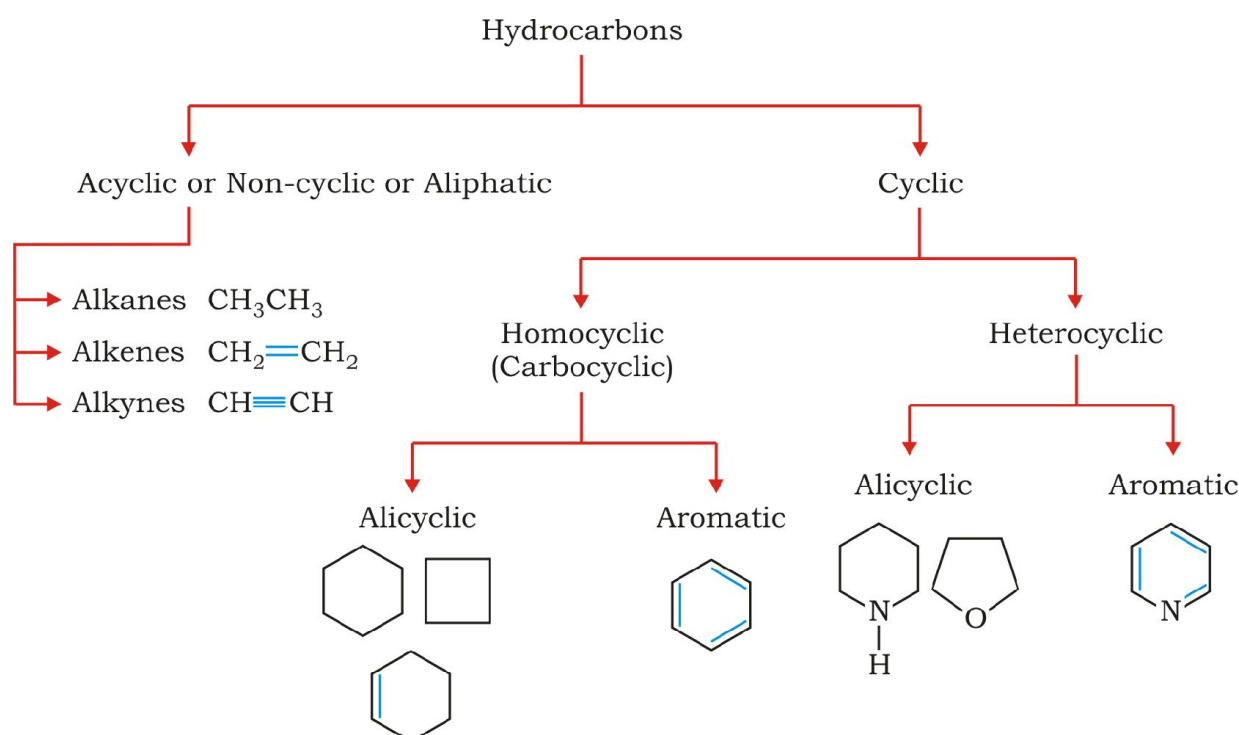
14. NOMENCLATURE-I

14.1 INTRODUCTION

In 1949, an organization was set up to formulate the rules for naming organic compounds. This organization is **IUPAC - International Union for Pure and Applied Chemistry**. Prior to this, the organic compounds were known by their **trivial/common names** which generally indicated their source. For example, formic acid gets its name from *formica* (Latin, red ants) and acetic acid gets its name from *acetum* (Latin, vinegar).

Important : Systematic nomenclature is required in order to have unambiguous names for all compounds.

14.2 CLASSIFICATION OF HYDROCARBONS



Important :

1. **Aliphatic Compounds** do not have rings and exist as a chain of carbon atoms.
2. **Alicyclic Compounds** have similar properties as aliphatic compounds except that the carbon chain exists in the form of ring.
3. **Aromatic Compounds** are based on the structure of benzene. These compounds are also called **arenes**.

14.3 HOMOLOGOUS SERIES

A homologous series (Greek *homos* = "the same as") is a family of compounds in which each member differs from the next by one **methylene (CH_2)** group. The members of the homologous series are called **homologues**. Example :

(a) Alkanes $\text{C}_n\text{H}_{2n+2}$

(b) Alkenes C_nH_{2n}

(c) Alkynes $\text{C}_n\text{H}_{2n-2}$

14.4 NOMENCLATURE OF UNBRANCHED ALKANES

Name	No. of Carbon Atoms	Structure	Substituent Name
Methane	1	CH ₄	Methyl
Ethane	2	CH ₃ CH ₃	Ethyl
Propane	3	CH ₃ CH ₂ CH ₃	Propyl
Butane	4	CH ₃ (CH ₂) ₂ CH ₃	Butyl
Pentane	5	CH ₃ (CH ₂) ₃ CH ₃	Pentyl
Hexane	6	CH ₃ (CH ₂) ₄ CH ₃	Hexyl
Heptane	7	CH ₃ (CH ₂) ₅ CH ₃	Heptyl
Octane	8	CH ₃ (CH ₂) ₆ CH ₃	Octyl
Nonane	9	CH ₃ (CH ₂) ₇ CH ₃	Nonyl
Decane	10	CH ₃ (CH ₂) ₈ CH ₃	Decyl
Undecane	11	CH ₃ (CH ₂) ₉ CH ₃	
Dodecane	12	CH ₃ (CH ₂) ₁₀ CH ₃	
Tridecane	13	CH ₃ (CH ₂) ₁₁ CH ₃	
Tetradecane	14	CH ₃ (CH ₂) ₁₂ CH ₃	
Pentadecane	15	CH ₃ (CH ₂) ₁₃ CH ₃	
Eicosane	20	CH ₃ (CH ₂) ₁₈ CH ₃	

Meth, Eth, Prop, But, etc. are called compound roots. Compound roots represent the number of carbon atoms.

14.5 IUPAC NAME TEMPLATE

Prefix	Compound Root	Suffix	
Substituents	No. of C in the Parent Chain	Unsaturation + Functional Group Index	Suffix (optional)
eg. Methyl	eg. Meth	eg. ane	eg. ol
Ethyl	Eth	ene	al
Propyl	Prop	yne	oic acid
Bromo			

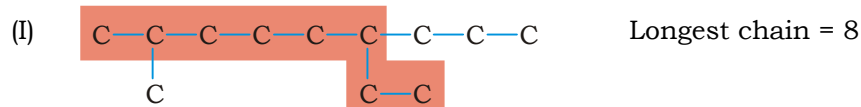
This template will be used to give the name for any organic compound.

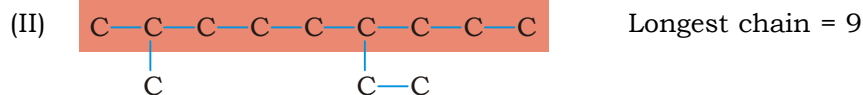
14.6 RULES FOR NOMENCLATURE OF SIMPLE HYDROCARBONS

Rule-1 :

Determine the **longest carbon chain in the molecule (parent chain)**. Where a double/triple bond is present, choose the chain which includes these bonds. If there is a cyclic structure present, the longest chain starts and stops within the cyclic structure.

EXAMPLE



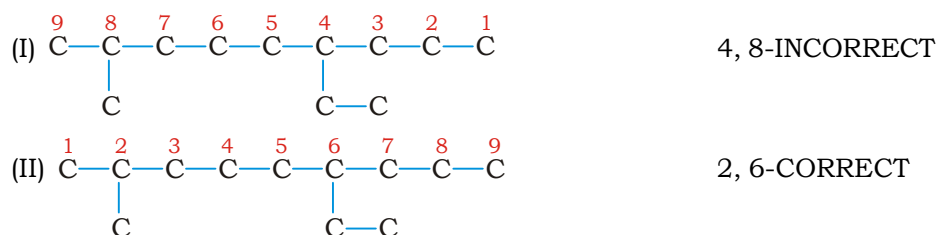


(II) is the correct chain selection.

Rule-2 :

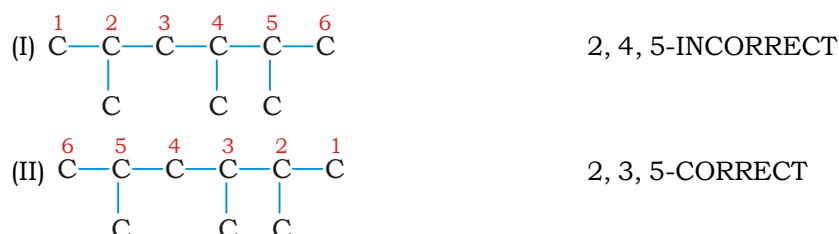
Assign numbers to each carbon of the parent chain. Numbering is done to identify the parent alkane (compound root) and to locate the positions of the carbon atoms at which branching takes place. The numbering is done in such a way that **the branched atoms get the lowest possible number**.

EXAMPLE



If there is a tie for the first branch, then go to the second and so on until a difference is observed.

EXAMPLE

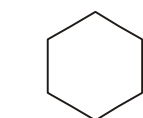
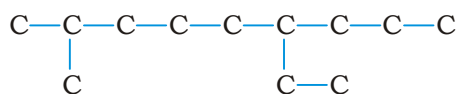


Rule-3 :

Determine the compound root and the unsaturation index. **Compound root corresponds to the number of carbon atoms in parent chain.** Add prefix cyclo if the parent chain is cyclic.

Unsaturation Index	Suffix
$\text{C}-\text{C}$	—ane
$\text{C}=\text{C}$	—ene
$\text{C}\equiv\text{C}$	—yne

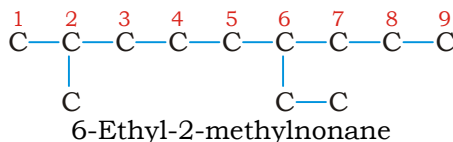
EXAMPLE



[but] [ene]
∴ Butene

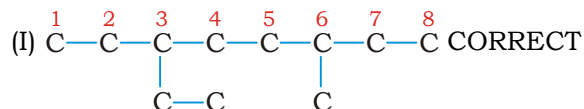
Rule-4 :

Determine the correct name for each branch for example, alkyl groups such as methyl, ethyl, etc. Attach the name of the branches alphabetically along with their positions to the parent chain as prefix. Separate numbers from letters with hyphens.

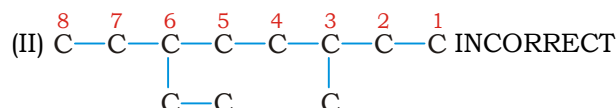
EXAMPLE

6-Ethyl-2-methylnonane

If two substituents are in equivalent positions, the lower number is given to the one coming first in the alphabetical listing.

EXAMPLE

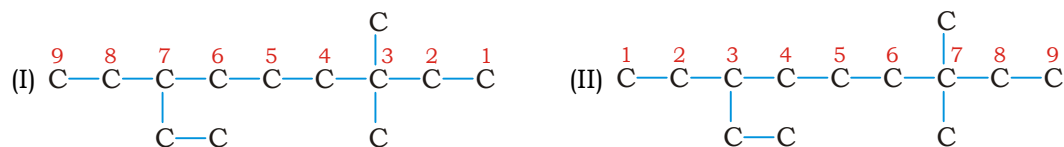
3-Ethyl-6-methyloctane



6-Ethyl-3-methyloctane

Rule-5 :

When two or more branches are identical, use prefixes di-, tri-, tetra-, etc. Numbers are separated with commas and prefixes are ignored while determining alphabetical order.

EXAMPLE

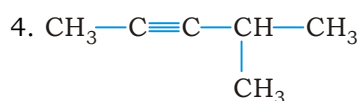
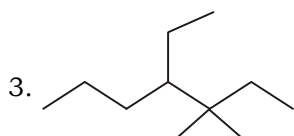
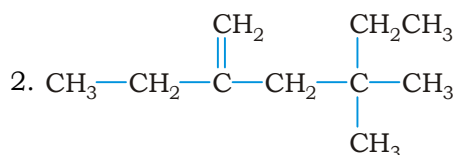
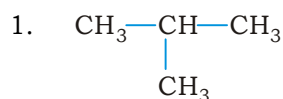
INCORRECT

3, 3-Dimethyl-7-ethylnonane INCORRECT
 7-Ethyl-3, 3-dimethylnonane CORRECT

This is an incorrect selection as lowest number rule is violated.

EXERCISE

Give the IUPAC names of :



Solution :

14.7 ALKYL SUBSTITUENTS - COMMON AND SYSTEMATIC NAMES

CH_3 – Methyl
 CH_3CH_2 – Ethyl

From three carbon onwards, alkyl substituents can be complicated for which common as well as systematic names are possible.

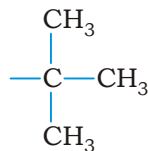
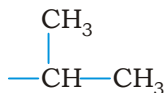
Rules for Systematic and Common Names

Rule-1 : For systematic names, the carbon atom of the branch that attaches to the parent chain is numbered one.

Rule-2 : Substituents which have branching at the first carbon can be classified as secondary (*sec-*) or tertiary (*tert-*) depending upon the number of carbons attached to it. '*sec-*' and '*tert-*' are italicized and can also be abbreviated as '*s-*' or '*t-*' respectively.

Rule-3 : Substituents which have a CH_3 - group on second last carbon are called **iso** and which have two methyl groups on second last carbon are called **neo**.

Rule-4 : **iso** and **neo** are considered while determining alphabetical order of substituents. '*sec-*' and '*tert-*' are not considered.



iso

neo

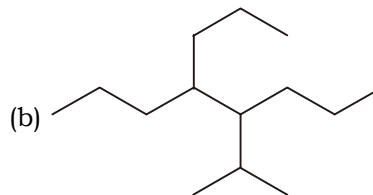
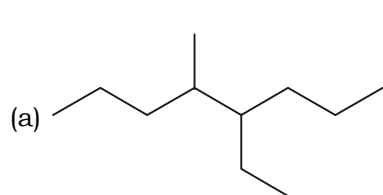
Alkyl Substituent	Systematic Name	Common Name
1. $-\text{CH}_2\text{CH}_2\text{CH}_3$	propyl	propyl
2. $\begin{array}{c} \text{CH}_3 \\ \\ -\text{CH}-\text{CH}_3 \end{array}$	1-methyl ethyl	isopropyl (not s-propyl)
3. $\begin{array}{c} \text{CH}_3 \\ \\ -\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3 \end{array}$		
4. $\begin{array}{c} \text{CH}_3 \\ \\ -\text{CH}-\text{CH}_2\text{CH}_3 \end{array}$		
5. $\begin{array}{c} \text{CH}_3 \\ \\ -\text{CH}_2-\text{CH}-\text{CH}_3 \\ \\ \text{CH}_3 \end{array}$		
6. $\begin{array}{c} \text{CH}_3 \\ \\ -\text{C}-\text{CH}_3 \\ \\ \text{CH}_3 \end{array}$		
7. $-\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$		
8. $\begin{array}{c} \text{CH}_3 \\ \\ -\text{CHCH}_2\text{CH}_2\text{CH}_3 \end{array}$		
9. $\begin{array}{c} \text{CH}_3 \\ \\ -\text{CH}_2-\text{CH}-\text{CH}_2\text{CH}_3 \\ \\ \text{CH}_3 \end{array}$		
10. $\begin{array}{c} \text{CH}_3 \\ \\ -\text{CH}_2-\text{CH}_2-\text{CH}-\text{CH}_3 \\ \\ \text{CH}_3 \end{array}$		
11. $\begin{array}{c} \text{CH}_2\text{CH}_3 \\ \\ -\text{CH}-\text{CH}_2\text{CH}_3 \end{array}$		
12. $\begin{array}{c} \text{CH}_3 \quad \text{CH}_3 \\ \quad \\ -\text{CH}-\text{CH}-\text{CH}_3 \\ \quad \\ \text{CH}_3 \quad \text{CH}_3 \end{array}$		
13. $\begin{array}{c} \text{CH}_3 \\ \\ -\text{C}-\text{CH}_2\text{CH}_3 \\ \\ \text{CH}_3 \end{array}$		
14. $\begin{array}{c} \text{CH}_3 \\ \\ -\text{CH}_2-\text{C}-\text{CH}_3 \\ \\ \text{CH}_3 \end{array}$		

15. NOMENCLATURE-II

15.1 ALKYL SUBSTITUENTS – COMMON AND SYSTEMATIC NAMES

EXERCISE

Give IUPAC names for :

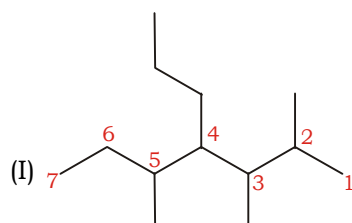


Solution :

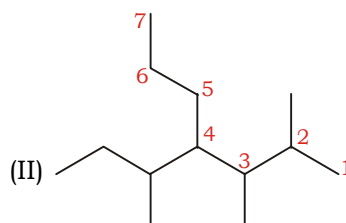
15.1.1 Corollary to Longest Chain Rule

If there happen to be two chains of equal length, then that chain is selected which contains more number of side chains

EXAMPLE

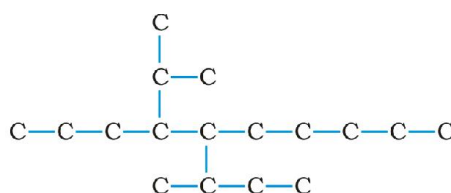
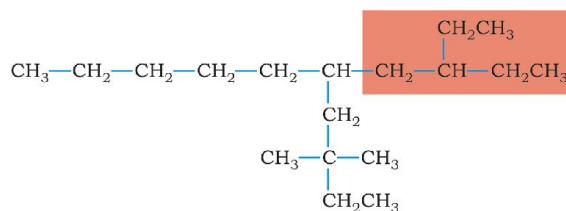
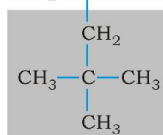


CORRECT (4 side chains)



INCORRECT (3 side chains)

IUPAC Name : 2,3,5-Trimethyl-4-propyloctane

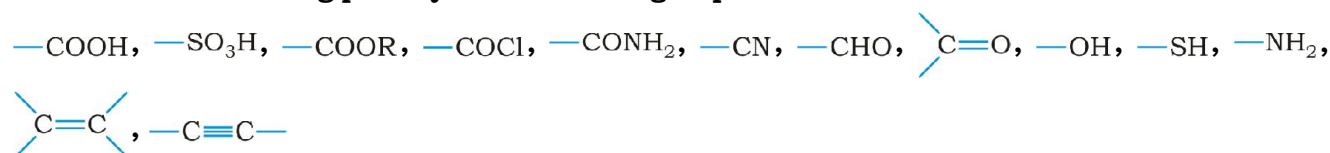


15.2 FUNCTIONAL GROUPS

A functional group is an atom or a group of atoms which characterizes the chemical reactivity of a molecule. The longest chain of atoms containing the functional group is numbered in such a way that the carbon to which functional group is attached is assigned the lowest number. In case of polyfunctional compounds, one of the functional groups is chosen as the principal functional group and the compound is named treating other functional groups as substituents.

Class	Formula	Prefix	Suffix
Acid Halide	— COX — (C)OX	Halocarbonyl —	— carbonyl halide — oyl halide
Alcohols	— OH	Hydroxy	— ol
Aldehydes	— CHO — (C)HO	Formyl Oxo	— carbaldehyde — al
Amides	— CONH ₂ — (C)ONH ₂	Carbamoyl —	— carboxamide — amide
Amines	— NH ₂	Amino	— amine
Carboxylates	— COO [⊖] — (C)OO [⊖]	Carboxylato —	— carboxylate — oate
Carboxylic Acid	— COOH — (C)OOH	Carboxy— —	— carboxylic acid — oic acid
Ethers	— OR	Alkoxy—	—
Esters of Carboxylic Acid	— COOR — (C)OOR	Alkoxy carbonyl —	Alkyl .. carboxylate Alkyl ... oate
Ketones		Oxo—	— one
cyanides (Nitriles)	— C≡N — (C)≡N	Cyano —	— carbonitrile — nitrile
Salts of Carboxylic	— COO [⊖] M [⊕]	—	Cation ... carboxylate
Acids	— (C)OO [⊖] M [⊕]	—	Cation ... oate
Sulphonic Acids	— SO ₂ —OH	Sulfo—	— sulfonic acid
Thiols	— SH	Sulfanyl	— thiol

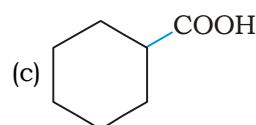
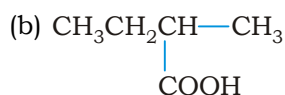
The order of decreasing priority for functional groups is :

**Important :**

- The —R, —C₆H₅, halogens, nitro and alkoxy groups are always treated as substituents.
- If more than one functional group of same type is present, its number is indicated by adding di-, tri- etc.
- When C is marked in parenthesis, the carbon of the functional group is counted as part of the parent chain and the numbering will always begin from this carbon and its position need not be marked in the name.

EXERCISE

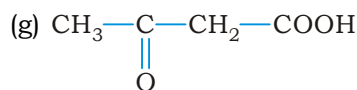
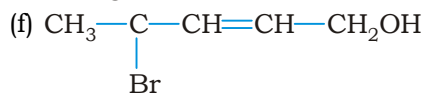
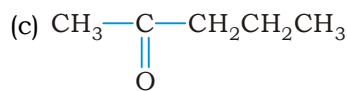
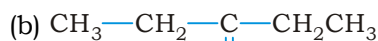
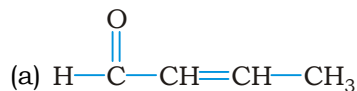
Write IUPAC names of :



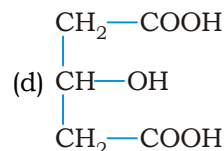
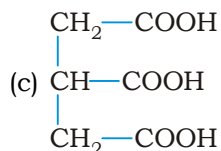
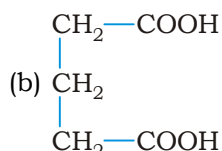
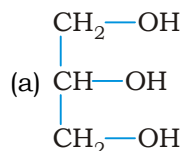
Solution :

EXERCISE

Give IUPAC names for :

**Solution :****EXERCISE**

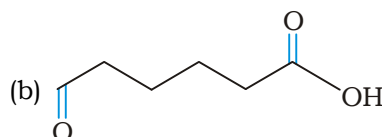
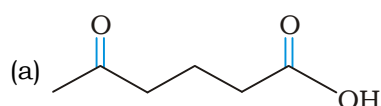
Give IUPAC names for :

**Solution :**

Important : Pentane-1, 5-dioic acid is incorrect name. There is no need to give positions of COOH because if we include their carbons in the parent chain, they will always be present at the end of the chain.

EXERCISE

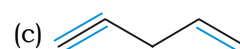
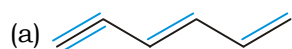
Give IUPAC names of :

**Solution :****15.3 PRIORITY BETWEEN ALKENE AND ALKYNE**

When there is a choice between alkene and alkyne, alkene is given lower number.

EXERCISE

Give IUPAC names of :

**Solution :**

Important : 'e' of 'ene' is dropped as 'yne' sounds like starting from 'i' (vowel).

15.3.1 Common Alkenyl/Alkynyl Substituents

Structure	Common Name	Systematic Name
$\text{CH}_2=\text{CH}-$	Vinyl	Ethenyl
$\text{CH}_2=\text{CH}-\text{CH}_2-$	Allyl	Prop-2-enyl
$\text{CH}\equiv\text{C}-$	-	Ethynyl

15.3.2 Chain selection Rules for Alkene and Alkyne Priority

Rule-1 :

Select that chain which contains maximum number of double and triple bonds.

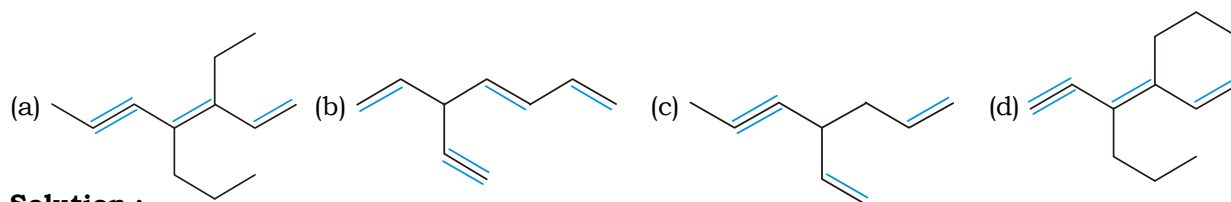
Rule-2 :

If two chains are competing for selection as the chain with maximum number of unsaturated bonds, then the choice goes to

- The one with the greatest number of carbon atoms.
- The number of carbon atoms being equal, the one containing maximum number of double bonds.

EXERCISE

Give IUPAC names for :

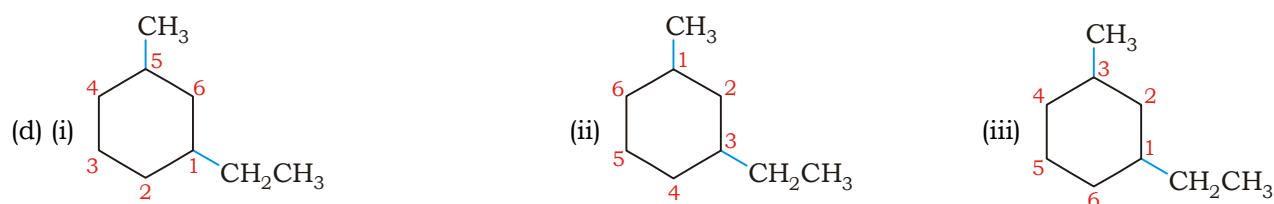


Solution :

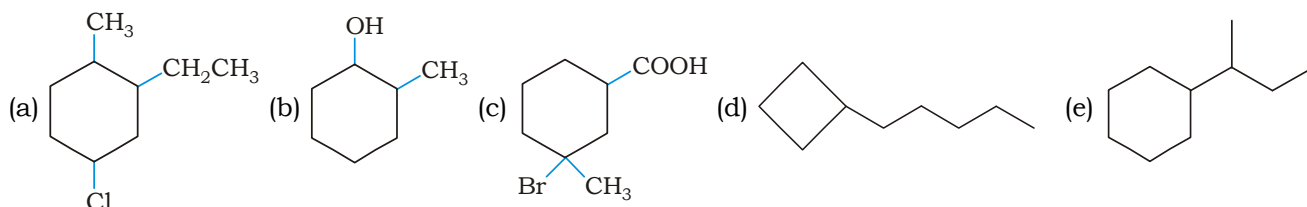
15.4 CYCLIC COMPOUNDS

Corollary to Longest Chain Rule

A ring is treated as a substituent only when the number of carbon atoms are less in the ring than in the chain.

EXAMPLE**EXERCISE**

Give IUPAC names for :

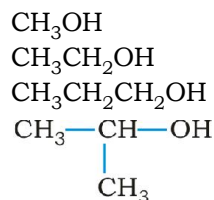
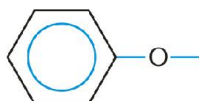
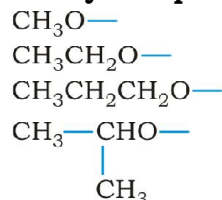


Solution :

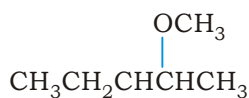
15. NOMENCLATURE-III

15.1 SUBSTITUENTS – ALKOXY, NITRO AND HALOGEN GROUPS

Alkoxy, nitro and halogen groups are always treated as substituents

Alcohol**Alkoxy Group****Substituent Name**

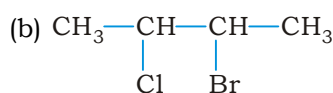
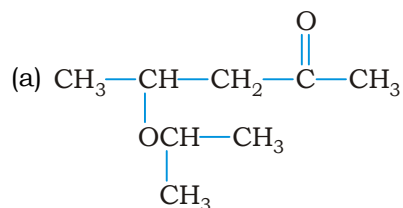
Methoxy
Ethoxy
Propoxy
Isopropoxy
Phenoxy

EXAMPLE

2-Methoxypentane

EXAMPLE

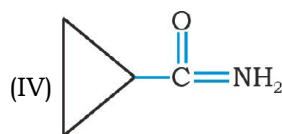
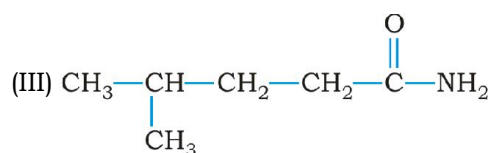
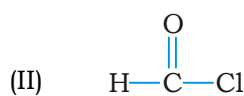
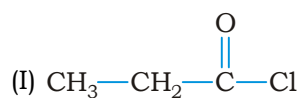
Give IUPAC names of :



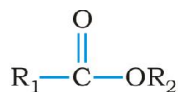
Solution :

15.2 CARBOXYLIC ACID DERIVATIVES

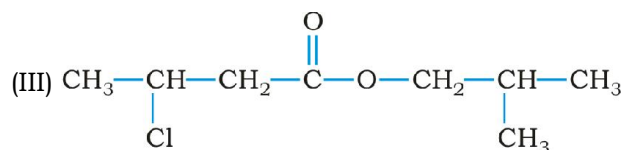
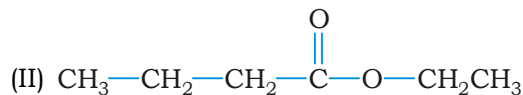
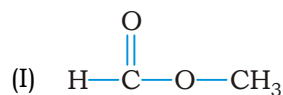
Nomenclature of acid halides and amides is very straight forward.

EXAMPLE

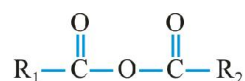
Esters nomenclature is done in the following manner :



Alkyl (R_2) Alkanoate ($\text{R}_1-\overset{\text{O}}{\underset{\text{O}}{\text{C}}}-\text{O}-$)

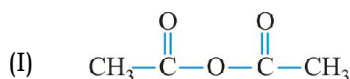
**Solution :**

2-Methylpropyl-3-chlorobutanoate
Acid Anhydrides

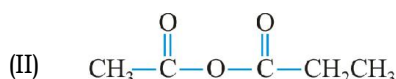


Alkanoic (R_1) + alkanoic (R_2) anhydride

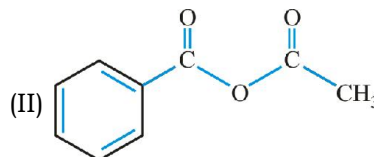
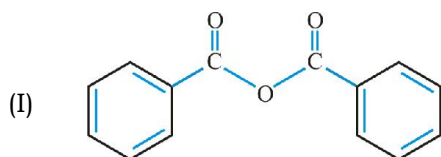
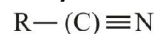
If $\text{R}_1 = \text{R}_2$, then name is Alkanoic anhydride

EXAMPLE :

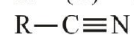
Ethanoic anhydride



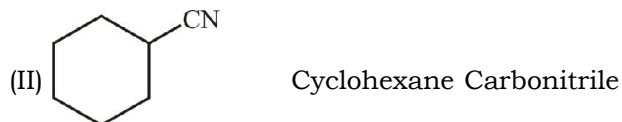
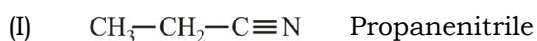
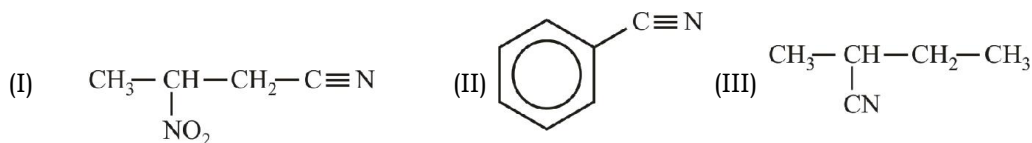
Ethanoic Propanoic anhydride

EXERCISE :**Cyanides/Nitriles**

Alkane nitrile (C is counted as part of parent chain)

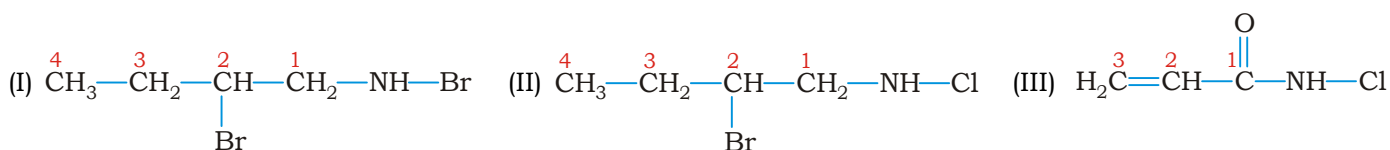


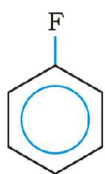
Alkane carbonitrile (C is not counted as part of parent chain)

EXAMPLE :**EXERCISE :**

15.3 N-SUBSTITUTED AMINES/AMIDES

The substituents on nitrogen atom is written as *N*-<substituent name>. Here, '*N*'(italic) indicates the position of substituent.

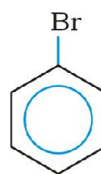
EXAMPLE

15.4 NOMENCLATURE OF BENZENE DERIVATIVES**15.4.1 Common Names and IUPAC names of Important Compounds**

Fluorobenzene



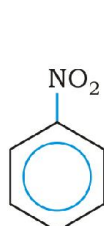
Chlorobenzene



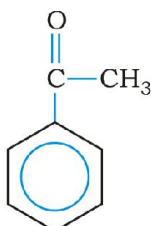
Bromobenzene



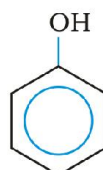
Iodobenzene



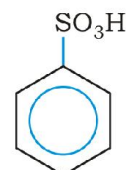
Nitrobenzene



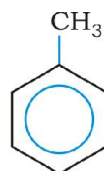
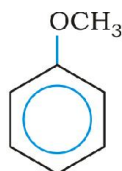
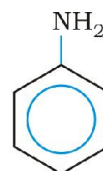
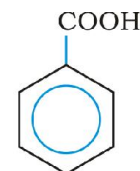
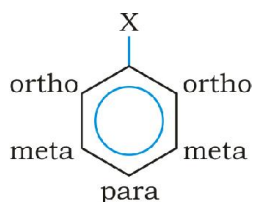
Acetophenone



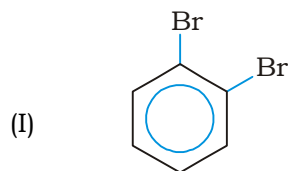
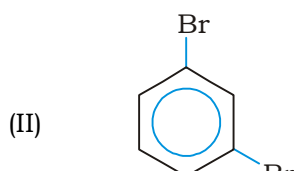
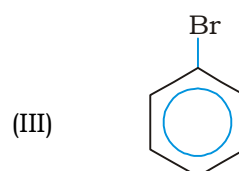
Phenol



Benzenesulphonic acid

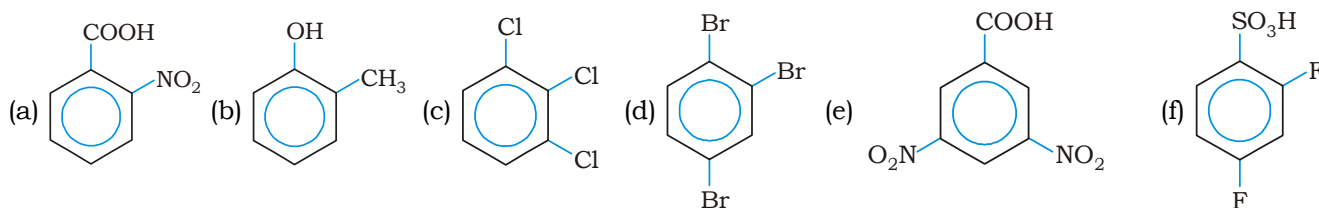
Methylbenzene
(Toluene)Methoxybenzene
(Anisole)Benzenamine
(Aniline)Benzene carboxylic acid
(Benzoic Acid)**15.4.2 Ortho, Meta and Para Positions**

With respect to substituent X, we define three positions on the benzene ring as shown in the figure. Ortho, meta, para positions are used for writing common names of disubstituted benzenes. The positions are also abbreviated as o-, m-, p-.

EXAMPLEo-Dibromobenzene
1,2-Dibromobenzenem-Dibromobenzene
1,3-Dibromobenzenep-Dibromobenzene
1,4-Dibromobenzene

EXERCISE

Give the IUPAC and common name (o/m/p wherever possible).



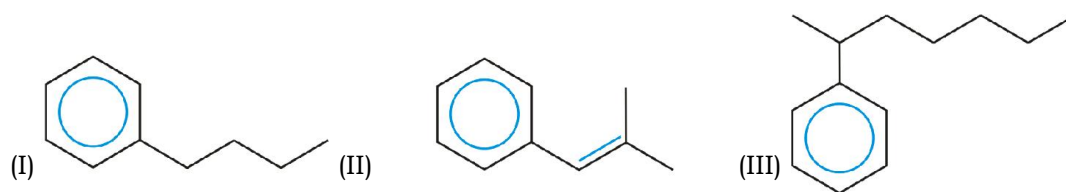
Solution :

15.4.3 Benzene as substituent-Phenyl Group

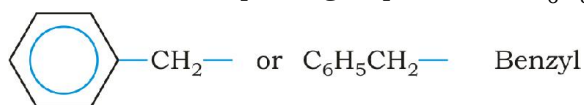
When C_6H_5- group is treated as substituent, it's called phenyl group.

Important :

- A saturated chain containing benzene ring is named as derivative of the larger structural unit. (Use corollary for longest chain selection for cyclic compounds).
- If the chain is unsaturated, the compound is always named as a derivative of that chain.

EXAMPLE

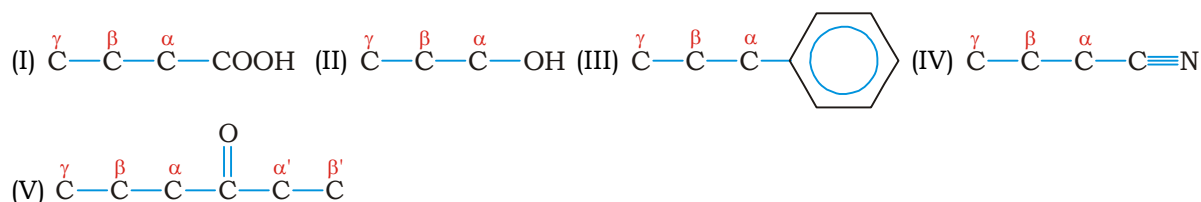
There is another special group in which C_6H_5- is present.

**15.5 Alpha (α), Beta (β), Gamma (γ) Nomenclature**

Prior to IUPAC, greek symbols were used to indicate positions of the substituents or functional groups. The carbons attached to the principal functional group are numbered as illustrated.



Principal Functional Group

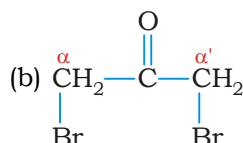
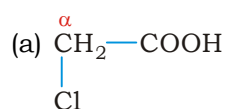
EXAMPLE**EXERCISE**

Draw the structure of

- (a) α -chloroacetic acid

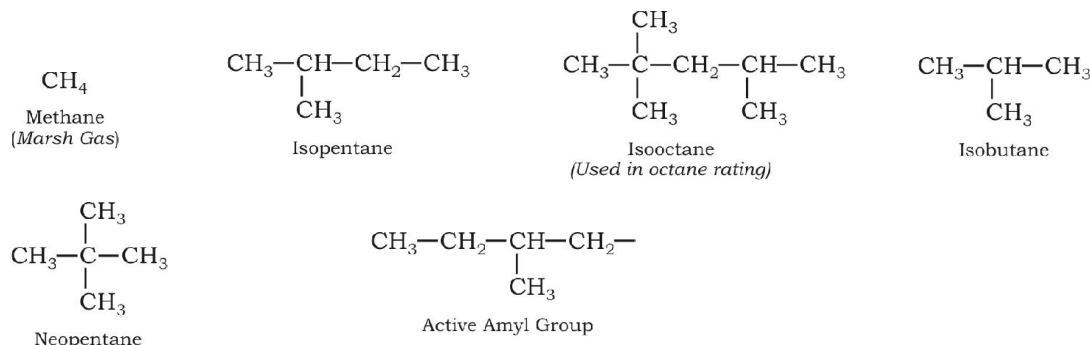
- (b) α, α' -Dibromoacetone

Solution :

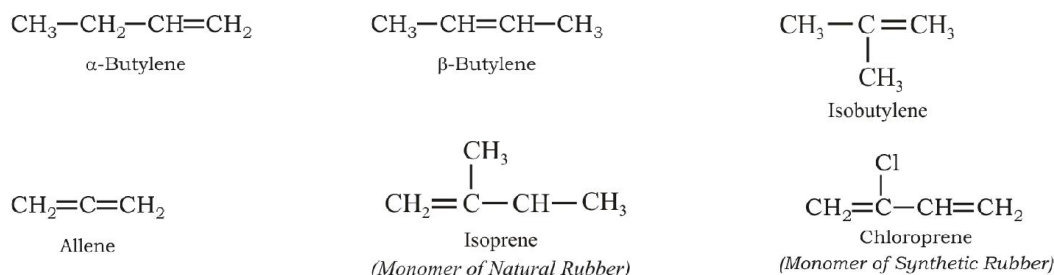


16. COMMON NAMES IN ORGANIC CHEMISTRY

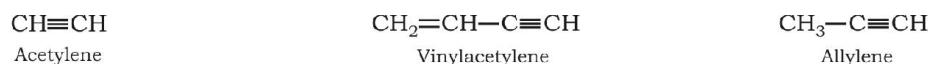
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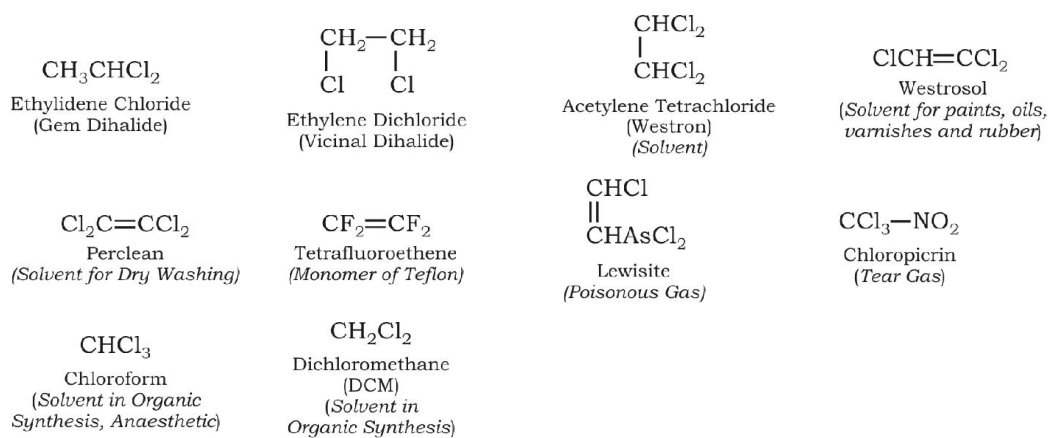
B. ALKENES



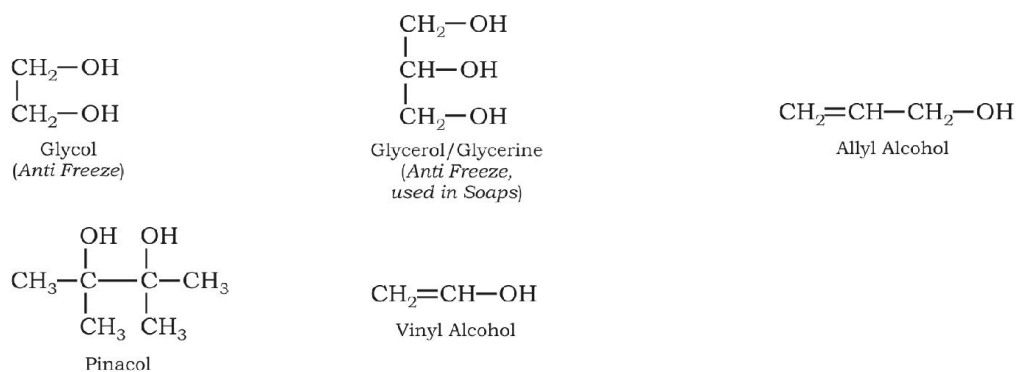
C. ALKYNES



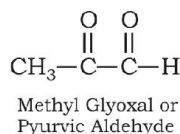
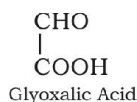
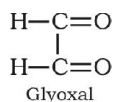
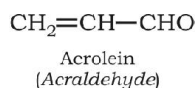
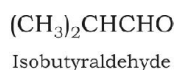
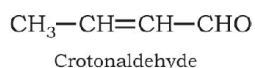
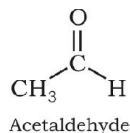
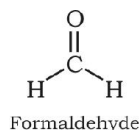
D. ALKYL HALIDES



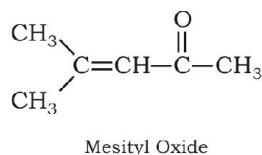
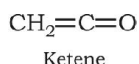
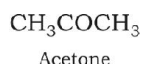
E. ALCOHOL



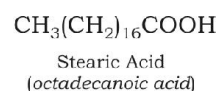
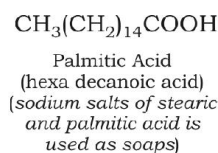
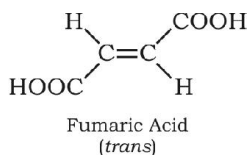
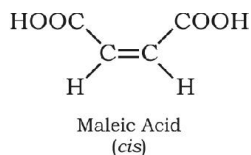
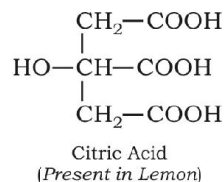
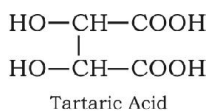
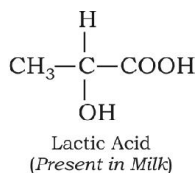
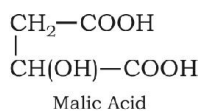
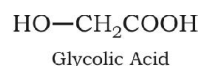
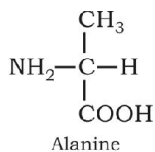
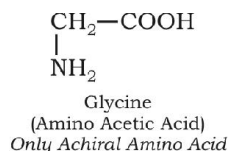
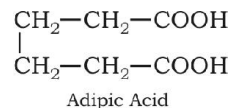
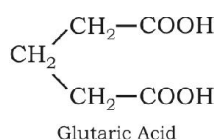
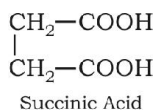
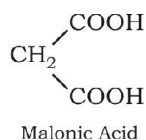
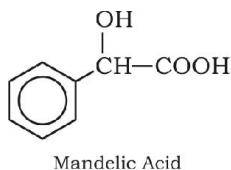
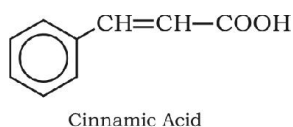
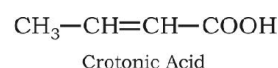
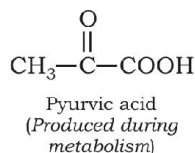
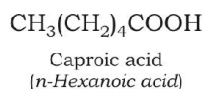
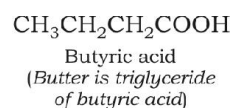
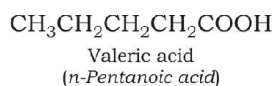
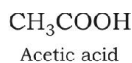
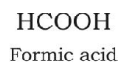
F. ALDEHYDE



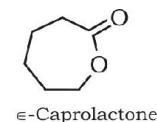
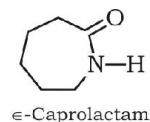
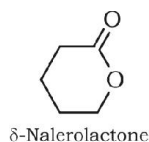
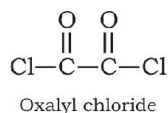
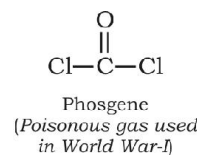
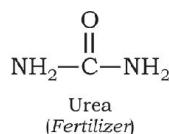
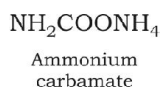
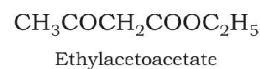
G. KETONE



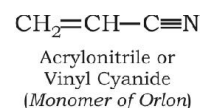
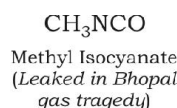
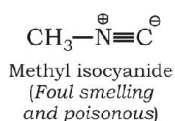
H. CARBOXYLIC ACID



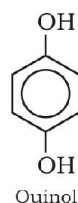
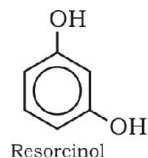
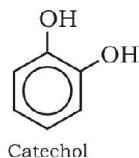
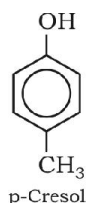
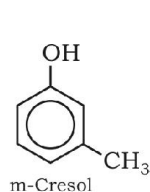
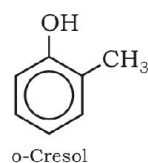
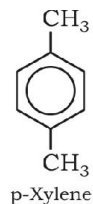
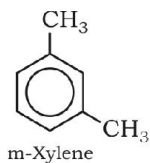
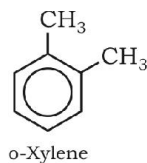
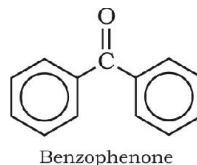
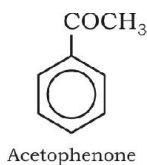
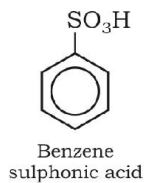
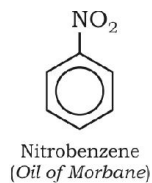
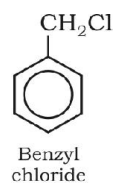
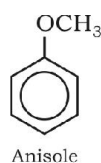
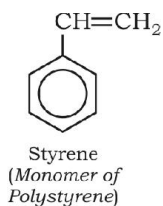
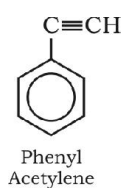
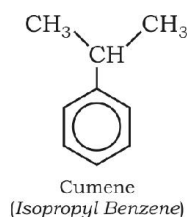
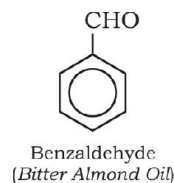
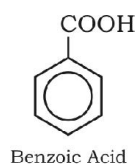
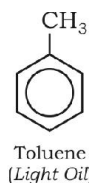
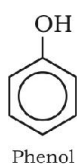
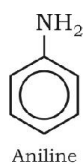
I. ACID DERIVATIVES

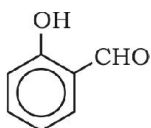


J. N-DERIVATIVES

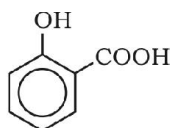


K. AROMATIC COMPOUNDS

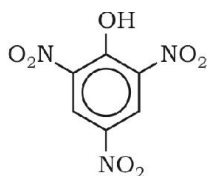




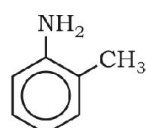
Salicylaldehyde



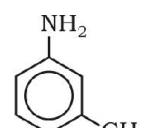
Salicylic Acid



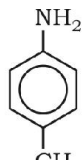
Picric Acid



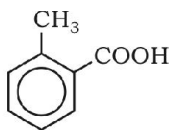
o-toluidine



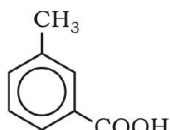
m-toluidine



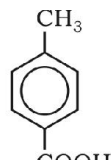
p-toluidine



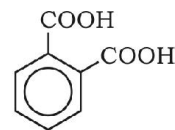
o-toluic acid



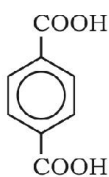
m-toluic acid



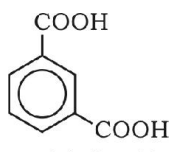
p-toluic acid



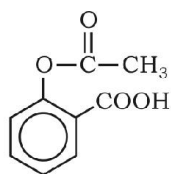
Phthalic Acid



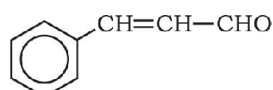
Terephthalic Acid



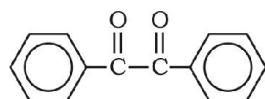
Isophthalic Acid



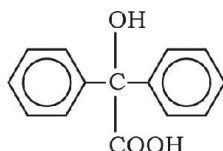
Acetyl Salicylic Acid
(Aspirin)



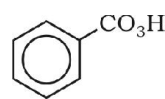
Cinnamaldehyde
(Cinnamon)



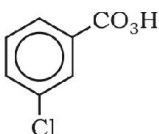
Benzil



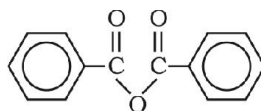
Benzoic Acid



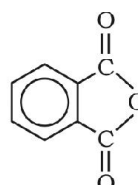
Perbenzoic Acid



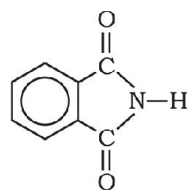
m-chloroperbenzoic acid
(m-CPBA)
(Used for Epoxidation
of Alkene)



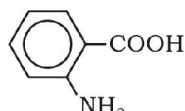
Benzoic anhydride



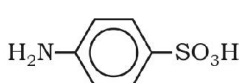
Phthalic anhydride



Phthalamide
(Used in Gabriel's
Phthalamide Reaction)



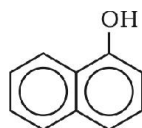
Anthranilic acid



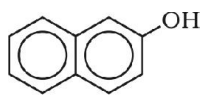
Sulphanilic Acid
(used in S-drugs)



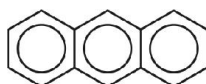
Naphthalene
(Middle or
Carbolic Oil)



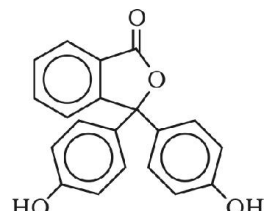
α-naphthol



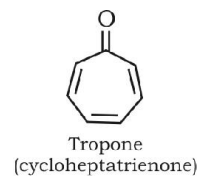
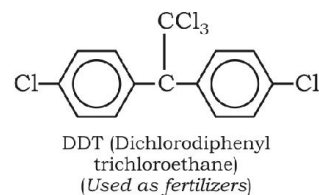
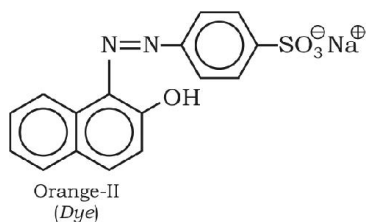
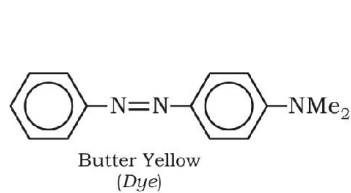
β-naphthol



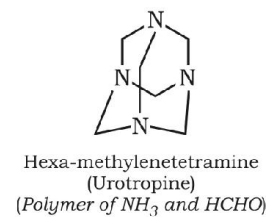
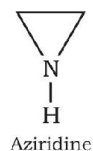
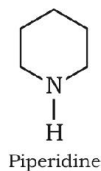
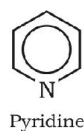
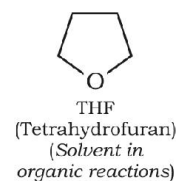
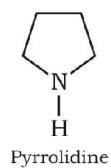
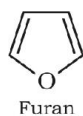
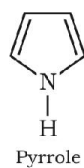
Anthracene
(Green Oil)



Phenolphthalein
(Indicator)



K. HETEROCYCLIC COMPOUNDS



METHODS OF PURIFICATION

1. Sublimation
2. Crystallization
3. Distillation
4. Differential Extraction
5. Chromatography

Sublimation

1. Some solid substances change from solid to vapour state without passing through liquid state.
2. The purification technique is used to **separate sublimable compounds from nonsublimable impurities**.

Crystallization

1. It is **based on the difference in the solubility** of the compound and the impurities in a suitable solvent.
2. The impure compound is dissolved in a solvent in which it is sparingly soluble at room temperature but appreciably soluble at higher temperature.
3. The solution is concentrated to get a nearly saturated solution. On cooling the solution, pure compound crystallises out and is removed by filtration. The filtrate (mother liquor) contains impurities and small quantity of the compound.
4. The process of inducing crystallization by adding a crystal of pure substance into its saturated solution is called **seeding**.
5. Impurities, which impart colour to the solution, are removed by adsorbing over activated charcoal.
6. Repeated crystallisation becomes necessary for the purification of compounds containing impurities of comparable solubility.
7. Sugar and benzoic acid are separated by this process.

Simple Distillation

This important method is used to separate

1. volatile liquids from non-volatile impurities
2. liquids having sufficient difference in their boiling points.

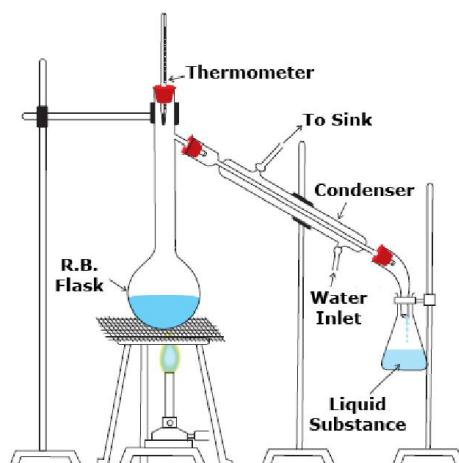
Liquids having different boiling points vaporise at different temperatures. The vapours are cooled and the liquids so formed are collected separately.

Example

Chloroform (b.p. 334 K) and Aniline (b.p. 457 K)

MECHANISM

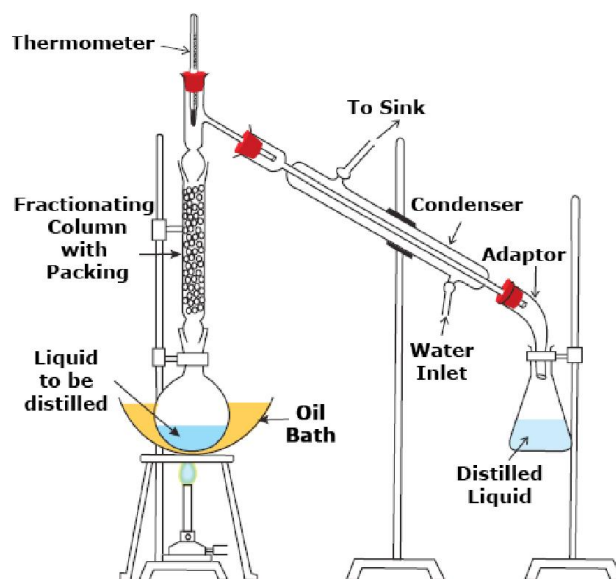
1. On boiling, the vapours of lower boiling component are formed first. The vapours are condensed by using a condenser and the liquid is collected in a receiver.
2. The vapours of higher boiling component form later and the liquid can be collected separately.

**Fractional Distillation**

Fractional distillation is required if the **difference in boiling points of two liquids is not much**. The vapours of such liquids are formed within the same temperature range and are condensed simultaneously.

MECHANISM

1. Vapours of a liquid mixture are passed through a fractionating column before condensation.
2. Vapours of the liquid with higher boiling point condense before the vapours of the liquid with lower boiling point.
3. The vapours rising up in the fractionating column become richer in more volatile component.
4. By the time the vapours reach to the top of the fractionating column, these are rich in the more volatile component.

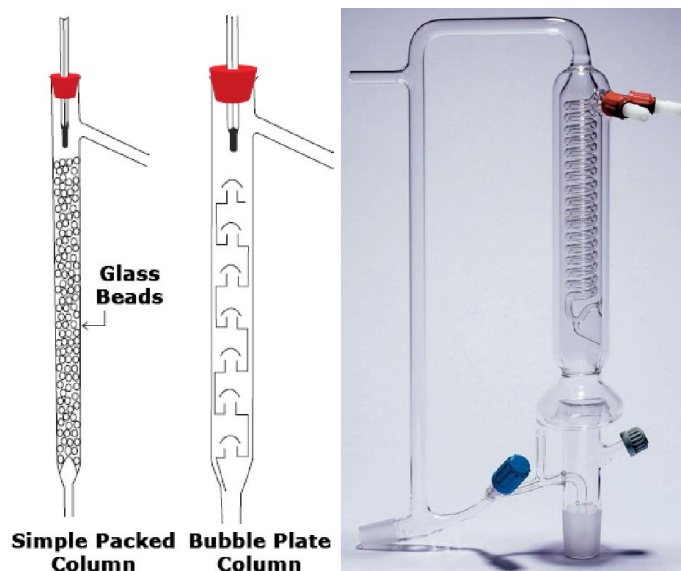


Fractionating columns

1. A fractionating column provides many surfaces for heat exchange between the ascending vapours and the descending condensed liquid.
2. The vapours thus become richer in low boiling component. The vapours of low boiling component ascend to the top of the column.
3. On reaching the top, **the vapours become pure in low boiling component** and pass through the condenser and the pure liquid is collected in a receiver.
4. After a series of successive distillations, the remaining liquid in the distillation flask gets enriched in high boiling component.
5. Each successive condensation and vaporisation unit in the fractionating column is called a **theoretical plate**.

Application: Crude oil in petroleum industry.

Types of Fractionating Columns

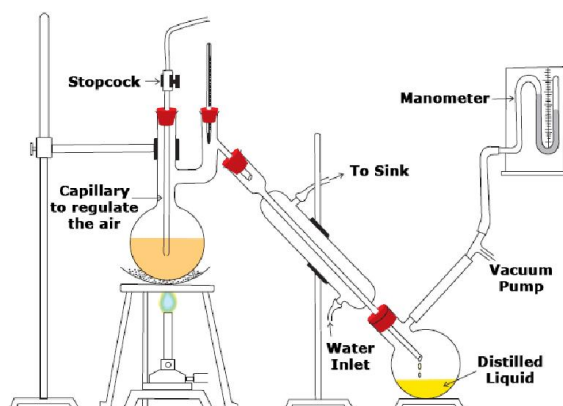


Distillation under Reduced Pressure

This method is used to **purify liquids** having **very high boiling points** and those, which **decompose at or below their boiling points**.

Such liquids are made to boil at a temperature lower than their normal boiling points by reducing the pressure on their surface. A liquid boils at a temperature at which its vapour pressure is equal to the external pressure. The pressure is reduced with the help of a water pump or vacuum pump.

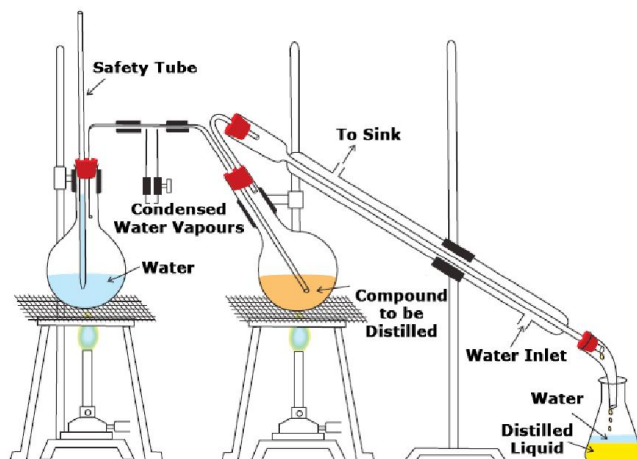
Application: Glycerol can be separated from spent-lye in soap industry.



Steam Distillation

1. It is used to separate substances which are **steam volatile** and are **immiscible with water**.
2. In steam distillation, steam from a steam generator is passed through a heated flask containing the liquid to be distilled. The mixture of steam and the volatile organic compound is condensed and collected.
3. In steam distillation, the liquid boils when the sum of vapour pressures due to the organic liquid (p_1) and that due to water (p_2) becomes equal to the atmospheric pressure (p), i.e. $p = p_1 + p_2$. Since p_1 is lower than p , the organic liquid vaporises at lower temperature than its boiling point.
4. Thus, if one of the substances in the mixture is water and the other, a water insoluble substance, then the mixture will boil close to but below, 373K.
5. A mixture of water and the substance is obtained which can be separated by using a separating funnel.

Example: Aniline–Water Mixture



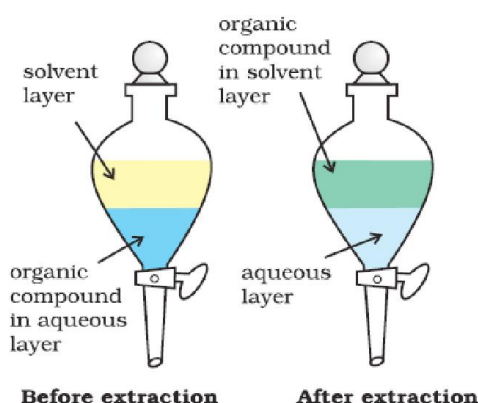
Differential Extraction

When an organic compound is present in an aqueous medium, it is separated by shaking it with an **organic solvent** in which it is **more soluble than in water**.

The organic solvent and the aqueous solution should be immiscible with each other so that they form two distinct layers which can be separated by separatory funnel. The organic solvent is later removed by distillation or by evaporation to get back the compound.

If the organic compound is less soluble in the organic solvent, a very large quantity of solvent would be required to extract even a very small quantity of the compound. **Continuous extraction** is employed in such cases. In this technique same solvent is repeatedly used for extraction of the compound.

Differential Extraction Apparatus



Chromatography

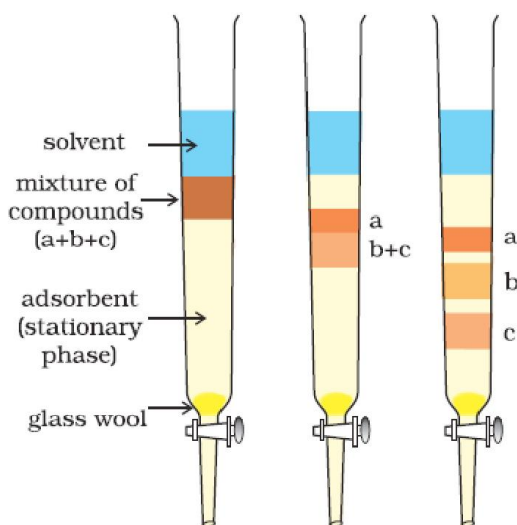
Chromatography is used to separate mixtures into their components, purify compounds and also to test the purity of compounds. It is based on **adsorption**.

Greek word *chroma* - colour.

MECHANISM

1. The mixture of substances is applied onto a stationary phase, which may be a solid or a liquid. A pure solvent, a mixture of solvents, or a gas is allowed to move slowly over the stationary phase.
2. The components of the adsorbent are placed on the top of the adsorbent column packed in a glass tube. An appropriate eluant, which is a liquid or a mixture of liquids is allowed to flow down the column slowly.
3. Depending upon the degree to which the compounds are adsorbed, complete separation takes place. The most readily adsorbed substances are retained near the top and others come down to various distances in the column.

Column Chromatography



Different stages represent the progressive separation of the mixture into three bands.

Thin Layer Chromatography

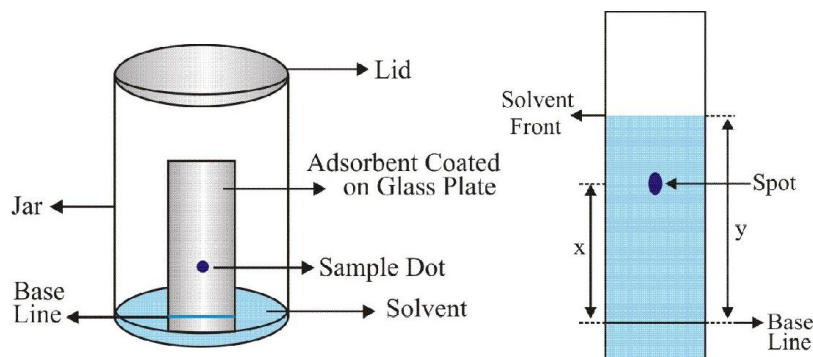
It involves separation of substances of a mixture over a thin layer of an adsorbent coated on glass plate. A thin layer of an adsorbent (silica gel or alumina) is spread over a glass plate of suitable size. The plate is known as **thin layer chromatography plate** or **chromaplate**.

MECHANISM

1. The solution of the mixture to be separated is applied as a small spot about 2 cm above one end of the TLC plate. The glass plate is then placed in a closed jar containing the eluant.

- As the eluant rises up the plate, the components of the mixture move up along with the eluant to different distances depending on their degree of adsorption and separation takes place.
- The relative adsorption of each component of the mixture is expressed in terms of its **retardation factor** i.e. **R_f value**.

TLC Apparatus



$$R_f = \frac{\text{Distance travelled by substance from the baseline (x)}}{\text{Distance travelled by solvent from the baseline (y)}}$$

Detection of Movement

The spots of coloured compounds are visible on TLC plate due to their original colour.

Other Detection Techniques

- Colourless compounds** can be detected by putting the plate under **ultraviolet light**.
- Place the plate in a covered jar containing a few crystals of iodine. Spots of compounds, which adsorb iodine, will show up as brown spots.
- Spraying Agents** - Amino acids may be detected by spraying the plate with **ninhydrin solution**.

Partition Chromatography

It is based on continuous differential partitioning of components of a mixture between stationary and mobile phases.

Paper chromatography is a type of partition chromatography. A special quality paper known as chromatography paper is used.

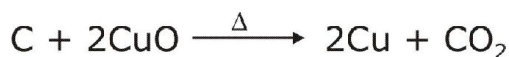
MECHANISM

- Chromatography paper contains water trapped in it, which acts as the stationary phase. A strip of chromatography paper spotted at the base with the solution of the mixture is suspended in a suitable solvent or a mixture of solvents. This solvent acts as the mobile phase.
- The solvent rises up the paper by capillary action and flows over the spot. The paper selectively retains different components according to their differing partition in the two phases. The paper strip so developed is known as a **chromatogram**.

Detection of Carbon and Hydrogen

Carbon and hydrogen are detected by heating the compound with CuO. Carbon present in the compound is oxidised to carbon dioxide and hydrogen to water.

CO₂ is tested with lime-water, which develops turbidity. H₂O is tested with anhydrous copper sulphate, which turns blue.



Detection of Other Elements

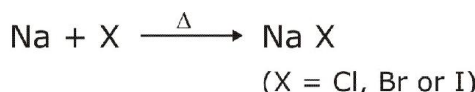
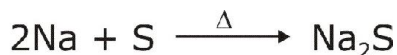
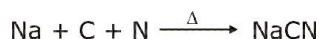
Nitrogen, sulphur, halogens and phosphorus present in an organic compound are detected by **Lassaigne's test**.

The elements present in the compound are converted from covalent form into the ionic form by **fusing the compound with sodium metal**.

Detection of Other Elements

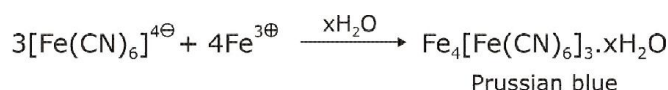
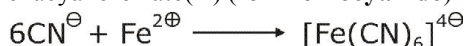
Nitrogen, sulphur, halogens and phosphorus present in an organic compound are detected by **Lassaigne's test**.

The elements present in the compound are converted from covalent form into the ionic form by **fusing the compound with sodium metal**.

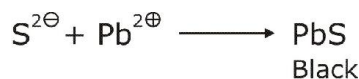
**(A) Test for Nitrogen**

The sodium fusion extract is boiled with FeSO_4 and then acidified with concentrated sulphuric acid. Sodium cyanide first reacts with FeSO_4 and forms sodium hexacyanoferrate(II).

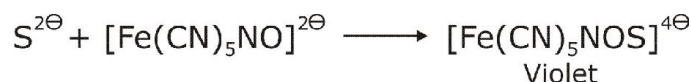
On heating with concentrated sulphuric acid, some iron(II) ions are oxidised to iron(III) ions which react with sodium hexacyanoferrate(II) to produce iron(III) hexacyanoferrate(II) (**ferriferrocyanide**) which is **Prussian blue** in colour.

**(B) Test for Sulphur**

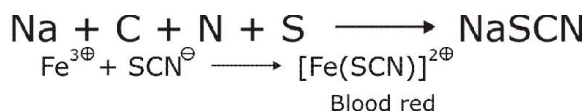
(1) The sodium fusion extract is acidified with **acetic acid** and **lead acetate** is added to it. A **black precipitate** of lead sulphide indicates the presence of sulphur.



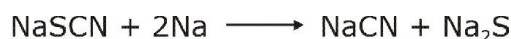
(2) On treating sodium fusion extract with **sodium nitroprusside**, appearance of a **violet colour** further indicates the presence of sulphur.

**Nitrogen and Sulphur Together**

In this case, **sodium thiocyanate** is formed. It gives **blood red colour** and no Prussian blue since there are no free cyanide ions.

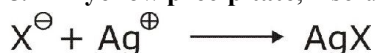


If sodium fusion is carried out with excess of sodium, the thiocyanate decomposes to yield cyanide and sulphide. These ions give their usual tests.

**(C) Test for Halogens**

The sodium fusion extract is acidified with nitric acid and then treated with **silver nitrate**.

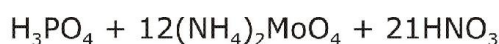
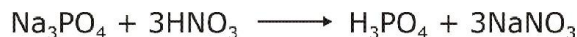
1. A **white precipitate**, soluble in ammonium hydroxide shows the presence of **chlorine**.
2. A **yellowish precipitate**, sparingly soluble in ammonium hydroxide shows the presence of **bromine**.
3. A **yellow precipitate**, insoluble in ammonium hydroxide shows the presence of **iodine**.



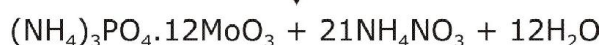
If nitrogen or sulphur is also present in the compound, the sodium fusion extract is first boiled with concentrated nitric acid to decompose cyanide or sulphide of sodium formed during Lassaigne's test. These ions would otherwise interfere with silver nitrate test for halogens.

(D) Test for Phosphorus

The compound is heated with an oxidising agent (sodium peroxide). The phosphorus present in the compound is oxidised to phosphate. The solution is boiled with nitric acid and then treated with **ammonium molybdate**. A **yellow colouration or precipitate** indicates the presence of phosphorus.



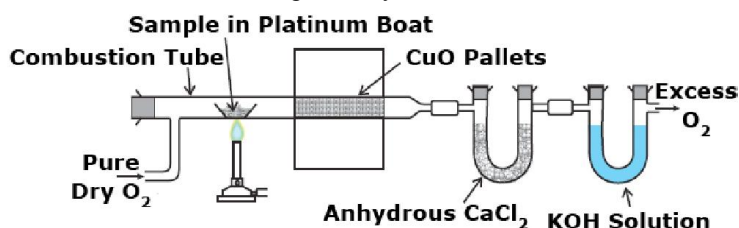
Ammonium
molybdate



Ammonium
phosphomolybdate

Carbon and Hydrogen

A known mass of an organic compound is burnt in the presence of excess of oxygen and CuO. Carbon and hydrogen in the compound are oxidised to carbon dioxide and water respectively.



1. H₂O is absorbed by anhydrous CaCl₂.
2. CO₂ is absorbed by KOH solution.
3. The increase in masses of CaCl₂ and KOH gives the amounts of H₂O and CO₂.

Percentage of C and H



$$\text{Percentage of C} = \frac{12 \times m_2 \times 100}{44 \times m}$$

$$\text{Percentage of H} = \frac{2 \times m_1 \times 100}{18 \times m}$$

Example

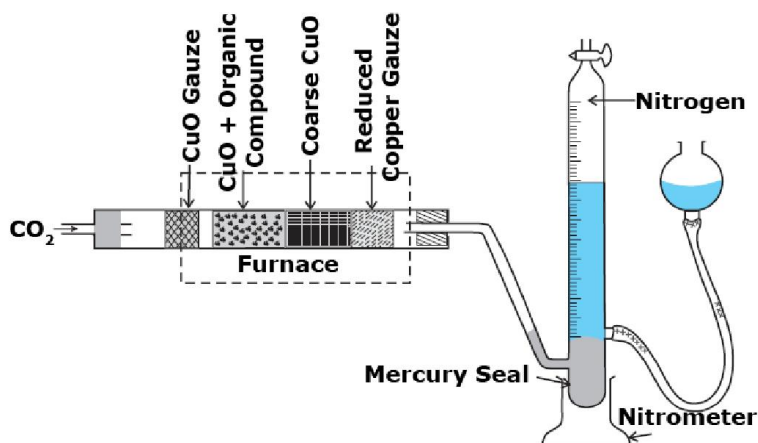
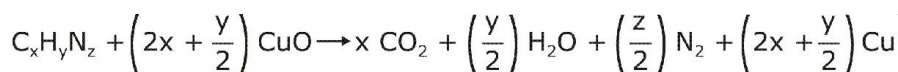
On complete combustion, 0.246 g of an organic compound gave 0.198 g of carbon dioxide and 0.1014 g of water. Determine the percentage composition of carbon and hydrogen in the sample.

% of carbon = 21.95

% of hydrogen = 4.58

Nitrogen - Dumas Method

The nitrogen containing organic compound, when heated with CuO in an atmosphere of CO₂, yields free N₂ in addition to CO₂ and H₂O.

**Calculations**

Mass of organic compound = m g

Volume of N_2 collected = V_l ml

Room temperature = T_l K

Volume of N_2 at STP $V =$

$$\frac{p_l \times V_l \times 273}{760 \times T_l} \text{ ml}$$

p_l = Barometric pressure – Aqueous tension

Calculations

22400 mL N_2 at STP weighs 28 g.

V ml of N_2 at STP weighs =

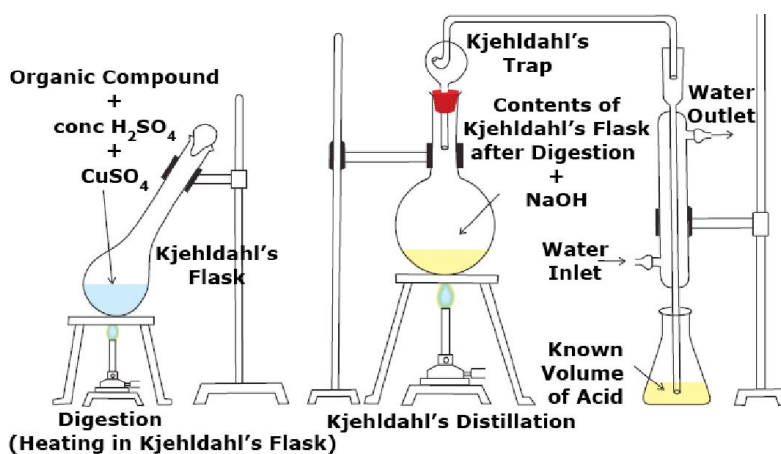
$$\frac{28 \times V}{22400} \text{ g}$$

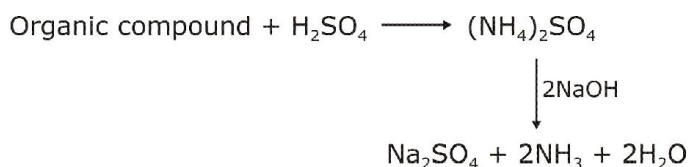
Percentage of nitrogen =

$$\frac{28 \times V \times 100}{22400 \times m} \%$$

Kjeldahl's method

The compound containing nitrogen is heated with conc H_2SO_4 . Nitrogen in the compound gets converted to $(NH_4)_2SO_4$. It is done by estimating unreacted conc H_2SO_4 left after the absorption of NH_3 by titrating it with standard alkali solution.





Let the mass of organic compound taken = m g

Volume of H_2SO_4 (1M) taken = V ml

Volume of NaOH (1M) for titration of excess of H_2SO_4 = V_1 ml

Moles of H_2SO_4 = $2 \times$ Moles of NaOH

Therefore, V_1 ml of NaOH = $V_1/2$ ml of H_2SO_4

Volume of H_2SO_4 unused = $V - V_1/2$ ml

$(V - V_1/2)$ ml of H_2SO_4 = $2(V - V_1/2)$ ml of NH_3 solution

1000 ml of 1M NH_3 solution contains 17 g of NH_3 or 14 g of N.

$2(V - V_1/2)$ ml of NH_3 solution of molarity M contains:

$$\frac{14 \times M \times 2(V - V_1/2)}{1000} \text{ gN}$$

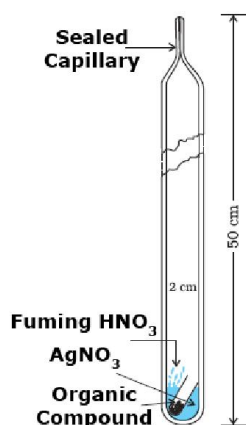
Percentage of Nitrogen =

$$\frac{14 \times M \times 2(V - V_1/2)}{1000} \times \frac{100}{m}$$

Carius Method

A known mass of an organic compound is heated with fuming nitric acid in the presence of silver nitrate contained in a hard glass tube known as Carius tube, in a furnace.

The halogen present forms the corresponding silver halide (AgX). It is filtered, washed, dried and weighed.



Mass of organic compound taken = m g

Mass of AgX formed = m_1 g

1 mol of AgX contains 1 mol of X

Mass of halogen in m_1 g of AgX =

$$\frac{\text{Atomic Mass of X} \times m_1}{\text{Molecular Mass of AgX}} \text{ g}$$

Percentage of halogen =

$$\frac{\text{Atomic Mass of X} \times m_1 \times 100}{\text{Molecular Mass of AgX} \times m} \%$$

Sulphur

A known mass of an organic compound is heated in a Carius tube with sodium peroxide or fuming nitric acid. Sulphur present in the compound is oxidised to sulphuric acid. It is precipitated as barium sulphate by adding excess of barium chloride solution in water.

Mass of organic compound taken = m g

Mass of BaSO_4 formed = m_1 g

1 mol of BaSO_4 (233 g) contains 1 mol sulphur (32 g)

m_1 g BaSO_4 contains

$$\frac{32 \times m_1}{233} \text{ g sulphur}$$

Percentage of Sulphur =

$$\frac{32 \times m_1 \times 100}{233 \times m} \%$$

Phosphorus

A known mass of an organic compound is heated with fuming nitric acid whereupon phosphorus present in the compound is oxidised to phosphoric acid. It is precipitated as ammonium phosphomolybdate, $(\text{NH}_4)_3\text{PO}_4 \cdot 12\text{MoO}_3$, by adding ammonia and ammonium molybdate. Alternatively, phosphoric acid may be precipitated as $\text{Mg}(\text{NH}_4)\text{PO}_4$ by adding magnesia mixture which on ignition yields $\text{Mg}_2\text{P}_2\text{O}_7$.

Mass of organic compound taken = m g

Mass of ammonium phosphomolybdate = m_1 g

Molar mass of $(\text{NH}_4)_3\text{PO}_4 \cdot 12\text{MoO}_3 = 1877$ g

Percentage of P =

$$\frac{31 \times m_1 \times 100}{1877 \times m} \%$$

If phosphorus is estimated as $\text{Mg}_2\text{P}_2\text{O}_7$,

Percentage of P =

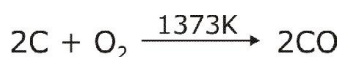
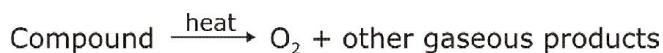
$$\frac{62 \times m_1 \times 100}{222 \times m} \%$$

Oxygen

Percentage of oxygen in an organic compound =

100 – percentage of other elements

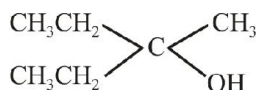
A definite mass of an organic compound is decomposed by heating in a stream of nitrogen gas. The mixture of gaseous products containing oxygen is passed over red-hot coke when all the oxygen is converted to carbon monoxide. This mixture is passed through warm iodine pentoxide (I_2O_5) when carbon monoxide is oxidised to carbon dioxide producing iodine.

**CHN Elemental Analyzer**

The estimation of elements in an organic compound is carried out by using microquantities of substances and automatic experimental techniques. The elements, carbon, hydrogen and nitrogen present in a compound are determined by an apparatus known as **CHN elemental analyser**.

EXERCISE - 1 : BASIC OBJECTIVE QUESTIONS**Nomenclature**

1. The correct IUPAC nomenclature for the following alcohol is :



- (a) 2-ethyl-2-butanol (b) 3-methyl-3-pentanol
(c) 3-ethyl-3-methyl-3-pentanol
(d) 1, 1-diethanol

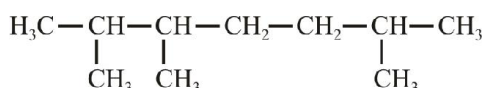
2. The IUPAC name of $\text{CH}_3\text{C}(\text{CH}_3)_2\text{CH}_2\text{CH}=\text{CH}_2$ is :

- (a) 2,2-dimethyl-4-pentene
(b) 4,4-dimethyl-1-pentene
(c) 1,1,1-trimethyl-3-butene
(d) 2-chloro-4,4-dimethylhexane

3. The IUPAC name of $\text{CH}_3-\text{C}\equiv\text{CCH}(\text{CH}_3)_2$ is :

- (a) 4-methyl-2-pentyne
(b) 4,4'-dimethyl-2-pentyne
(c) methyl isopropyl acetylene
(d) 4,4,4-trimethyl-1-butene

4. The IUPAC name of the following compound is :



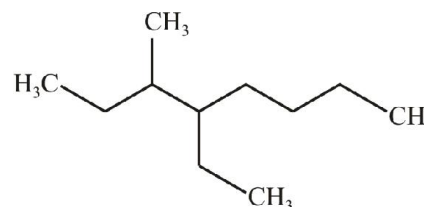
- (a) 2, 5, 6-trimethylheptane
(b) 1,3-isopropyl-3-methylpropane
(c) 2, 6, 3-trimethylheptane
(d) 2,3,6-trimethylheptane

5. The name of $\text{ClCH}_2-\text{C}(\text{Br})=\text{C}(\text{Br})-\text{CH}_2\text{Cl}$ according to IUPAC

nomenclature system is :

- (a) 2,3-dibromo-1,4-dichlorobutene-2
(b) 1,4-dichloro-2,3-dibromobutene-2
(c) dichlorobromobutene
(d) dichlorobromobutane

6. Name of the compound given below is :



- (a) 2,3-diethylheptane (b) 3-methyl-4-ethyloctane
(c) 5-ethyl-6-methyloctane (d) 4-ethyl-3-methyloctane

7. The IUPAC name of $\text{CH}_3\text{CH}_2\text{C}(\text{Br})=\text{CHCl}$ is :

- (a) 4-chloro-3-bromo but-3-ene
(b) 2-bromo-1-chloro butane
(c) 2-bromo-1-chloro but-1-ene
(d) 2-bromo-2-ethyl-3-chloro propene

8. The IUPAC name of $\text{CH}_3-\text{CH}(\text{C}_2\text{H}_5)-\text{CH}_2-\text{CH}(\text{CH}_3)-\text{COOH}$

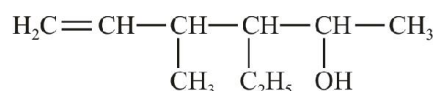
is :

- (a) 2,2-diethyl pentanoic acid
(b) 2,4-dimethyl hexanoic acid
(c) 2-methyl-4-ethyl pentanoic acid
(d) 4-ethyl-2-methyl pentanoic acid

9. The IUPAC name of $\text{H}_2\text{C}=\text{CH}-\text{CH}(\text{CH}_2\text{CH}_2\text{CH}_3)_2$ is :

- (a) 3-propylpent-1-ene (b) 3-ethylpent-1-ene
(c) 4-ethylhex-1-ene (d) 3-ethylhex-1-ene

10. The IUPAC name of the compound



is :

- (a) 3-ethyl-4-methylhex-5-en-2-ol
(b) 3-ethyl-4-ethylhex-1-en-5-ol
(c) 3-ethyl-2-hydroxy-4-methylhex-5-ene
(d) none of the above

11. The IUPAC name of the compound $(\text{CH}_3)_2\text{CH}-\text{CH}=\text{CH}-\text{CHOH}-\text{CH}_3$ is :
 (a) 5-methylhex-3-en-2-ol
 (b) 2-methylhex-3-en-5-ol
 (c) 2-hydroxy-5-methyl-3-hexene
 (d) 5-hydroxy-2-methyl-3-hexene
12. The suffixes for alcohols, aldehydes and ketones according to IUPAC system are respectively :
 (a) -alc, -ald, -ket (b) -ol, -al, -ket
 (c) -ol, -al, -one (d) -coh, -ald, -one

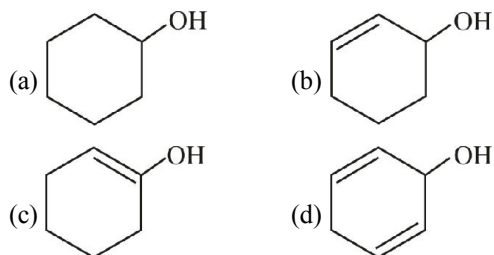
Hybridisation

13. The hybridization of carbon atoms in $\text{C}-\text{C}$ single bond of $\text{H}-\text{C}\equiv\text{C}-\text{CH}=\text{CH}_2$ is :
 (a) sp^3-sp^3 (b) sp^2-sp^3
 (c) sp^3-sp (d) $\text{sp}-\text{sp}^2$
14. Hybridized carbon atom used in the formation of the compound $\text{CH}_3-\text{CH}=\text{CH}-\text{CH}_2-\text{CH}_3$ are :
 (a) sp and sp^3 (b) sp and sp^2
 (c) only sp (d) sp^2 and sp^3
15. In the hydrocarbon,

$$\text{H}_3\text{C}-\underset{6}{\text{CH}}=\underset{5}{\text{CH}}-\underset{4}{\text{CH}}-\underset{3}{\text{CH}_2}-\underset{2}{\text{C}}\equiv\underset{1}{\text{CH}}$$
 The state of hybridization of carbons 1,3,5 are in the following sequence :
 (a) $\text{sp}, \text{sp}^2, \text{sp}^3$ (b) $\text{sp}^3, \text{sp}^2, \text{sp}$
 (c) $\text{sp}^2, \text{sp}, \text{sp}^3$ (d) $\text{sp}, \text{sp}^3, \text{sp}^2$
16. Which of the following statements is true about hybridization ?
 (a) Hybrid orbitals frequently undergo linear overlaps making sigma bonds.
 (b) Hybrid orbitals frequently undergo lateral overlaps making π -bonds. In other words, there are several compounds in which π -bonds are formed using hybrid orbitals.
 (c) Hybrid orbitals are molecular orbitals.
 (d) A hybrid orbital bigger in size makes shorter bond.

17. $\text{CH}_2=\underset{3}{\text{CH}}-\underset{2}{\text{CH}}-\underset{1}{\text{CN}}$
 (a) $\text{C1}-\text{C2}$ bond of this molecule is formed by
 (a) sp^3-sp^2 overlap (b) sp^2-sp^3 overlap
 (c) $\text{sp}-\text{sp}^2$ overlap (d) sp^2-sp^2 overlap

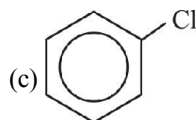
18. In which of the following compounds, C has used sp^2 orbital in making $\text{C}-\text{O}$ bond ?



19. $\text{CH}_2=\text{C}=\text{CH}_2$
 In this molecule (allene)
 (a) all three C-atoms are sp^2 -hybridized
 (b) both terminal C-atoms are sp^2 -hybridized while central C-atom is sp -hybridized
 (c) both terminal C-atoms are sp -hybridized while central C-atom is sp^2 -hybridized
 (d) none of these
20. Carbon atoms in the compound $(\text{CN})_4\text{C}_2$ are
 (a) sp hybridised (b) sp^2 hybridised
 (c) sp and sp^2 hybridised
 (d) sp, sp^2 and sp^3 hybridised
21. Number of π electrons present in naphthalene is
 (a) 4 (b) 6
 (c) 10 (d) 14
22. In which bond angle is the highest
 (a) sp^3 (b) sp^2
 (c) sp (d) sp^3d

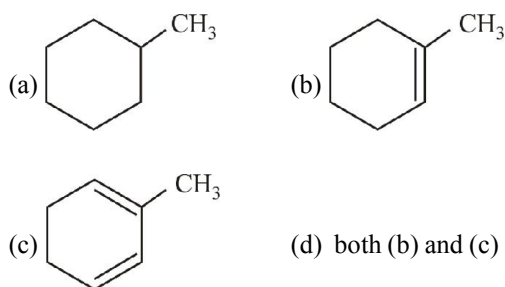
Inductive Effect

23. Inductive effect of Cl atom operates in
 (a) $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{Cl}$
 (b) $\text{CH}_2=\text{CH}-\text{CH}=\text{CH}-\text{Cl}$

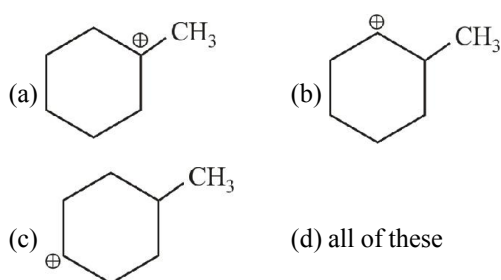


- (d) all of these

24. Inductive effect of a —CH_3 group operates



25. Inductive effect of —CH_3 group operates in



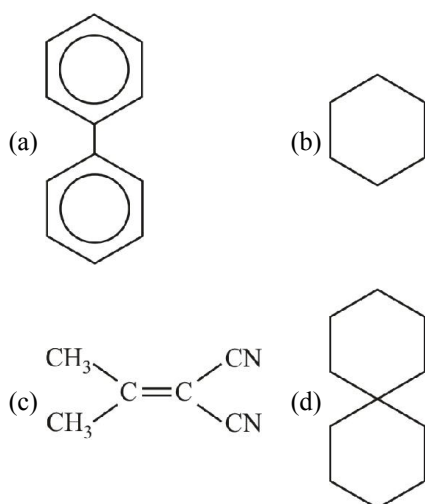
26. —NO_2 I —Cl II —I III

Which of these groups has —I effect ?

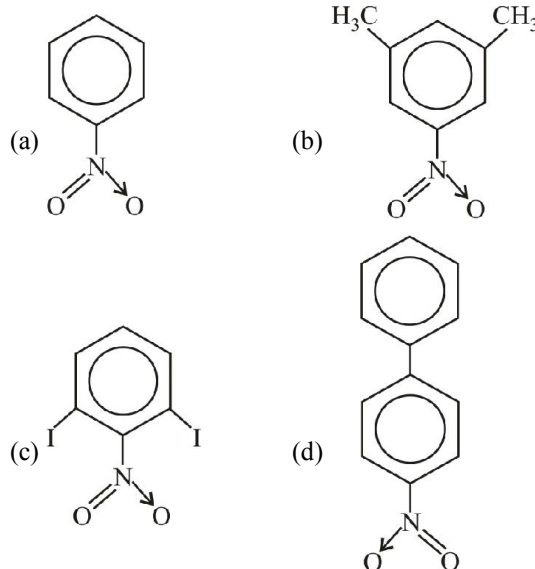
- (a) I (b) II
(c) III (d) all of these

Resonance

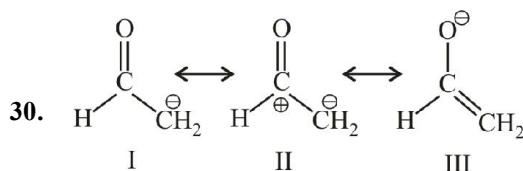
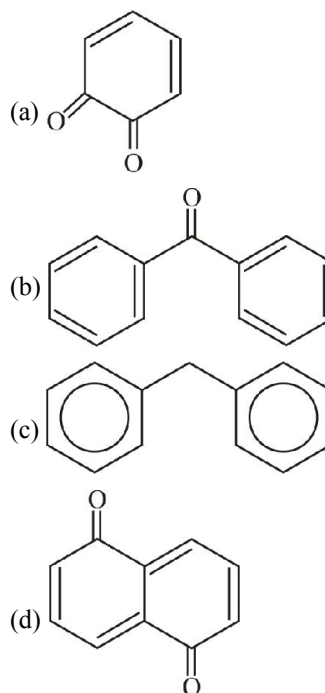
27. In which of the following molecules, all atoms are coplanar?



28. In which of the following molecules, —NO_2 group is not coplanar with phenyl ring ?

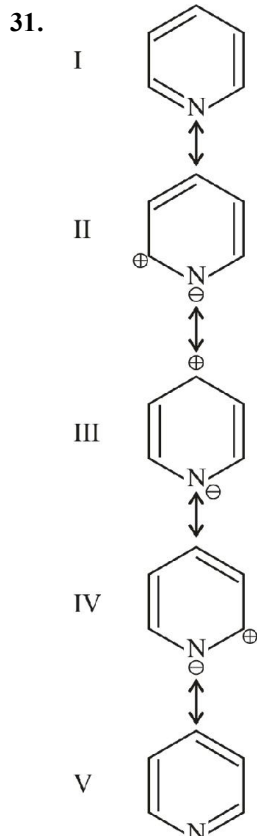


29. In which of the following molecules, all atoms are not coplanar ?



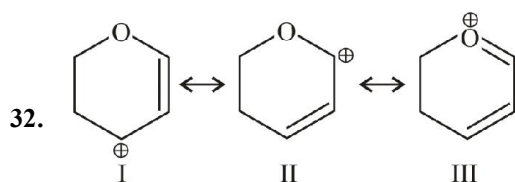
The correct order of stability among these canonical structures is

- (a) $\text{I} > \text{III} > \text{II}$ (b) $\text{III} > \text{I} > \text{II}$
(c) $\text{II} > \text{III} > \text{I}$ (d) $\text{II} > \text{I} > \text{III}$



Among these canonical structures of pyridine, the correct order of stability is

- (a) (I = V) > (II = IV) > III (b) (II = IV) > (I = V) > III
(c) (I = V) > III > (II = IV) (d) III > (II = IV) > (I = V)

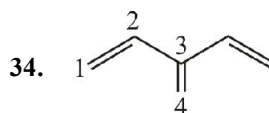


The most stable canonical structure among these is

- (a) I (b) II
(c) III (d) all are equally stable

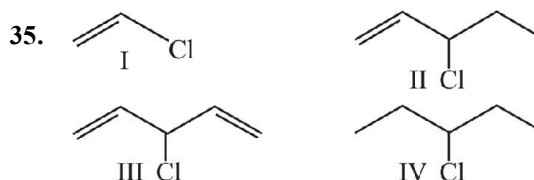
33. Heat of hydrogenation of cyclohexene is -28.6 kcal/mol and that of benzene is -49.8 kcal/mol. Then resonance energy of benzene will be

- (a) 46 kcal/mol (b) 39 kcal/mol
(c) 36 kcal/mol (d) 30 kcal/mol



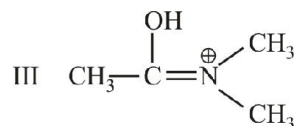
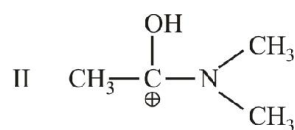
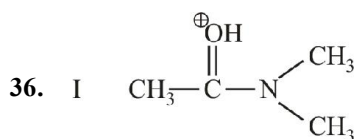
Which of the following statements is true about this molecule ?

- (a) C1—C2 and C3—C4 bonds are of same length
(b) C1—C2 bond is shorter than C3—C4 bond
(c) C1—C2 bond is longer than C3—C4 bond
(d) C1—C2 and C2—C3 bonds are of same length



Among these compounds, which one has shortest C—Cl bond ?

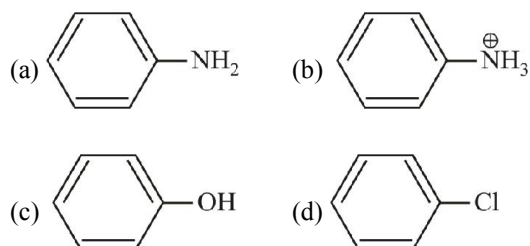
- (a) I (b) II
(c) III (d) IV



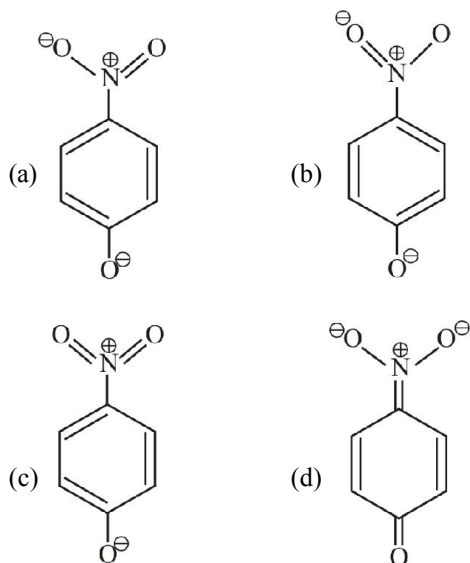
The stability order of these canonical structures is

- (a) I > II > III (b) III > I > II
(c) I > III > II (d) II > III > I

37. In which of the following molecule the resonance effect is not present ?



38. The most unlikely representation of resonance structure of p-nitrophenoxide ion is :



39. Resonance structure of a molecules should have
(a) Identical arrangement of atoms
(b) Nearly the same energy content
(c) The same number of paired electrons
(d) All the above

Electrophile/Nucleophile

40. Which of the following species is an electrophile ?
(a) $\text{H}^{\oplus}$ (b) $\text{H}^{\ominus}$
(c) both (a) and (b) (d) none of these
41. Which of the following species is a nucleophile ?
(a) H_2O (b) $\text{H}_3\text{O}^{\oplus}$
(c) both (a) and (b) (d) none of these
42. Which of the following species is a nucleophile ?
(a) $\text{CH}_3\text{O}^{\ominus}$ (b) CH_3OH
(c) both (a) and (b) (d) none of these

43.
pyrrole (I) pyridine (II) aniline (III)
Which is/are correct statements ?
(a) I is more basic than II

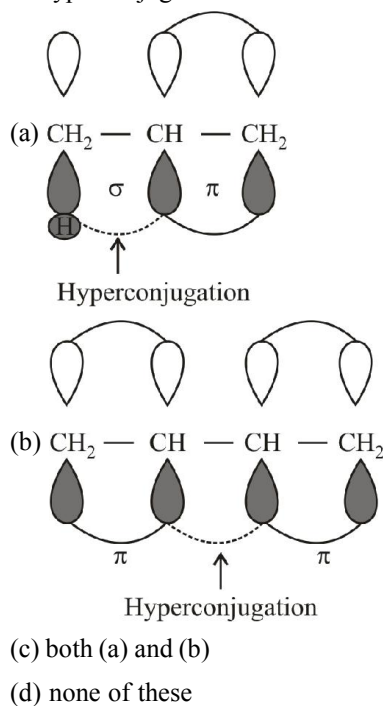
- (b) II is more basic than I and III
(c) III is more basic than II
(d) None of these

Hyperconjugation

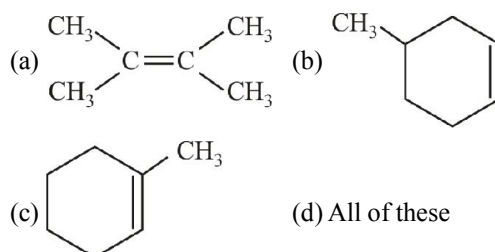
44. Which of the following represents the case of hyperconjugation ?
(a) $\text{CH}=\text{CH}=\text{CH}_2$ (b) $\text{CH}_2-\text{CH}=\text{CH}_2$

(c) $\text{CH}_2-\text{CH}=\text{O}$ (d) both (b) and (c)

45. Which of the following orbital diagrams represents the case of hyperconjugation ?

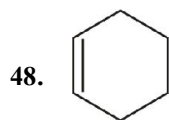


46. Hyperconjugation occurs in



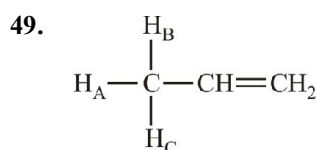
47. Hyperconjugation occurs in

- (a) $\text{CH}_3-\overset{\oplus}{\text{C}}\text{H}_2$ (b) $\text{CH}_3-\overset{\oplus}{\text{C}}\text{H}-\text{CH}_3$
(c) $\text{CH}_3-\text{CH}_2-\overset{\oplus}{\text{C}}\text{H}_2$ (d) all of these

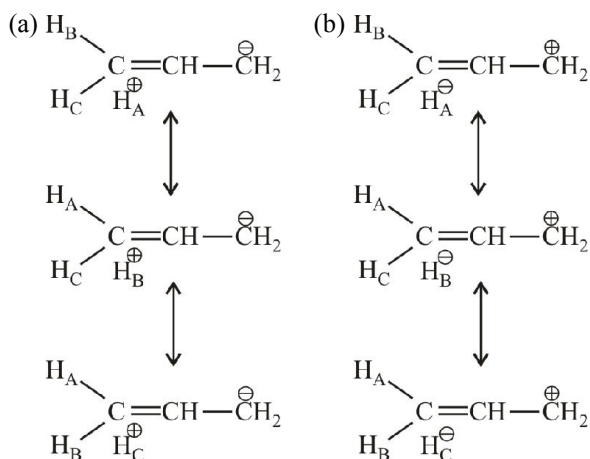


How many hyperconjugable H-atoms does this molecule has ?

- (a) 0 (b) 2
(c) 3 (d) 4



Canonical structures due to hyperconjugation in this molecule can be written as



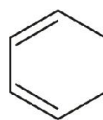
- (c) both (a) and (b)
(d) none of these

Diene	Heat of Hydrogenation
$\text{CH}_2=\text{CH}-\text{CH}=\text{CH}-\text{CH}_3$ I	-54.1 kcal/mol
$\text{CH}_3-\text{CH}=\text{C}=\text{CH}-\text{CH}_3$ II	-70.5 kcal/mol
$\text{CH}_2=\text{CH}-\text{CH}_2-\text{CH}=\text{CH}_2$ III	-60.2 kcal/mol

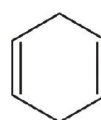
Examine these dienes and their heat of hydrogenations. Now, what will be the stability order among these dienes ?

- (a) $\text{I} > \text{II} > \text{III}$ (b) $\text{II} > \text{III} > \text{I}$
(c) $\text{I} > \text{III} > \text{II}$ (d) $\text{II} > \text{I} > \text{III}$

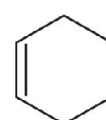
51.



I



II



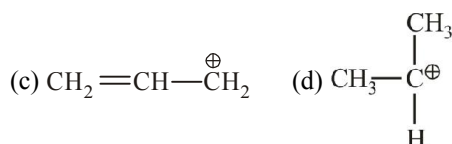
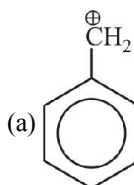
III

Among these compounds, the order of heats of hydrogenation is :

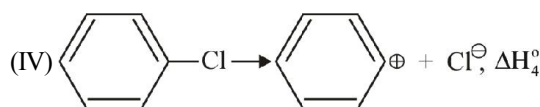
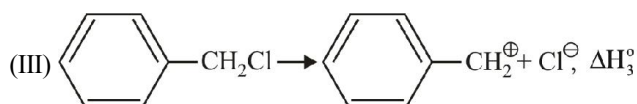
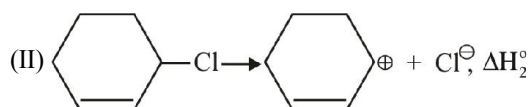
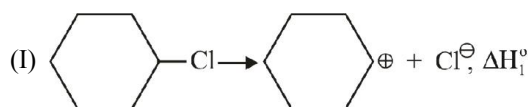
- (a) $\text{I} > \text{II} > \text{III}$ (b) $\text{III} > \text{II} > \text{I}$
(c) $\text{II} > \text{I} > \text{III}$ (d) $\text{II} > \text{III} > \text{I}$

Carbocation

52. Which of the following carbocation is least stable ?



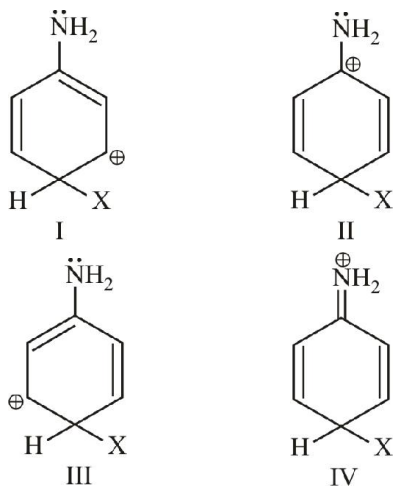
53. For the reactions



The correct decreasing order of enthalpies of reaction for producing carbocation is

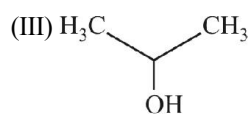
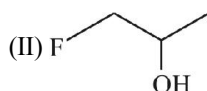
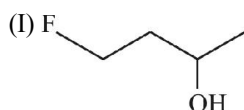
- (a) $\Delta H_1^\circ > \Delta H_2^\circ > \Delta H_3^\circ > \Delta H_4^\circ$
(b) $\Delta H_4^\circ > \Delta H_1^\circ > \Delta H_2^\circ > \Delta H_3^\circ$
(c) $\Delta H_3^\circ > \Delta H_2^\circ > \Delta H_1^\circ > \Delta H_4^\circ$
(d) $\Delta H_2^\circ > \Delta H_1^\circ > \Delta H_4^\circ > \Delta H_3^\circ$

54. Which of the following is most stable



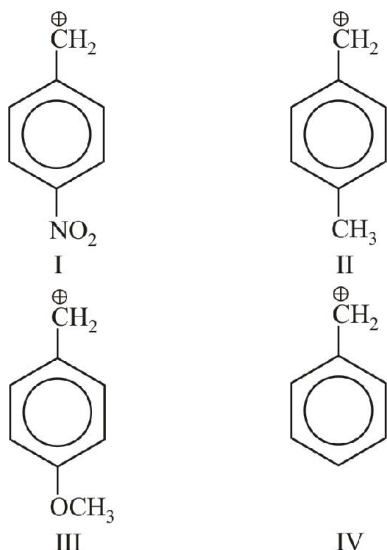
- (a) I (b) II
(c) III (d) IV

55. The order of reactivity of the following alcohols towards conc. HCl is :



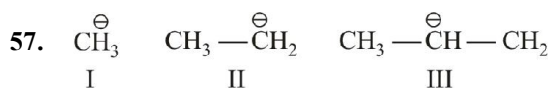
- (a) I > II > III (b) I > III > II
(c) III > II > I (d) III > I > II

56. What is the correct order of stability of the given cations :



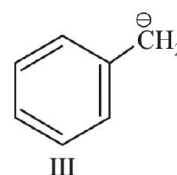
- (a) III > II > IV > I (b) I > II > III > IV
(c) III > I > II > IV (d) IV > II > III > I

Carbanion



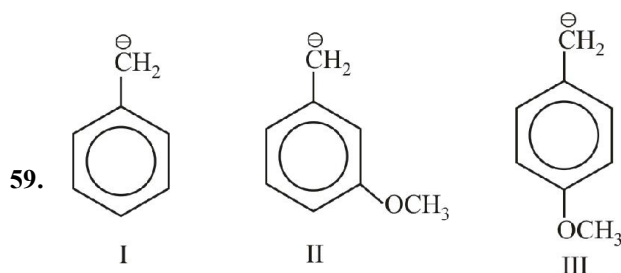
Which of the following orders is correct for the stability of these carbanions ?

- (a) I > II > III (b) III > II > I
(c) II > I > III (d) I $\approx$ II $\approx$ III



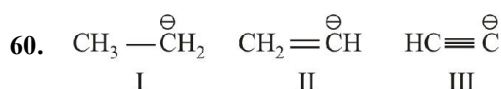
Which of the following orders is correct for the stability of these carbanions ?

- (a) I > II > III (b) III > II > I
(c) II > III > I (d) II > I > III



Which of the following orders is correct for the stability of these carbanions ?

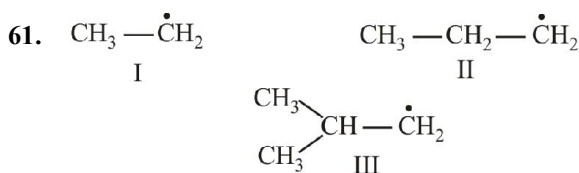
- (a) I > II > III (b) III > II > I
(c) III > I > II (d) II > I > III



Which of the following orders is correct for the stability of these carbanions ?

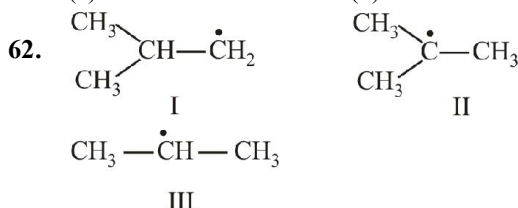
- (a) I > II > III (b) III > II > I
(c) II > I > III (d) III > I > II

Carbon free Radical



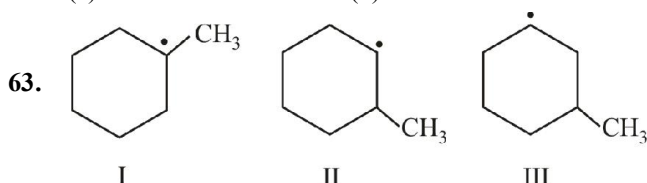
Which of the following orders is correct for no-bond-resonance energy of these radicals ?

- (a) I > II > III (b) III > II > I
(c) III > I > II (d) II > III > I



Which of the following orders is correct for no-bond-resonance energy of these radicals ?

- (a) I > II > III (b) III > II > I
(c) II > III > I (d) II > I > III

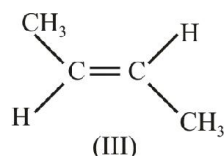
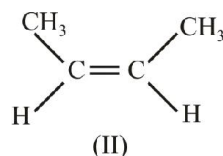
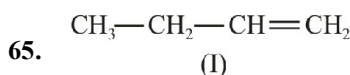


Which of the following orders is correct for no-bond-resonance energy of these radicals ?

- (a) I > II > III (b) III > II > I
(c) II > III > I (d) II > I > III

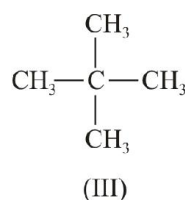
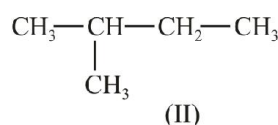
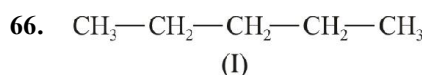
64. Which of the following statements is correct about methyl group ?
- (a) It stabilizes both carbocation and free radical equally
(b) it stabilizes a free radical more than a carbocation
(c) it stabilizes a carbocation more than a free radical
(d) none of these

Isomerism



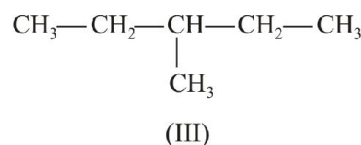
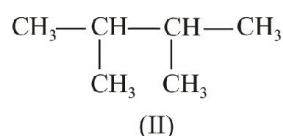
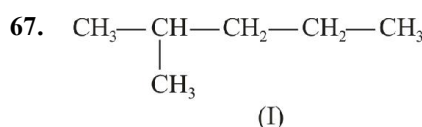
Which among these are stereo-isomers ?

- (a) I and II (b) I and III
(c) II and III (d) all of these



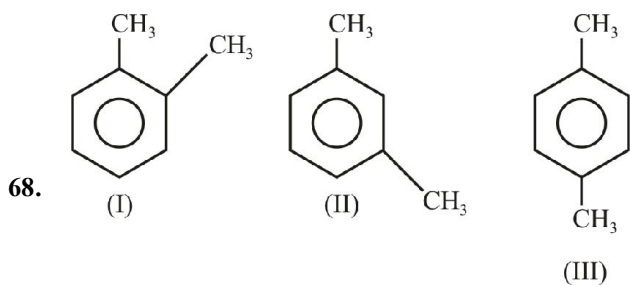
Among these, chain isomers are

- (a) I and II (b) I and III
(c) II and III (d) all of these



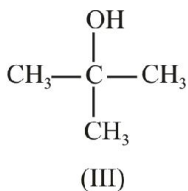
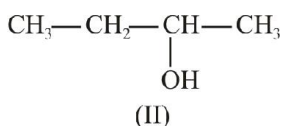
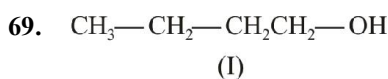
Among these, I is the chain isomer of

- (a) II only (b) III only
(c) both (d) none of these



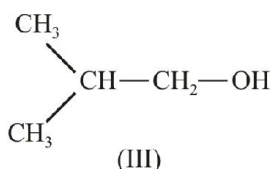
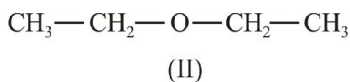
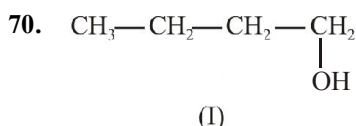
Among these, positional isomers are

- (a) I and II (b) I and III
(c) II and III (d) all of these



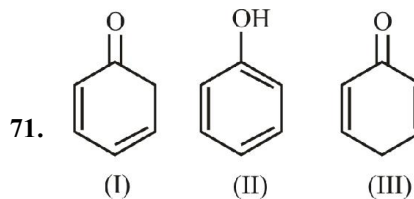
Among these III is the chain isomer of

- (a) I only (b) II only
(c) both I and II (d) none of these



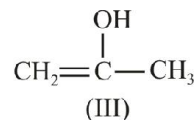
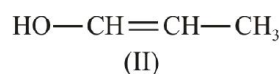
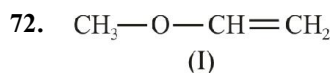
Among these functional isomers are

- (a) I and II (b) I and III
(c) II and III (d) Both (a) and (c)



The tautomer of II is

- (a) I (b) III
(c) both I and III (d) none of these



Which among these are tautomers ?

- (a) I and II (b) I and III
(c) II and III (d) None of these

Isomerism

73. Which of the following will exhibit geometrical isomerism ?

- (a) 1-Phenylbutene-2-ene
(b) 3-Phenylbut-1-ene
(c) 2-Phenylbut-1-ene
(d) 1, 1-Diphenylprop-1-ene

74. Geometrical isomerism will be exhibited by

- (a) Pent-1-ene
(b) 3-Methylbut-1-ene
(c) Pent-2-ene
(d) All of these compounds

75. Which of the following compounds can show geometrical isomerism ?

- (a) Vinyl chloride (b) 1, 1-Dichloroethene
(c) 1, 2-Dichloroethene (d) Trichloroethene

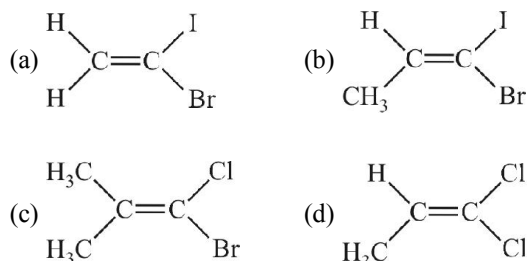
76. The simplest alkene which is capable of exhibiting geometrical isomerism has

- (a) 3 carbon atoms (b) 5 carbon atoms
(c) 6 carbon atoms (d) 4 carbon atoms

77. Which of the following compounds can exhibit geometrical isomerism ?

- (a) Hex-1-ene (b) 2-Methylpent-2-ene
(c) 3-Methylpent-1-ene (d) Hex-2-ene

78. Which one of the following compound will show geometrical isomerism ?



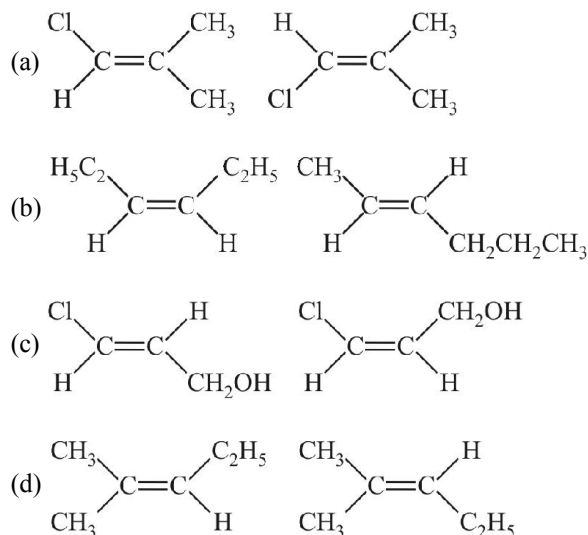
79. How many open chain structural isomers of compound with molecular formula C_6H_{12} show geometrical isomerism ?

- (a) 2 (b) 3
(c) 4 (d) 5

80. Which of the following can exist as cis and trans isomers ?

- (a) $CH_2=CH-Cl$ (b) $CH_2=CH-CH=CH_2$
(c) $CH_3CH_2CH=CHCl$ (d) $CHCl=CCl_2$

81. Which of the following pair of structures represent geometrical isomers of each other?



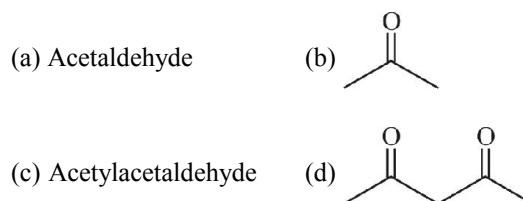
82. The compound $C_4H_{10}O$ can show

- (a) metamerism (b) positional isomerism
(c) functional isomerism (d) all types

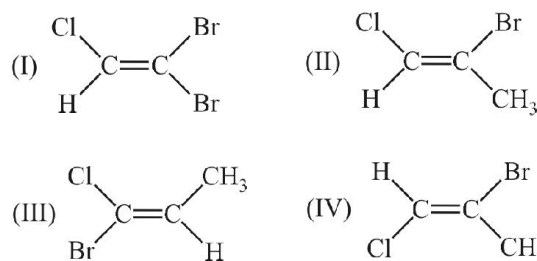
83. Isomers are the compounds having the

- (a) same molecular formula but different physical and/or chemical properties.
(b) same structural formula but different molecular formulae
(c) same chemical properties and physical properties
(d) same physical properties but different chemical properties.

84. Which of the following has maximum enol content ?

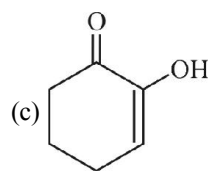
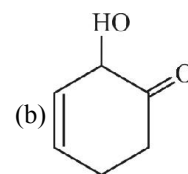
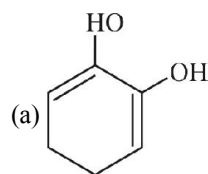
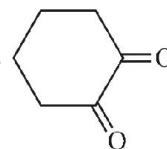


85. Which is a pair of geometrical isomers ?



- (a) I and II (b) I and III
(c) II and IV (d) III and IV

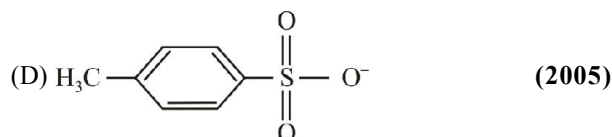
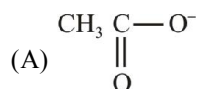
86. Enolic form of the given keto compound is



(d) It will not show tautomerism

(d) C1

11. The decreasing order of nucleophilicity among the nucleophiles



- (a) (C), (B), (A), (D) (b) (B), (C), (A), (D)
(c) (D), (C), (B), (A) (d) (A), (B), (C), (D)

12. Among the following acids which has the lowest pK_a value? (2005)



13. Due to the presence of an unpaired electron, free radicals are (2005)

- (a) cations (b) anions
(c) chemically inactive (d) chemically reactive

14. The number and type of bonds between two carbon atoms in calcium carbide are (2005)

- (a) two sigma, two pi (b) two sigma, one pi
(c) one sigma, two pi (d) one sigma, one pi

15. The correct order of increasing acid strength of the compounds (2006)

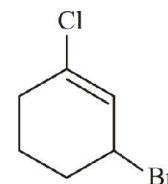


- (a) $\text{B} < \text{D} < \text{A} < \text{C}$ (b) $\text{D} < \text{A} < \text{C} < \text{B}$
(c) $\text{D} < \text{A} < \text{B} < \text{C}$ (d) $\text{A} < \text{D} < \text{C} < \text{B}$

16. The increasing order of stability of the following free radicals is (2006)

- (a) $(\text{CH}_3)_2\dot{\text{C}}\text{H} < (\text{CH}_3)_3\dot{\text{C}} < (\text{C}_6\text{H}_5)_2\dot{\text{C}}\text{H} < (\text{C}_6\text{H}_5)_3\dot{\text{C}}$
(b) $(\text{C}_6\text{H}_5)_3\dot{\text{C}} < (\text{C}_6\text{H}_5)_2\dot{\text{C}}\text{H} < (\text{CH}_3)_3\dot{\text{C}} < (\text{CH}_3)_2\dot{\text{C}}\text{H}$
(c) $(\text{C}_6\text{H}_5)_2\dot{\text{C}}\text{H} < (\text{C}_6\text{H}_5)_3\dot{\text{C}} < (\text{CH}_3)_3\dot{\text{C}} < (\text{CH}_3)_2\dot{\text{C}}\text{H}$
(d) $(\text{CH}_3)_2\dot{\text{C}}\text{H} < (\text{CH}_3)_3\dot{\text{C}} < (\text{C}_6\text{H}_5)_3\dot{\text{C}} < (\text{C}_6\text{H}_5)_2\dot{\text{C}}\text{H}$

17. The IUPAC name of the compound shown below is (2006)



- (a) 2-bromo-6-chlorocyclohex-1-ene
(b) 6-bromo-2-chlorocyclohexene
(c) 3-bromo-1-chlorocyclohexene
(d) 1-bromo-3-chlorocyclohexene

18. The IUPAC name of is (2007)

- (a) 1, 1-diethyl-2, 2-dimethylpentane
(b) 4, 4-dimethyl-5, 5-diethylpentane
(c) 5, 5-diethyl-4, 4-dimethylpentane
(d) 3-ethyl-4, 4-dimethylheptane

19. The correct decreasing order of priority for the functional groups of organic compounds in the IUPAC system of nomenclature is (2008)

- (a) $-\text{COOH}$, $-\text{SO}_3\text{H}$, $-\text{CONH}_2$, $-\text{CHO}$
(b) $-\text{SO}_3\text{H}$, $-\text{COOH}$, $-\text{CONH}_2$, $-\text{CHO}$
(c) $-\text{CHO}$, $-\text{COOH}$, $-\text{SO}_3\text{H}$, $-\text{CONH}_2$
(d) $-\text{CONH}_2$, $-\text{CHO}$, $-\text{SO}_3\text{H}$, $-\text{COOH}$

20. The IUPAC name of neopentane is (2009)

- (a) 2-methylbutane (b) 2, 2-dimethylpropane
(c) 2-methylpropane (d) 2,2-dimethylbutane

21. Arrange the carbanions,

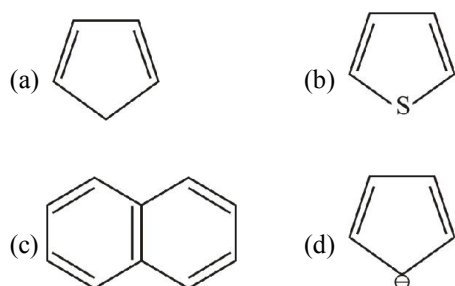
$(\text{CH}_3)_3\text{C}^-$, $\text{C}_6\text{H}_5\text{CH}_2^-$, $\text{C}_6\text{H}_5\text{CH}^-$, in order of their decreasing stability (2009)

- (a) $\text{C}_6\text{H}_5\text{CH}_2^- > \text{C}_6\text{H}_5\text{CH}^- > (\text{CH}_3)_3\text{C}^- > (\text{CH}_3)_2\text{CH}^-$
(b) $(\text{CH}_3)_2\text{CH}^- > \text{C}_6\text{H}_5\text{CH}_2^- > (\text{CH}_3)_3\text{C}^- > \text{C}_6\text{H}_5\text{CH}^-$
(c) $\text{C}_6\text{H}_5\text{CH}^- > \text{C}_6\text{H}_5\text{CH}_2^- > (\text{CH}_3)_2\text{CH}^- > (\text{CH}_3)_3\text{C}^-$
(d) $(\text{CH}_3)_3\text{C}^- > (\text{CH}_3)_2\text{CH}^- > \text{C}_6\text{H}_5\text{CH}_2^- > \text{C}_6\text{H}_5\text{CH}^-$

22. The correct order of increasing basicity of the given conjugate bases ($R = CH_3$) is (2010)

- (a) $RCO\bar{O} < HC \equiv \bar{C} < \bar{R} < \bar{N}H_2$
 (b) $\bar{R} < HC \equiv \bar{C} < RCO\bar{O} < \bar{N}H_2$
 (c) $RCO\bar{O} < \bar{N}H_2 < HC \equiv \bar{C} < \bar{R}$
 (d) $RCO\bar{O} < HC \equiv \bar{C} < \bar{N}H_2 < \bar{R}$

23. The non-aromatic compound among the following is (2011)

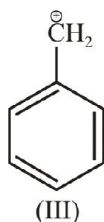
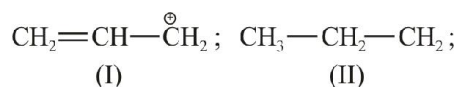


24. How many chiral compounds are possible on monochlorination of 2-methyl butane? (2012)

- (a) 2 (b) 4
 (c) 6 (d) 8

25. The order of stability of the following carbocations

(2013)



- (a) $III > II > I$ (b) $II > III > I$
 (c) $I > II > III$ (d) $III > I > II$

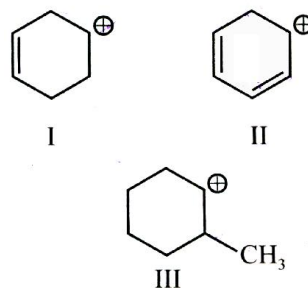
26. For the estimation of nitrogen, 1.4 g of an organic compound was digested by Kjeldahl method and the

evolved ammonia was absorbed in 60 mL of $\frac{M}{10}$ sulphuric

acid. The unreacted acid required 20 mL of $\frac{M}{10}$ sodium

hydroxide for complete neutralization. The percentage of nitrogen in the compound is (2014)

- (a) 5% (b) 6%
 (c) 10% (d) 3%



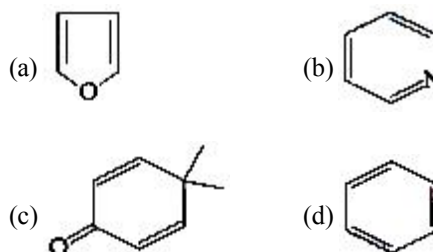
Among these cations, which of the following orders is correct for their no-bond-resonance energy? (2015)

- (a) $I > II > III$ (b) $III > II > I$
 (c) $I > III > II$ (d) $III > I > II$

28. 3-Methyl-pent-2-ene on reaction with HBr in presence of peroxide forms an addition product. The number of possible stereoisomers for the product is : (2017)

- (a) Zero (b) Two
 (c) Four (d) Six

29. Which of the following molecules is least resonance stabilized? (2017)



JEE MAINS ONLINE QUESTION

1. Which one of the following acids does not exhibit optical isomerism? **Online 2014 SET (3)**

(a) Maleic acid (b) Tartaric acid
(c) Lactic acid (d) α -amino acid

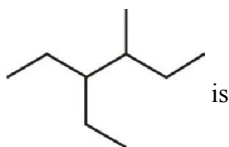
2. Which one of the following substituents at para-position is most effective in stabilizing the phenoxide ions ? **Online 2014 SET (4)**

(a) $-\text{OCH}_3$ (b) $-\text{CH}_2\text{OH}$
(c) $-\text{CH}_3$ (d) $-\text{COCH}_3$

3. Which one of the following compounds will not be soluble in sodium bicarbonate ? **Online 2014 SET (4)**

(a) 2, 4, 6 – Trinitrophenol
(b) Benzoic acid
(c) o – Nitrophenol
(d) Benzene sulphonic acid

4. The correct IUPAC name of the following compound :



is

Online 2014 SET (4)

(a) 4 – methyl – 3 – ethylhexane
(b) 3 – ethyl – 4 – methylhexane
(c) 3, 4 – ethylmethylhexane
(d) 4 – ethyl – 3 – methylhexane

5. The number of structural isomers for C_6H_{14} is :

Online 2015 SET (1)

(a) 4 (b) 3
(c) 6 (d) 5

6. Match the organic compounds in column-I with the Lassaigne's test results in column-II appropriately :

Column-I

(A) Aniline

(B) Benzenesulfonic acid

(C) Thiourea

Column-II

(i) Red colour with



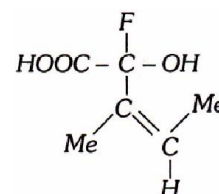
(ii) Violet colour with sodium nitroprusside

(iii) Blue colour with hot and acidic solution of FeSO_4

Online 2015 SET (1)

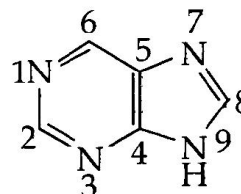
(a) A – (ii); B – (iii); C – (i) (b) A – (iii); B – (i); C – (ii)
(c) A – (iii); B – (ii); C – (i) (d) A – (ii); B – (i); C – (iii)

7. The configuration of the chiral centre and the geometry of the double bond in the following molecule can be described by **Online 2016 SET (1)**



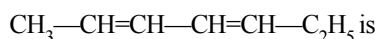
(a) R and E (b) S and E
(c) R and Z (d) S and Z

8. The "N" which does **not** contribute to the basicity for the compound is : **Online 2016 SET (2)**



(a) N 7 (b) N 9
(c) N 1 (d) N 3

9. The number of geometrical isomers in the following compound



Online 2017 SET (1)

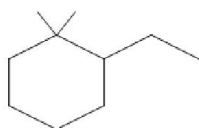
(a) 4 (b) 3
(c) 2 (d) 5

10. Which of the following statements is **not** true about partition chromatography ?

Online 2017 SET (2)

(a) Mobile phase can be a gas
(b) Stationary phase is a finely divided solid adsorbent
(c) Separation depends upon equilibration of solute between a mobile and a stationary phase
(d) Paper chromatography is an example of partition chromatography

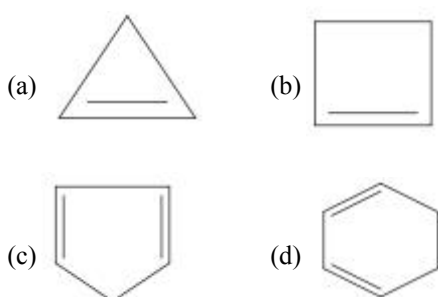
11. The IUPAC name of the following compound is :



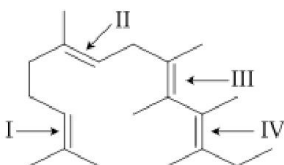
Online 2017 SET (3)

- (a) 1, 1-Dimethyl-2-ethylcyclohexane
(b) 2-Ethyl-1, 1-dimethylcyclohexane
(c) 1-Ethyl-2, 2-dimethylcyclohexane
(d) 2, 2-Dimethyl-1-ethylcyclohexane
12. Which of the following compounds is most reactive to an aqueous solution of sodium carbonate ?

Online 2018 SET (1)



13. In the following structure, the double bonds are marked as I, II, III and IV

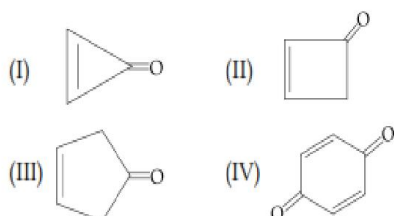


Geometrical isomerism is not possible at site (s) :

Online 2018 SET (2)

- (a) III (b) I
(c) I and III (d) III and IV
14. Which of the following compounds will show highest dipole moment

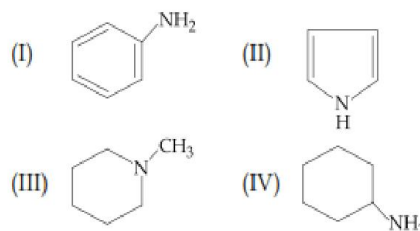
Online 2018 SET (3)



- (a) I (b) II
(c) III (d) IV

15. Among the following compounds, the increasing order of their basic strength is :

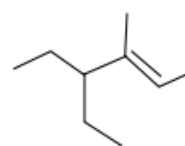
Online 2019 SET (1)



- (a) (I) < (II) < (IV) < (III) (b) (I) < (II) < (III) < (IV)
(c) (II) < (I) < (IV) < (III) (d) (II) < (I) < (III) < (IV)

16. The IUPAC name of the following compound is :

Online 2019 SET (1)



- (a) 4-methyl-3-ethylhex-4-ene
(b) 3-ethyl-4-methylhex-4-ene
(c) 4-ethyl-3-methylhex-2-ene
(d) 4, 4-diethyl-3-methylbut-2-ene

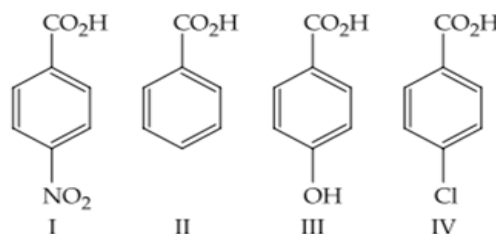
17. Two compounds I and II are eluted by column chromatography (adsorption of I > II). Which one of following is a correct statement ?

Online 2019 SET (2)

- (1) I moves faster and has higher R_f value than II
(2) II moves faster and has higher R_f value than I
(3) I moves slower and has higher R_f value than II
(4) II moves slower and has higher R_f value than I

18. The increasing order of the acidity of the following carboxylic acids is :

Online 2019 SET (3)

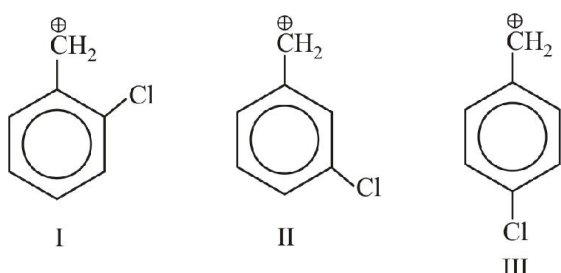


- (1) I < III < II < IV (2) IV < II < III < I
(3) II < IV < III < I (4) III < II < IV < I

EXERCISE - 3 : ADVANCED OBJECTIVE QUESTIONS

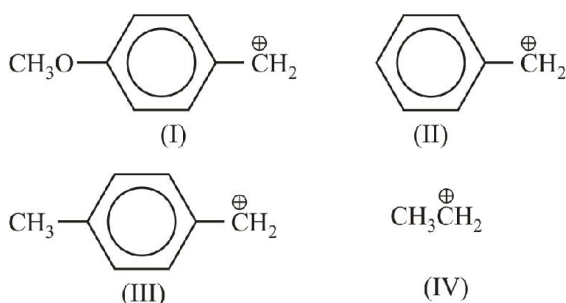
1. All questions marked “S” are single choice questions
2. All questions marked “M” are multiple choice questions
3. All questions marked “C” are comprehension based questions
4. All questions marked “A” are assertion–reason type questions
 - (A) If both assertion and reason are correct and reason is the correct explanation of assertion.
 - (B) If both assertion and reason are true but reason is not the correct explanation of assertion.
 - (C) If assertion is true but reason is false.
 - (D) If reason is true but assertion is false.
5. All questions marked “X” are matrix–match type questions
6. All questions marked “I” are integer type questions

1. (S) The correct arrangement of the relative stability of the three carbocation is :



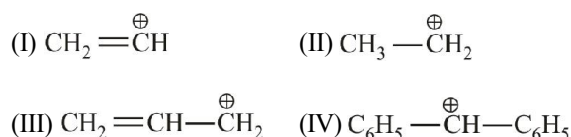
- (a) $I > II > III$ (b) $III > II > I$
(c) $I > III > II$ (d) $III > I > II$

2. (S) Relative stabilities of the following carbocation will be in order :



- (a) $I < II < III < IV$ (b) $IV < III < II < I$
(c) $IV < II < III < I$ (d) $II < IV < III < I$

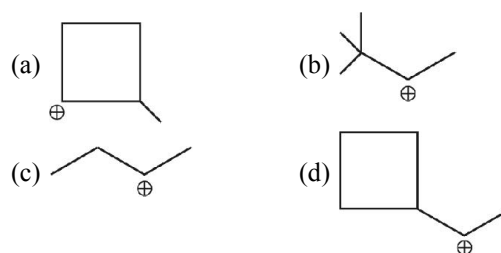
3. (S) Consider the following structures



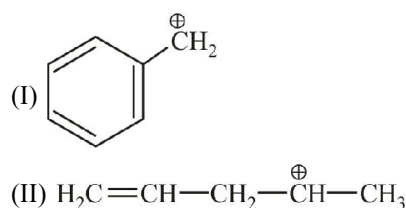
The correct sequence of these carbocations in the decreasing order of their stability is

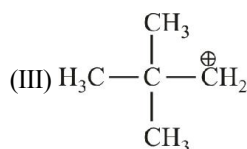
- (a) IV, III, II, I (b) I, II, III, IV
(c) IV, II, III, I (d) I, III, II, IV

4. (S) Which of the following is most likely to undergo a favourable hydride shift ?



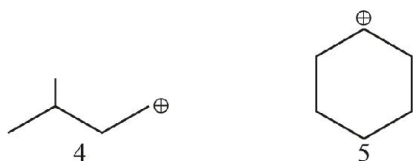
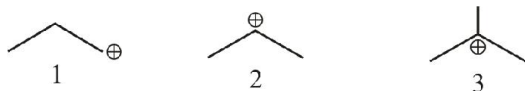
5. (S) Which of the following options is the correct order of relative stabilities of cations as written below (most stable first) ?





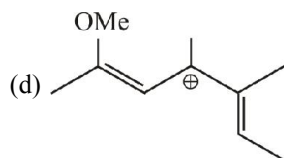
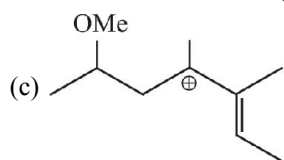
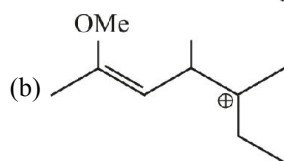
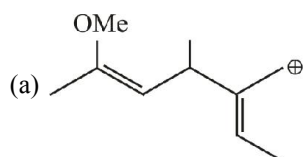
- (a) $\text{I} > \text{II} > \text{III}$ (b) $\text{II} > \text{III} > \text{I}$
(c) $\text{III} > \text{I} > \text{II}$ (d) $\text{I} > \text{III} > \text{II}$

6. (S) What is the decreasing order of stability (most stable $\rightarrow$ least stable) of the following carbocations ?

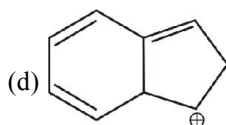
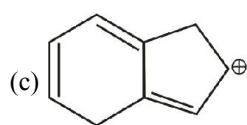
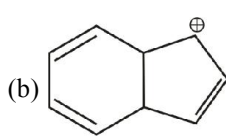
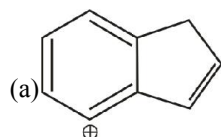


- (a) $3 > 2 > 1 > 4 > 5$ (b) $3 > 2 > 5 > 4 > 1$
(c) $1 \approx 4 > 2 \approx 5 > 3$ (d) $3 > 1 \approx 4 > 2 \approx 5$

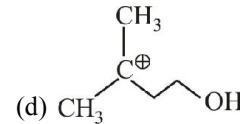
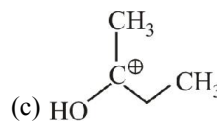
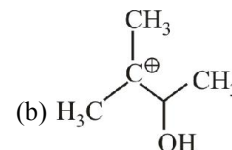
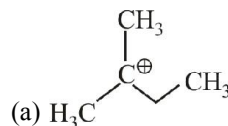
7. (S) Which of the following is the most stabilized carbocation?



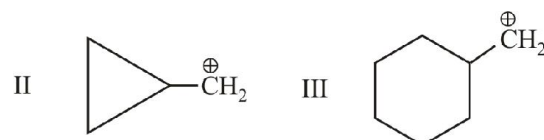
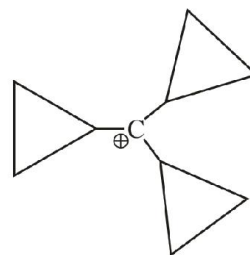
8. (S) Which carbocation is the most stabilized ?



9. (S) Which of the following carbocation is most stable ?



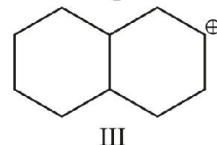
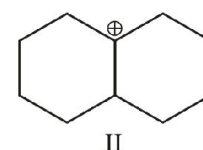
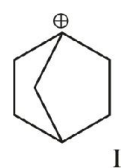
10. (S) I



Which of the following orders is correct for the stability of these carbocations ?

- (a) $\text{I} > \text{II} > \text{III}$ (b) $\text{III} > \text{II} > \text{I}$
(c) $\text{II} > \text{III} > \text{I}$ (d) $\text{III} > \text{I} > \text{II}$

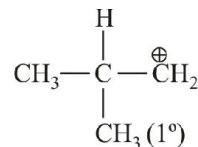
11. (S)



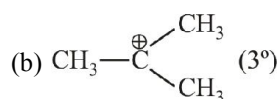
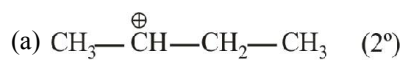
Which of the following orders is correct for the stability of these carbocations ?

- (a) $\text{I} > \text{II} > \text{III}$ (b) $\text{III} > \text{II} > \text{I}$
(c) $\text{II} > \text{I} > \text{III}$ (d) $\text{II} > \text{III} > \text{I}$

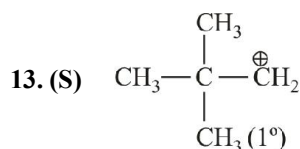
12. (S)



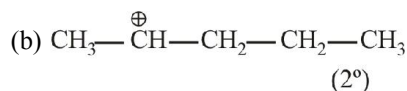
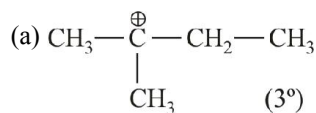
Rearrangement in this carbocation will produce



- (c) both (a) and (b) (d) none of these

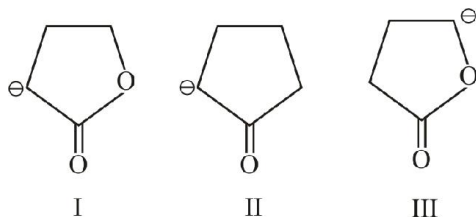


Regarrangement in this carbocation will produce



(c) both (a) and (b) (d) none of these

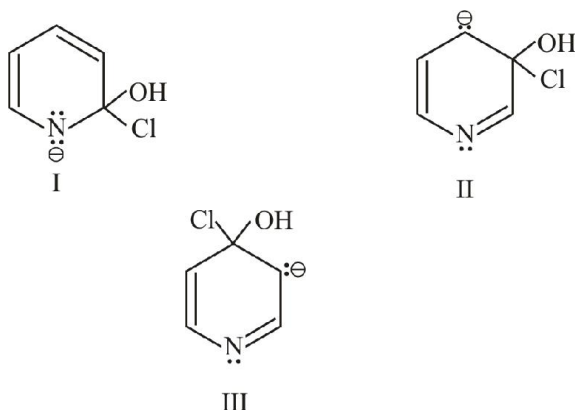
14. (S)



Which of the following orders is correct for the stability of these carbanions ?

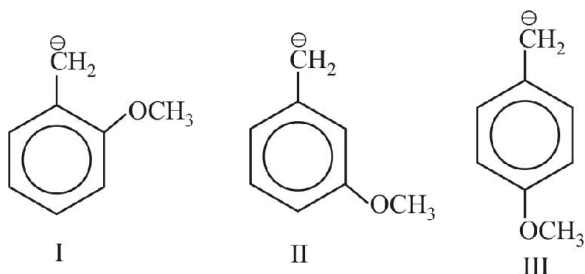
- (a) $\text{I} > \text{II} > \text{III}$ (b) $\text{III} > \text{II} > \text{I}$
(c) $\text{II} > \text{III} > \text{I}$ (d) $\text{II} > \text{I} > \text{III}$

15. (S) Which of the following is least stable ?



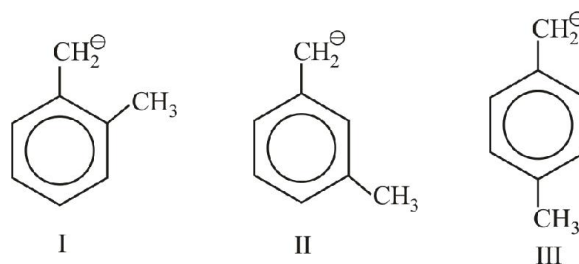
- (a) I (b) II
(c) III (d) None

16. (S) Arrange the following in decreasing order of stability.



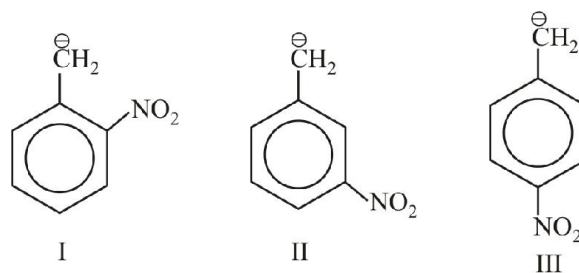
- (a) $\text{I} > \text{II} > \text{III}$ (b) $\text{I} > \text{III} > \text{II}$
(c) $\text{II} > \text{I} > \text{III}$ (d) $\text{II} > \text{III} > \text{I}$

17. (S) The correct order of stability of the three isomeric carbanions is:



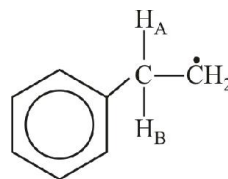
- (a) $\text{I} > \text{II} < \text{III}$ (b) $\text{I} > \text{III} > \text{II}$
(c) $\text{II} > \text{I} > \text{III}$ (d) $\text{II} > \text{III} > \text{I}$

18. (S) The three carbanions follows which order of stability

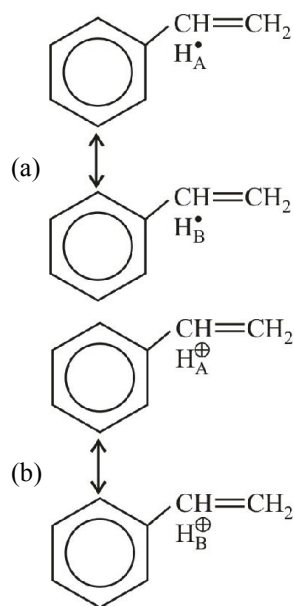


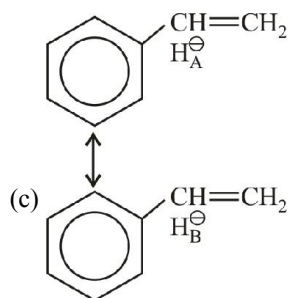
- (a) $\text{I} > \text{II} > \text{III}$ (b) $\text{I} > \text{III} > \text{II}$
(c) $\text{II} > \text{I} > \text{III}$ (d) $\text{II} > \text{III} > \text{I}$

19. (S)



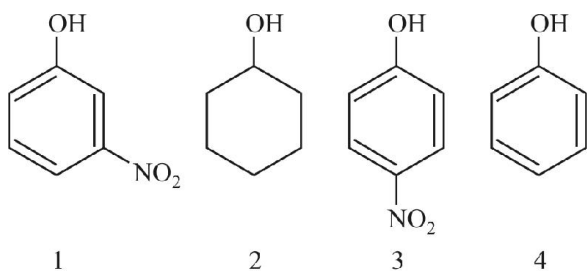
Which of the following no-bond-canonical structures is correct for this radical ?





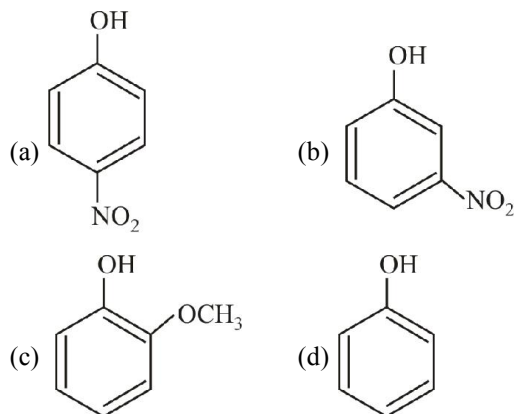
(d) none of these

20. (S) Rank the following compounds in order of decreasing acid strength (most acidic $\rightarrow$ least acidic).

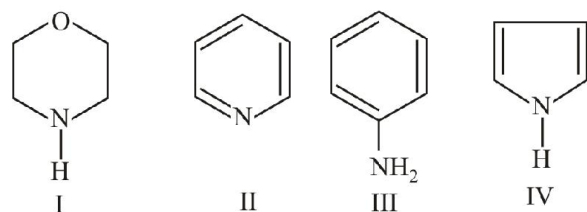


- (a) $2 > 4 > 1 > 3$ (b) $1 > 3 > 4 > 2$
(c) $3 > 1 > 2 > 4$ (d) $3 > 1 > 4 > 2$

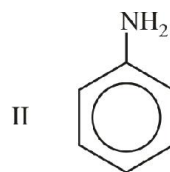
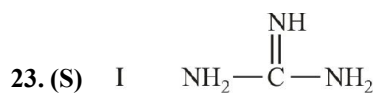
21. (S) Which of the following is strongest acid ?



22. (S) Arrange following compounds in decreasing order of basicity.



- (a) $IV > I > III > II$ (b) $III > I > IV > II$
(c) $II > I > III > IV$ (d) $I > II > III > IV$



Which of the following orders is correct for the basic-strength of these compounds ?

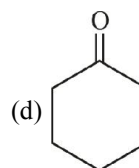
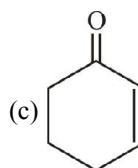
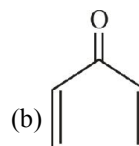
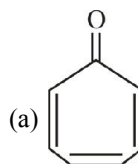
- (a) $I > II > III$ (b) $III > II > I$
(c) $I > III > II$ (d) $II > I > III$

24. (M) Br has low reactivity in $\text{CH}_2=\text{CH—Br}$ because :

- (a) the C—Br bond has a partial double bond character
(b) of the +M effect of bromine
(c) Br is electronegative
(d) None of the above

25. (M) Which of the following have larger C—O bond length

than C—O bond length of $\text{CH}_3\text{—C(=O)—CH}_3$?



26. (A) **Assertion :** Tropylium cation is more stable than $(\text{CH}_3)_3\text{C}^\oplus$

Reason : It is stabilized by both resonance effect and inductive effect.

- (a) A (b) B
(c) C (d) D

27. (A) **Assertion :** The carbocation $\text{CF}_3\text{—}\overset{\oplus}{\text{CH}}_2$ is less stable than $\overset{\oplus}{\text{CF}}_3$.

Reason : In case of $\text{CF}_3\text{—}\overset{\oplus}{\text{CH}}_2$, CF_3 is strong electron withdrawing, therefore increases +ve charge whereas in $\overset{\oplus}{\text{CF}}_3$, lone pair of 'F' overlap with vacant p-orbital of carbon reducing +ve charge by $p\pi - p\pi$ bonding or back bonding.

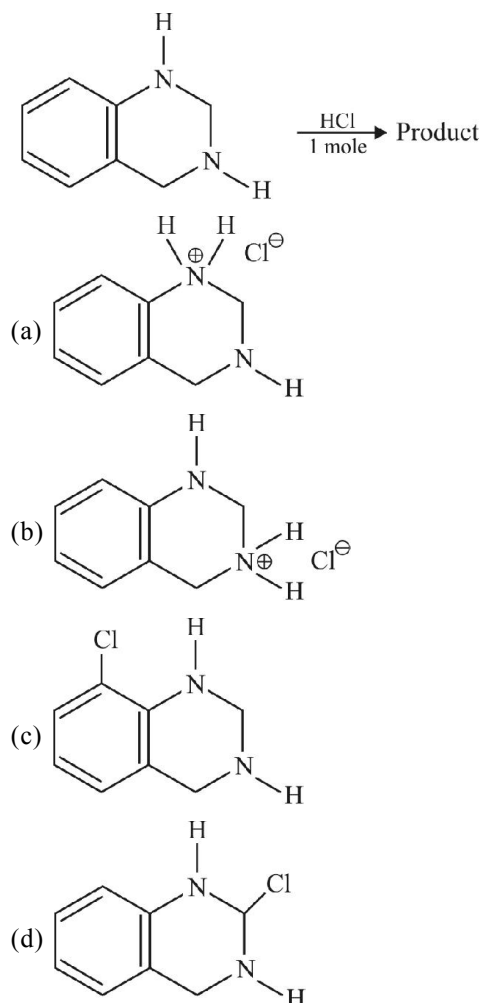
- (a) A (b) B
(c) C (d) D

28. (A) **Assertion :** The acetate ion is a weaker base than the ethoxide ion.

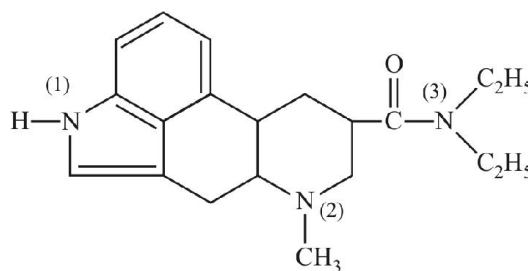
Reason : In carboxylic acids, the carbonyl group is polarized and so the carbon of the carbonyl group bears a partial +ve charge.

- (a) A (b) B
(c) C (d) D

29. (S) Which is the major product of the following reaction ?



30. (S) Correct order of basicity of various nitrogen in

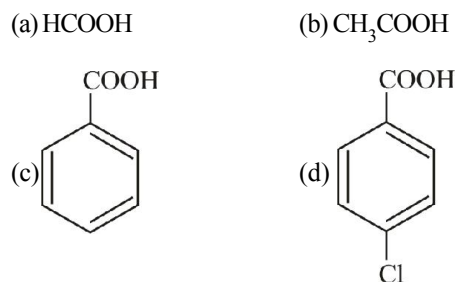


- (a) $1 > 2 > 3$ (b) $2 > 1 > 3$
(c) $2 > 3 > 1$ (d) $3 > 2 > 1$

31. (S) Aniline is weaker base than ethyl amine. This is due to

- (a) – I effect of NH_2 in aniline
(b) – R effect of NH_2 in aniline
(c) + I effect of NH_2 in aniline
(d) + R effect of NH_2 in aniline

32. (S) Which of the following is strongest acid ?



33. (S) (I) CH_3NH_2 (II) $(\text{CH}_3)_2\text{NH}$ (III) $(\text{CH}_3)_3\text{N}$

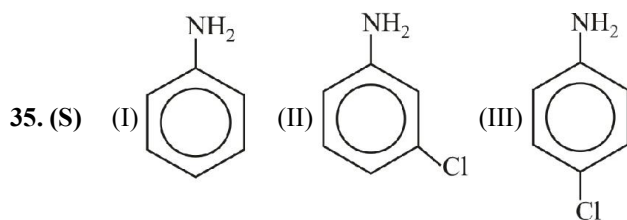
Which of the following orders is correct for the basic-strength of these 1°, 2° and 3° amines in gas phase ?

- (a) $\text{I} > \text{II} > \text{III}$ (b) $\text{III} > \text{II} > \text{I}$
(c) $\text{II} > \text{I} > \text{III}$ (d) $\text{II} > \text{III} > \text{I}$

34. (S) (I) CH_3NH_2 (II) $(\text{CH}_3)_2\text{NH}$
(III) $(\text{CH}_3)_3\text{N}$

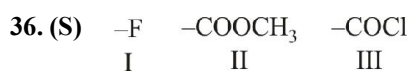
Which of the following orders is correct for the basic-strength of these amines in polar solvents (e.g., $\text{—H}_2\text{O}$) ?

- (a) $\text{I} > \text{II} > \text{III}$ (b) $\text{III} > \text{II} > \text{I}$
(c) $\text{II} > \text{I} > \text{III}$ (d) $\text{II} > \text{III} > \text{I}$



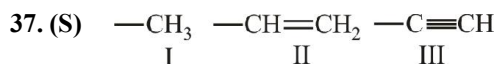
Which of the following orders is correct for the basic-strength of these anilines ?

- (a) I > II > III (b) III > II > I
(c) II > I > III (d) I > III > II



Which of these groups has -I effect ?

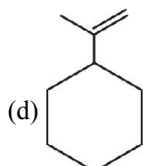
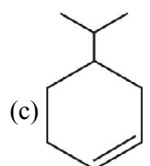
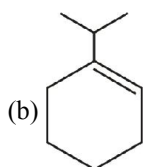
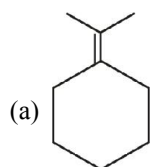
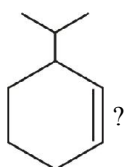
- (a) I (b) II
(c) III (d) all of these



Which of these groups has +I effect ?

- (a) I (b) II
(c) III (d) all of these

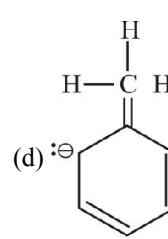
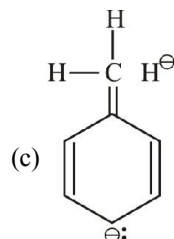
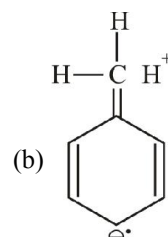
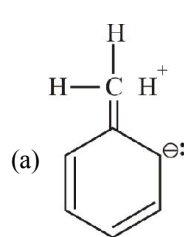
38. (M) Which of the following alkenes are more stable than



Comprehension

Hyperconjugation is no bond resonance. It involves σ electrons of C—H bond. Greater the number of C—H bonds (α -hydrogen atoms w.r.t. double bond) more will be hyperconjugative structures, more will be stability. Heats of hydrogenation of alkenes are affected by hyperconjugative effects. Greater the hyperconjugative structures, lower will be the heat of hydrogenation.

39. (C) Which of the following is incorrect hyperconjugative structure ?



40. (C) Which of the following has highest heat of hydrogenation?

- (a) $CH_2=CH_2$ (b) But-1-ene
(c) Isobutylene (d) Tetramethyl ethylene

41. (C) Which of the following has highest heat of combustion ?

- (a) Isobutylene (b) But-1-ene
(c) Tetramethyl ethylene (d) But-2-ene

Comprehension

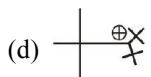
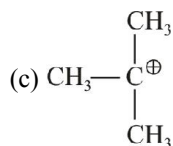
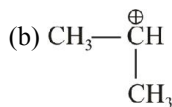
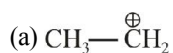
When (C—H) σ electrons are in conjugation to pi bond, this conjugation is known as hyperconjugation : For any compound to show hyperconjugation

- (i) Compound should have one sp^2 -hybridised carbon.
(ii) α -carbon with respect to sp^2 should be sp^3

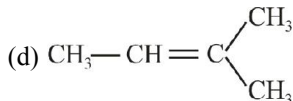
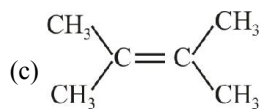
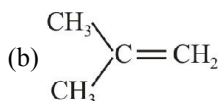
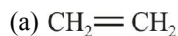
(iii) α -carbon should contain at least one hydrogen atom.

No. of α -carbon $\propto$ stability of cation and alkene.

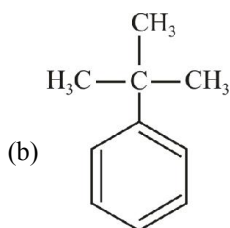
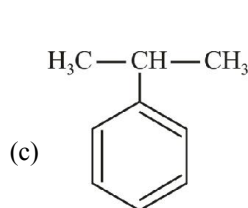
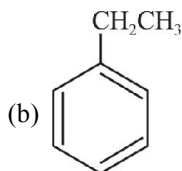
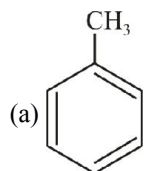
42. (C) Which of following cations is hyperconjugation destabilized ?



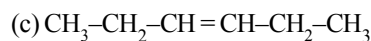
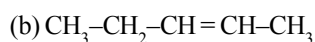
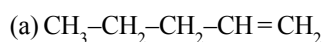
43. (C) Which of following alkenes is most stabilized ?



44. (C) Which of following alkyl benzene has maximum electron density ?



45. (S) Which of the following alkenes will exhibit geometrical isomerism ?



(d) both (b) and (c)

46. (S) $\text{CH}_2=\text{CH}-\text{CH}=\text{CH}-\text{CH}=\text{CH}-\text{CH}_3$

How many geometrical isomers of this compound are possible ?

(a) 2

(b) 3

(c) 4

(d) 8

47. (S) $\text{CH}_3-\text{CH}_2-\text{CH}=\text{CH}-\text{CH}=\text{CH}-\text{CH}_3$

How many geometrical isomers of this compound are possible ?

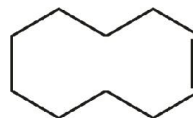
(a) 2

(b) 3

(c) 4

(d) 8

48. (S)



How many geometrical isomers of this cyclohexene are possible ?

(a) 0

(b) 2

(c) 3

(d) 4

49. (S) Number of which of the following isomers is always infinite ?

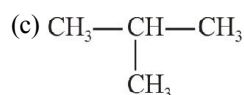
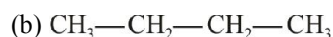
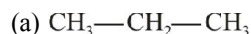
(a) Geometrical isomers

(b) Optical isomers

(c) Conformational isomers

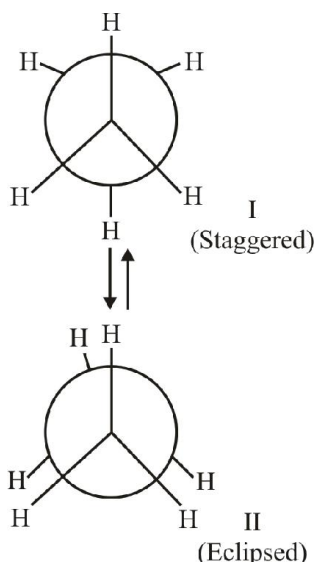
(d) Structural isomers

50. (S) Which of the following compounds will exhibit conformational isomerism ?



(d) all of these

51. (S)



These two conformers exist in a state of dynamic equilibrium. Which one is more stable ?

- (a) I (b) II
(c) Both are equally stable
(d) It is not possible to compare them for stability

52. (X) Match the Column I and II (Matrix)

- | | |
|------------------------|---------------|
| (A) —NO ₂ | (P) —M effect |
| (B) —O [−] | (Q) +M effect |
| (C) —O—CH ₃ | (R) +I effect |
| (D) —C≡N | (S) —I effect |

53. (X) Match the Column I and II (Matrix)

- | | |
|---|-------------------------|
| (A) CH ₃ — $\overset{\oplus}{\text{C}}\text{H}$ —CH ₃ | (P) Pyramidal structure |
| (B) $\overset{\ominus}{\text{C}}\text{H}_3$ | (Q) Planar geometry |
| (C) CH ₃ — $\dot{\text{C}}\text{H}$ —CH ₃ | (R) Electrophile |
| (D) Singlet carbene | (S) Nucleophile |

54. (A) **Assertion :** A mixture of o-nitrophenol and p-nitrophenol can be separated by steam distillation.

Reason : p-Nitrophenol is steam volatile while o-nitrophenol is not steam volatile.

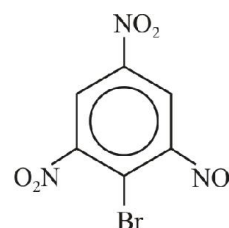
- (a) A (b) B
(c) C (d) D

55. (A) **Assertion :** Tertiary carbocations are generally formed more easily than primary carbocations.

Reason : Hyperconjugation as well as inductive effect due to additional alkyl groups stabilize tertiary carbocations.

- (a) A (b) B
(c) C (d) D

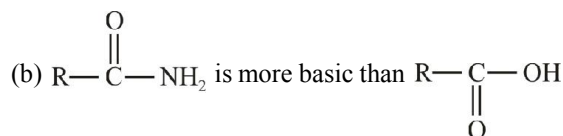
56. (M) Which of the statements are incorrect about the following compound ?



- (a) All three C—N bonds are of same length
(b) C₁—N and C₃—N bonds are of same length but shorter than C₅—N bond
(c) C₁—N and C₅—N bonds are of same length but longer than C₃—N bond
(d) C₁—N and C₃—N bonds are of different length but both are longer than C₅—N bond

57. (M) Which statements among following are correct ?

- (a) Hydration effect stabilise dimethyl ammonium ion more than trimethyl ammonium ion



- (c) $\overset{\ominus}{\text{C}}\text{H}_3 < \overset{\ominus}{\text{N}}\text{H}_2 < \overset{\ominus}{\text{O}}\text{H}$ nucleophilicity in DMSO

- (d) Phenol is more acidic than CH₃OH

58. (A) **Assertion :** The C≡C distance is longer than C=C distance

Reason : Both C atoms in alkenes are sp² hybridized.

- (a) A (b) B
(c) C (d) D

59. (A) Assertion : Ethylene is more reactive than ethane.

Reason : Ethylene possess π bond.

- (a) A (b) B
(c) C (d) D

60. (A) Assertion : The two structures, $\text{CH}_3 - \text{CH}_2 - \text{CH}_2 - \text{C} \equiv \text{CH}$ and $\text{CH}_3 - \underset{\text{CH}_3}{\text{CH}} - \text{C} \equiv \text{CH}$ are chain isomers.

Reason : The chain isomers differ in the chain of carbon atoms.

- (a) A (b) B
(c) C (d) D

61. (A) Assertion : Conjugated dienes are more stable than non-conjugated dienes.

Reason : 1,4-Pentadiene is a non-conjugated diene.

- (a) A (b) B
(c) C (d) D

62. (A) Assertion : All carbon atoms in 1,3-butadiene are sp^2 hybridized.

Reason : 1,3-Butadiene is a conjugated diene.

- (a) A (b) B
(c) C (d) D

63. (A) Assertion : $\text{CHCl} = \text{CHBr}$ exhibits geometrical isomerism.

Reason : Presence of $\text{C} = \text{C}$ is the condition for a compound to exhibit geometrical isomerism.

- (a) A (b) B
(c) C (d) D

EXERCISE - 4 : PREVIOUS YEAR JEE ADVANCED QUESTION

Objective Questions (Only one correct option)

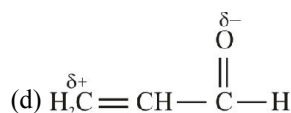
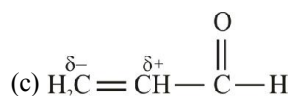
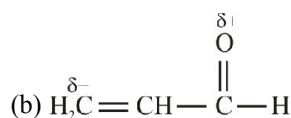
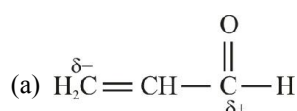
1. The bond between carbon atom (1) and carbon atom (2) in compound $\text{N} \equiv \underset{1}{\text{C}} - \underset{2}{\text{CH}} = \text{CH}_2$ involves the hybridisation as

(1987)

- (a) sp^2 and sp^2 (b) sp^3 and sp
(c) sp and sp^2 (d) sp and sp

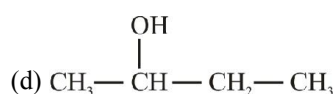
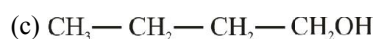
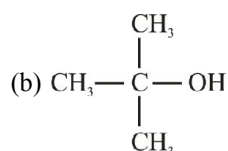
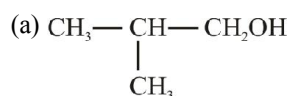
2. Polarisation of electrons in acrolein may be written as

(1988)



3. The compounds which gives the most stable carbonium ion on dehydration is

(1989)



4. The number of sigma and pi-bonds in 1-butene-3-yne are (1989)

- (a) 5σ and 5π (b) 7σ and 3π
(c) 8σ and 2π (d) 6σ and 4π

5. Amongst the following, the most basic compounds is

(1990)

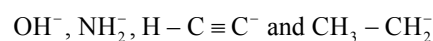
- (a) benzylamine (b) aniline
(c) acetanilide (d) p-nitroaniline

6. The hybridisation of carbon atoms in $\text{C}-\text{C}$ single bond $\text{H}-\text{C} \equiv \text{C}-\text{CH}=\text{CH}_2$ is

(1991)

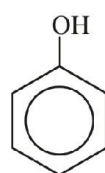
- (a) $\text{sp}^3 - \text{sp}^3$ (b) $\text{sp}^2 - \text{sp}^3$
(c) $\text{sp} - \text{sp}^3$ (d) $\text{sp}^3 - \text{sp}$

7. What is the decreasing order of strength of the bases ?



- (a) $\text{CH}_3 - \text{CH}_2^- > \text{NH}_2^- > \text{H}-\text{C} \equiv \text{C}^- > \text{OH}^-$
(b) $\text{H}-\text{C} \equiv \text{C}^- > \text{CH}_3 - \text{CH}_2^- > \text{NH}_2^- > \text{OH}^-$
(c) $\text{OH}^- > \text{NH}_2^- > \text{H}-\text{C} \equiv \text{C}^- > \text{CH}_3 - \text{CH}_2^-$
(d) $\text{NH}_2^- > \text{H}-\text{C} \equiv \text{C}^- > \text{OH}^- > \text{CH}_3 - \text{CH}_2^-$

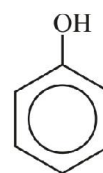
8. In the following compounds :



(I)



(II)



(III)



(IV)

The order of acidity is :

(1996)

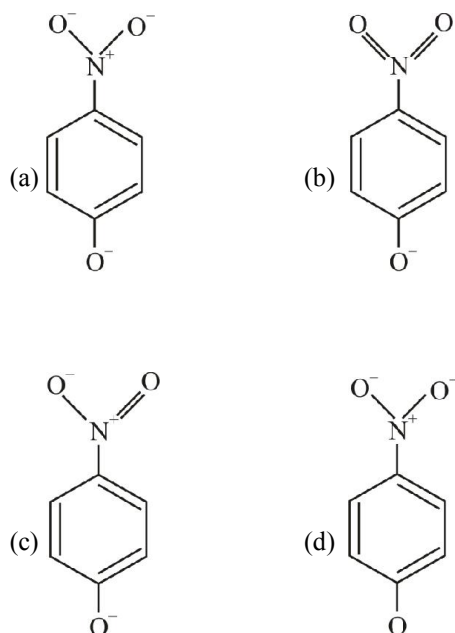
- (a) $\text{III} > \text{IV} > \text{I} > \text{II}$ (b) $\text{I} > \text{IV} > \text{III} > \text{II}$
(c) $\text{II} > \text{I} > \text{III} > \text{IV}$ (d) $\text{IV} > \text{III} > \text{I} > \text{II}$

9. Among the following compounds, the strongest acid is

(1988)

- (a) $\text{HC} \equiv \text{CH}$ (b) C_6H_6
(c) C_2H_6 (d) CH_3OH

10. The most unlikely representation of resonance structures of p-nitrophenoxide ion is (1999)



11. Which of the following, has the most acidic hydrogen ?

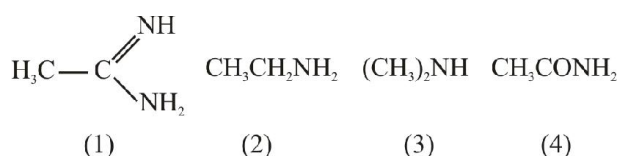
(2000)

- (a) 3-hexanone (b) 2, 4-hexanedione
(c) 2, 5-hexanedione (d) 2, 3-hexanedione

12. Among the following, the strongest base is (2000)

- (a) $C_6H_5NH_2$ (b) $p\text{-NO}_2C_6H_4NH_2$
(c) $m\text{-NO}_2-C_6H_4NH_2$ (d) $C_6H_5CH_2NH_2$

13. The correct order of basicities of the following compounds is (2001)

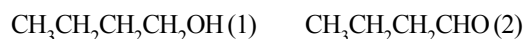


- (a) $2 > 1 > 3 > 4$ (b) $1 > 3 > 2 > 4$
(c) $3 > 1 > 2 > 4$ (d) $1 > 2 > 3 > 4$

14. Which of the following hydrocarbons has the lowest dipole moment ? (2002)

- (a) cis-2-butene (b) 2-butyne
(c) 1-butyne (d) $H_2C=CH-C\equiv CH$

15. Identify the correct order of boiling points of the following compounds

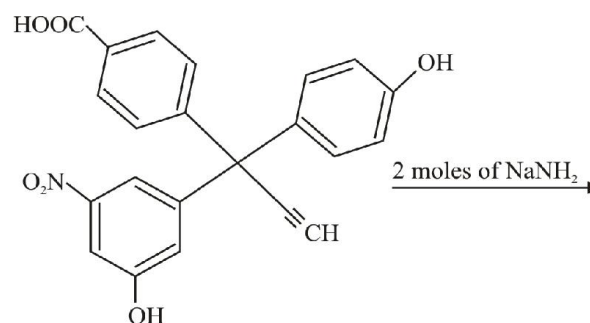


- (a) $1 > 2 > 3$ (b) $3 > 1 > 2$
(c) $1 > 3 > 2$ (d) $3 > 2 > 1$

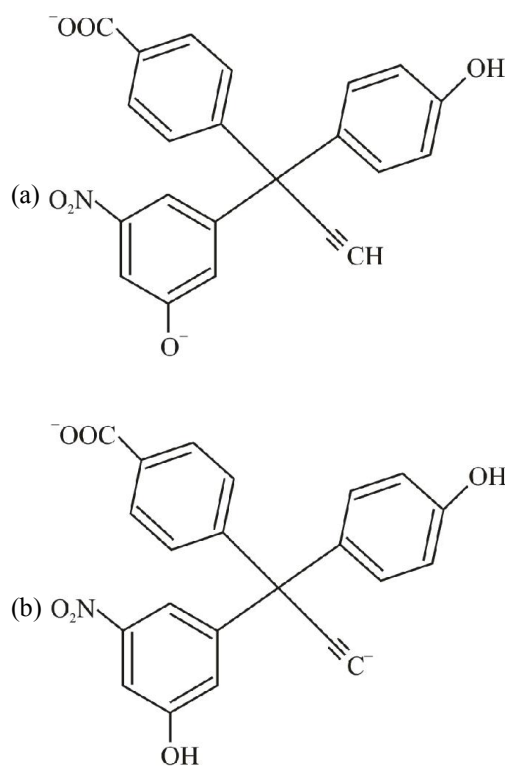
16. Which of the following acids has the smallest dissociation constant ? (2002)

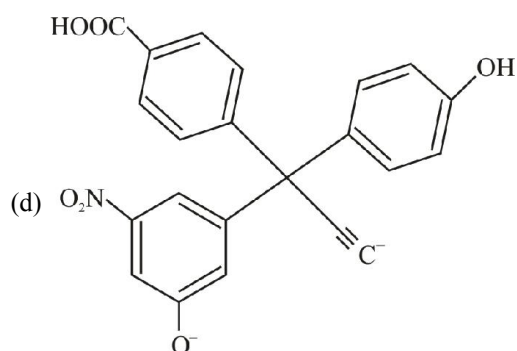
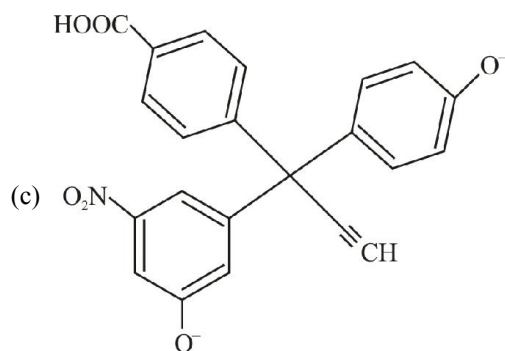
- (a) $CH_3CHFCOOH$ (b) FCH_2CH_2COOH
(c) $BrCH_2CH_2COOH$ (d) $CH_3CHBrCOOH$

- 17.



The product A will be





18. Which of the following represents the given mode of hybridisation $sp^2 - sp^2 - sp - sp$ from left to right ?

(2003)

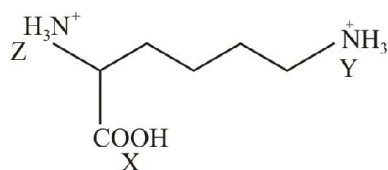


19. Among the following, the molecule with the highest dipole moment is

(2003)



20.

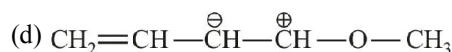
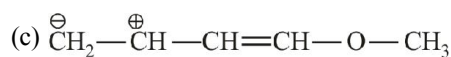


Arrange in order of increasing acidic strength (2004)



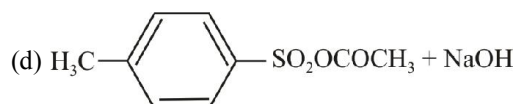
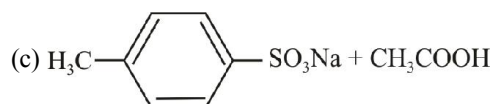
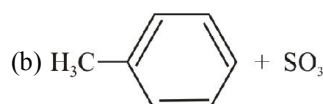
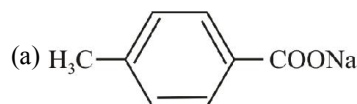
21. Which of the following resonating structures of 1-methoxy-1,3-butadiene is least stable ?

(2005)



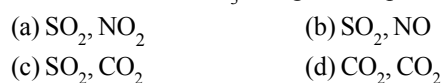
22. Which of the following is obtained when 4-methylbenzene sulphonic acid is hydrolysed with excess of sodium acetate ?

(2005)



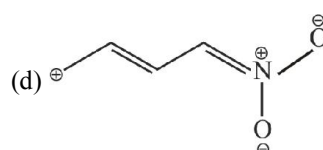
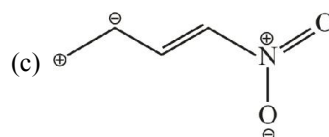
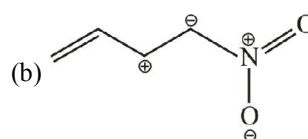
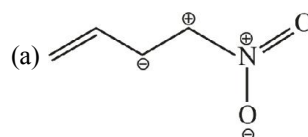
23. When benzene sulphonic acid and p-nitrophenol are treated with $NaHCO_3$, the gases respectively are

(2006)



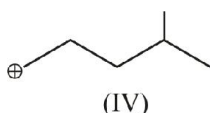
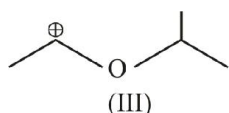
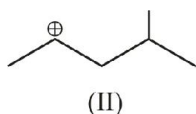
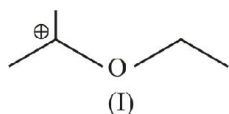
24. Among the following, the least stable resonance structure is

(2007)



25. The correct stability order for the following species is

(2008)

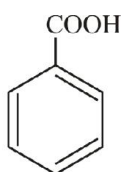
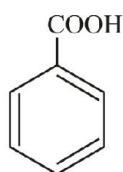
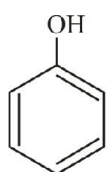
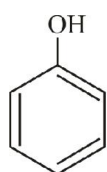


- (a) (II) > (IV) > (I) > (III) (b) (I) > (II) > (III) > (IV)
(c) (II) > (I) > (IV) > (III) (d) (I) > (III) > (II) > (IV)

26. Hyperconjugation involves overlap of the following orbitals (2008)

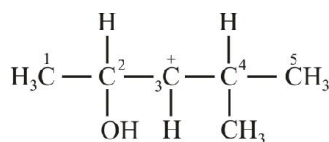
- (a) σ - σ (b) σ - π
(c) p-p (d) π - π

27. The correct acidity order of the following is



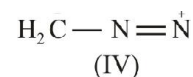
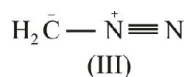
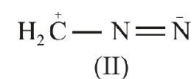
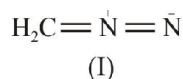
- (a) (III) > (IV) > (II) > (I) (b) (IV) > (III) > (I) > (II)
(c) (III) > (II) > (I) > (IV) (d) (II) > (III) > (IV) > (I)

28. In the following carbocation; H/CH₃ that is most likely to migrate to the positively charged carbon is (2009)



- (a) CH₃ at C-4 (b) H at C-4
(c) CH₃ at C-2 (d) H at C-2

29. The correct stability order of the following resonance structure is



(2009)

- (a) (I) > (II) > (IV) > (III) (b) (I) > (III) > (II) > (IV)
(c) (II) > (I) > (III) > (IV) (d) (III) > (I) > (IV) > (II)

30. Among the following compounds, the most acidic is

(2011)

- (a) p-nitrophenol (b) p-hydroxybenzoic acid
(c) o-hydroxybenzoic acid (d) p-toluic acid

Objective Questions (One or more than one correct option)

31. Phenol is less acidic than (1986)

- (a) acetic acid (b) p-methoxy phenol
(c) p-nitrophenol (d) ethanol

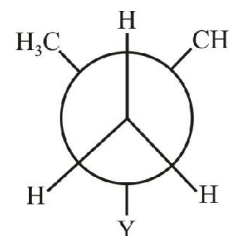
32. The molecules that will have dipole moment are (1992)

- (a) 2, 2-dimethyl propane (b) trans 2-pentene
(c) cis 3-hexene
(d) 2, 2, 3, 3-tetramethyl butane

33. The compound in which C uses its sp³-hybrid orbitals for bond formation is (2000)

- (a) HCOOH (b) (H₂N)₂CO
(c) (CH₃)₃COH (d) CH₃CHO

34. In the Newmann projection for 2,2-dimethylbutane

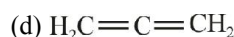
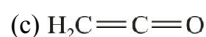
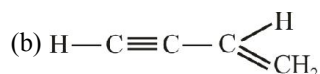
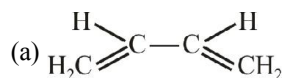


X and Y can respectively by

(2010)

- (a) H and H (b) H and C₂H₅
(c) C₂H₅ and H (d) CH₃ and CH₃

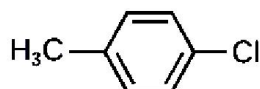
35. Amongst the given options, the compound(s) in which all the atoms are in one plane in all the possible conformations (if any), is (are) (2011)



36. With respect to graphite and diamond, which of the statements(s) given below is (are) correct? (2012)

- (a) Graphite is harder than diamond.
 (b) Graphite has higher electrical conductivity than diamond.
 (c) Graphite has higher thermal conductivity than diamond.
 (d) Graphite has higher C—C bond order than diamond.

37. The IUPAC name(s) of the following compound is (are) (2017)



- (a) 4-methylchlorobenzene
 (b) 4-chlorotoluene
 (c) 1-chloro-4-methylbenzene
 (d) 1-methyl-4-chlorobenzene

ASSERTION – REASON

In the following set of questions, two statements “Assertion” and “Reason” are given. Choose the correct answer from the following options :

- (a) Assertion and Reason are true and the Reason is the correct explanation of the Assertion.
 (b) Assertion and Reason are true but the Reason is not a correct explanation of the Assertion.
 (c) Assertion is true but the Reason is false.
 (d) Assertion is false Reason is true.

38. **Assertion :** p-nitrophenol is a stronger acid than o-nitrophenol.

Reason : Intramolecular hydrogen bonding make the o-isomer weaker acid than p-isomer.

- (a) A (b) B
 (c) C (d) D

39. **Assertion :** p-hydroxybenzoic acid has a lower boiling point than o-hydroxybenzoic acid.

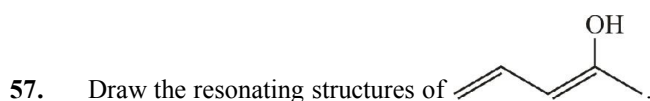
Reason : o-hydroxybenzoic acid has intramolecular hydrogen bonding.

- (a) A (b) B
 (c) C (d) D

Fill in the Blanks

40. Among the given cations, is most stable.
 (sec-butyl carbonium ion, tert-butyl carbonium ion, n-butyl carbonium ion) (1981)
41. In acidic medium, behaves as the strongest base.
 (nitrobenzene, aniline, phenol) (1981)
42. The compound having both sp and sp² hybridised carbon atoms is (propane, propene, propadiene). (1981)
43. ring is most strained. (cyclopropane, cyclobutane, cyclopentane) (1981)
44. The terminal carbon atom in butane is hybridised. (1985)
45. A diol has two hydroxyl groups on carbon atoms. (1985)
46. The shape of (CH₃)⁺ is (1990)
47. The structure of the enol form of CH₃—CO—CH₂—CO—CH₃ with intermolecular hydrogen bonding is (1993)
48. The bond dissociation energy needed to form the benzyl radical from toluene is than the formation of the methyl radical from methane. (1994)

49. The IUPAC name of succinic acid is (1990)
50. The kind of delocalisation involving sigma bond orbitals is called (1994)



(2003)

Subjective Questions

51. Arrange the following in the order of their increasing basicity. p-toluidine, N, N-dimethyl-p-toluidine, p-nitroaniline, aniline. (1985)

52. Although phenoxide ion has more number of resonating structures than benzoate ion, benzoic acid is a stronger acid than phenol. Why? (1997)

53. Give reasons for the following in one or two sentences.

The central carbon-carbon bond in 1, 3-butadiene is shorter than that of n-butane. (1998)

54. Discuss the hybridisation of carbon atoms in allene (C_3H_4) and show the π -orbital overlaps. (1999)

55. Give reasons for the following

$CH_2=CH^-$ is more basic than $HC\equiv C^-$. (2000)

56. Match the following with their K_a values (2003)

Benzoic acid 4.2×10^{-5}

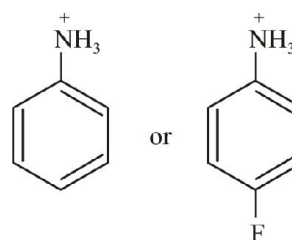
p-nitrobenzoic acid 3.3×10^{-5}

p-chlorobenzoic acid 6.4×10^{-5}

p-methylbenzoic acid 36.2×10^{-5}

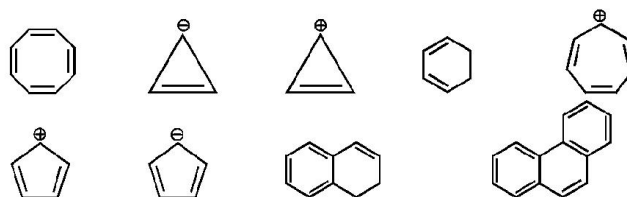
p-methoxybenzoic acid 10.2×10^{-5}

58. Which of the following is more acidic and why?



(2004)

59. Among the following, the number of aromatic compound(s) is (2017)



Answer Key**Exercise - 1 : (Basic Objective Questions)**

1. (b)	2. (b)	3. (a)	4. (d)	5. (a)	6. (d)	7. (c)	8. (b)	9. (d)	10. (a)
11. (a)	12. (c)	13. (d)	14. (d)	15. (d)	16. (a)	17. (c)	18. (c)	19. (b)	20. (c)
21. (c)	22. (c)	23. (d)	24. (d)	25. (d)	26. (d)	27. (a)	28. (c)	29. (c)	30. (b)
31. (a)	32. (c)	33. (c)	34. (b)	35. (a)	36. (b)	37. (b)	38. (c)	39. (d)	40. (a)
41. (a)	42. (c)	43. (b)	44. (b)	45. (a)	46. (d)	47. (d)	48. (d)	49. (a)	50. (c)
51. (c)	52. (b)	53. (b)	54. (d)	55. (d)	56. (a)	57. (a)	58. (b)	59. (d)	60. (b)
61. (a)	62. (c)	63. (c)	64. (c)	65. (c)	66. (d)	67. (a)	68. (d)	69. (c)	70. (d)
71. (c)	72. (d)	73. (a)	74. (c)	75. (c)	76. (d)	77. (d)	78. (b)	79. (c)	80. (c)
81. (c)	82. (d)	83. (a)	84. (c)	85. (c)	86. (a,c)				

Exercise - 2 : (Previous Year JEE Mains Questions)

1. (c)	2. (a)	3. (b)	4. (d)	5. (c)	6. (d)	7. (a)	8. (c)	9. (c)	10. (d)
11. (b)	12. (c)	13. (d)	14. (c)	15. (c)	16. (a)	17. (c)	18. (d)	19. (a)	20. (b)
21. (c)	22. (d)	23. (a)	24. (a)	25. (d)	26. (c)	27. (c)	28. (c)	29. (c)	

JEE Mains Online

1. (a)	2. (d)	3. (c)	4. (b)	5. (d)	6. (c)	7. (c)	8. (b)	9. (a)	10. (a)
11. (b)	12. (c)	13. (b)	14. (a)	15. (c)	16. (c)\	17. (b)	18. (d)		

Exercise - 3 : (Advanced Objective Questions)

1. (d)	2. (c)	3. (a)	4. (a)	5. (a)	6. (b)	7. (d)	8. (c)	9. (c)	10. (a)
11. (d)	12. (b)	13. (a)	14. (d)	15. (b)	16. (c)	17. (d)	18. (b)	19. (a)	20. (d)

21. (a) 22. (d) 23. (c) 24. (ab) 25. (abc) 26. (c) 27. (a) 28. (b) 29. (b) 30. (c)
31. (d) 32. (a) 33. (b) 34. (c) 35. (d) 36. (d) 37. (a) 38. (abcd) 39. (c) 40. (a)
41. (b) 42. (d) 43. (c) 44. (a) 45. (d) 46. (c) 47. (c) 48. (b) 49. (c) 50. (d)
51. (a) 52. A – P, S; B – Q, R; C – Q, S; D – P, S 53. A – Q, R; B – P, S; C – Q, R; D – Q, R
54. (c) 55. (a) 56. (abcd) 57. (abd) 58. (d) 59. (a) 60. (a) 61. (b) 62. (b) 63. (c)

Exercise - 4 : (Previous Year JEE Advanced Questions)

1. (c) 2. (d) 3. (b) 4. (b) 5. (a) 6. (d) 7. (a) 8. (d) 9. (d) 10. (b)
11. (b) 12. (d) 13. (b) 14. (b) 15. (b) 16. (c) 17. (a) 18. (a) 19. (a) 20. (a)
21. (c) 22. (c) 23. (d) 24. (a) 25. (d) 26. (b) 27. (a) 28. (d) 29. (b) 30. (c)
31. (ac) 32. (bc) 33. (cd) 34. (bd) 35. (bc) 36. (bd) 37. (bc) 38. (a) 39. (d)
40. ter-butyl carbonium ion, 41. aniline 42. propene 43. cyclopropane 44. sp^3
45. geminal, same 46. triangular planar 47. cyclic 48. less 49. butanedioic acid 50. hyperconjugation
59. (5)

Dream on !!

